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(54) **TREATMENT OF CARDIOVASCULAR DISEASE**

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(57) **ABSTRACT**

This disclosure relates to a nucleic acid comprising a double stranded RNA molecule comprising sense and antisense strands and further comprising a single stranded DNA molecule covalently linked to at least the 5' end of either the sense or antisense RNA part of the molecule wherein the double stranded inhibitory RNA targets genes associated with cardiovascular disease in the treatment hypercholesterolemia and diseases associated with hypercholesterolemia such as cardiovascular disease.

Specification includes a Sequence Listing.

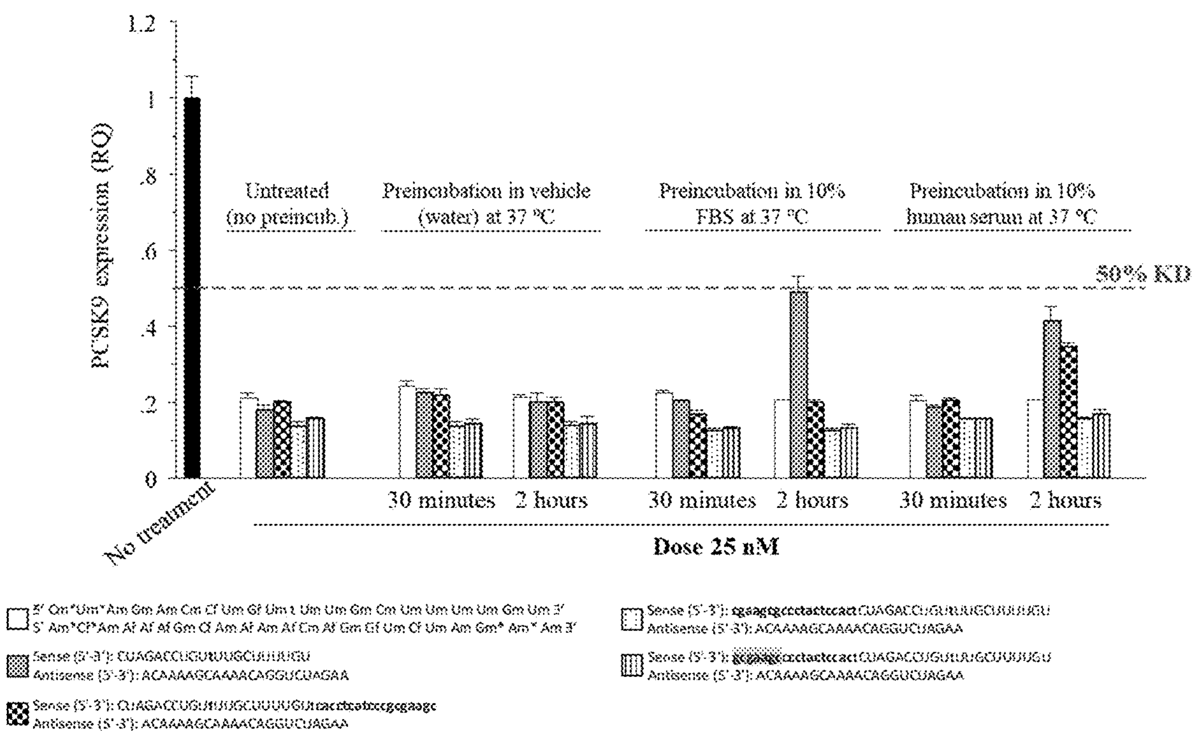


FIG. 1

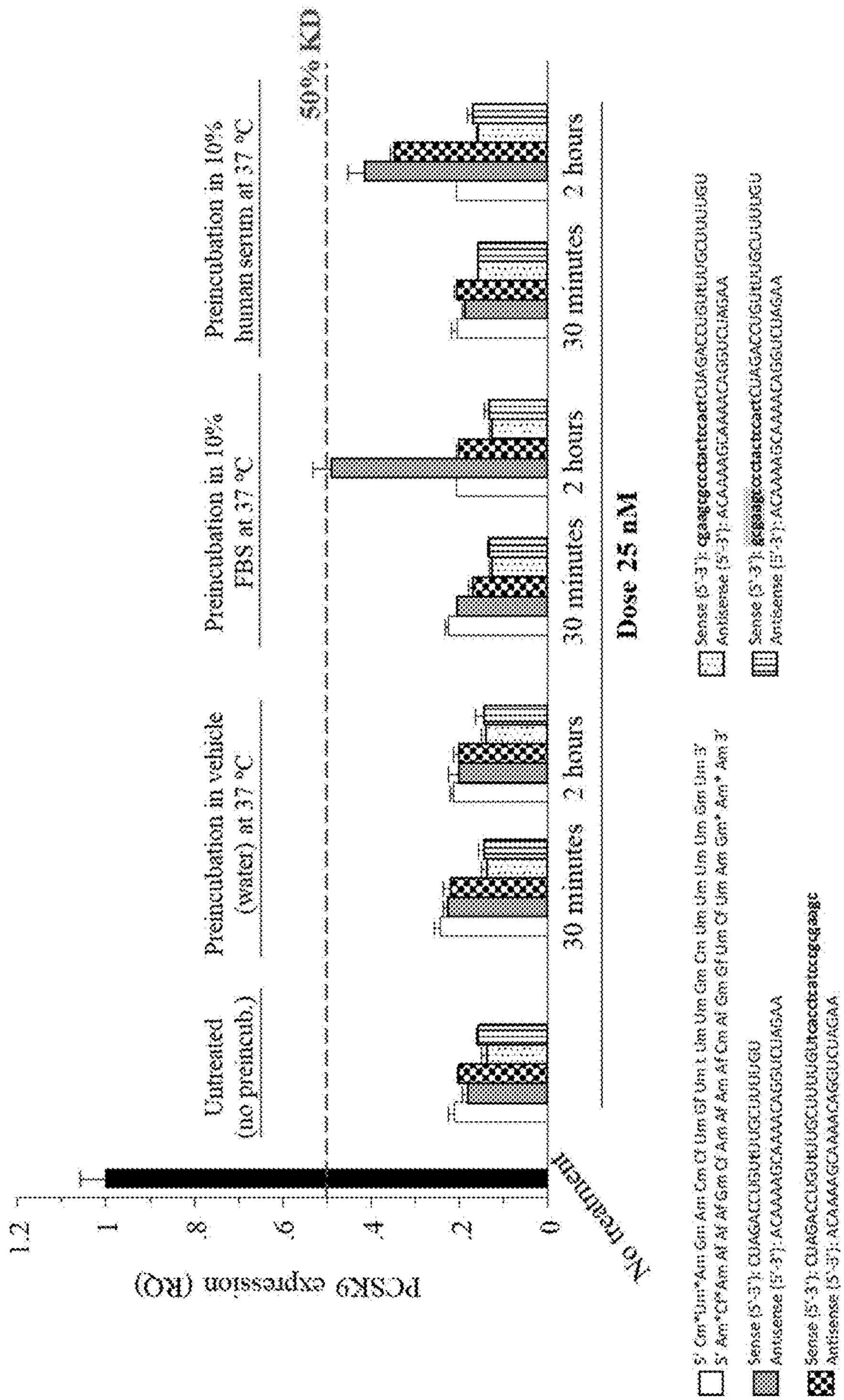

PCSK9 expression (RQ)
1.2
1
.8
.6
.4
.2
0
No treatment
Untreated (no preincub.)
Preincubation in vehicle (water) at 37 °C
Preincubation in 10% FBS at 37 °C
Preincubation in 10% human serum at 37 °C
50% KD
30 minutes
2 hours
30 minutes
2 hours
30 minutes
2 hours
Dose 25 nM

FIG. 2

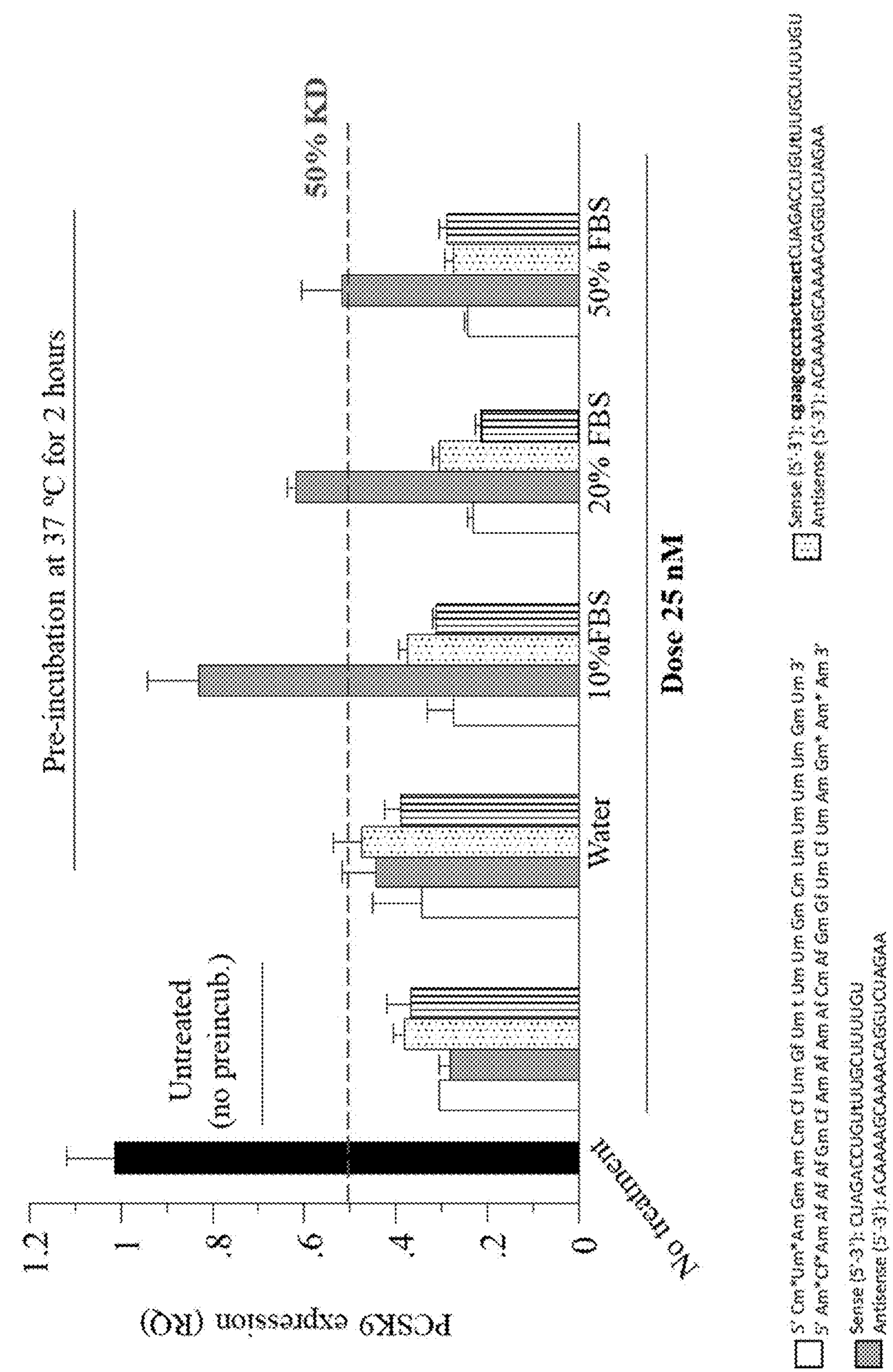

PCSK9 expression (RQ)
1.2
1
.8
.6
.4
.2
0
No treatment
Untreated
(no preincub.)
Pre-incubation at 37 °C for 2 hours
50% KD
Water
10%FBS
20% FBS
50% FBS
Dose 25 nM
5' Cm*Um*Am Gm Am Cm Cf Um Gf Um t Um Um Gm Cm Um Um Um Um Gm Um 3'
5' Am*Cf*Am Af Af Af Gm Cf Am Af Am Af Cm Af Gm Gf Um Cf Um Am Gm* Am* Am 3'
Sense (5'-3'): CUAGACCUGUtUUGCUUUUGU
Antisense (5'-3'): ACAAAAGCAAAACAGGUCUAGAA
Sense (5'-3'): cgaagcgcctactcactCUAGACCUGUtUUGCUUUUGU
Antisense (5'-3'): ACAAAAGCAAAACAGGUCUAGAA

FIG. 3

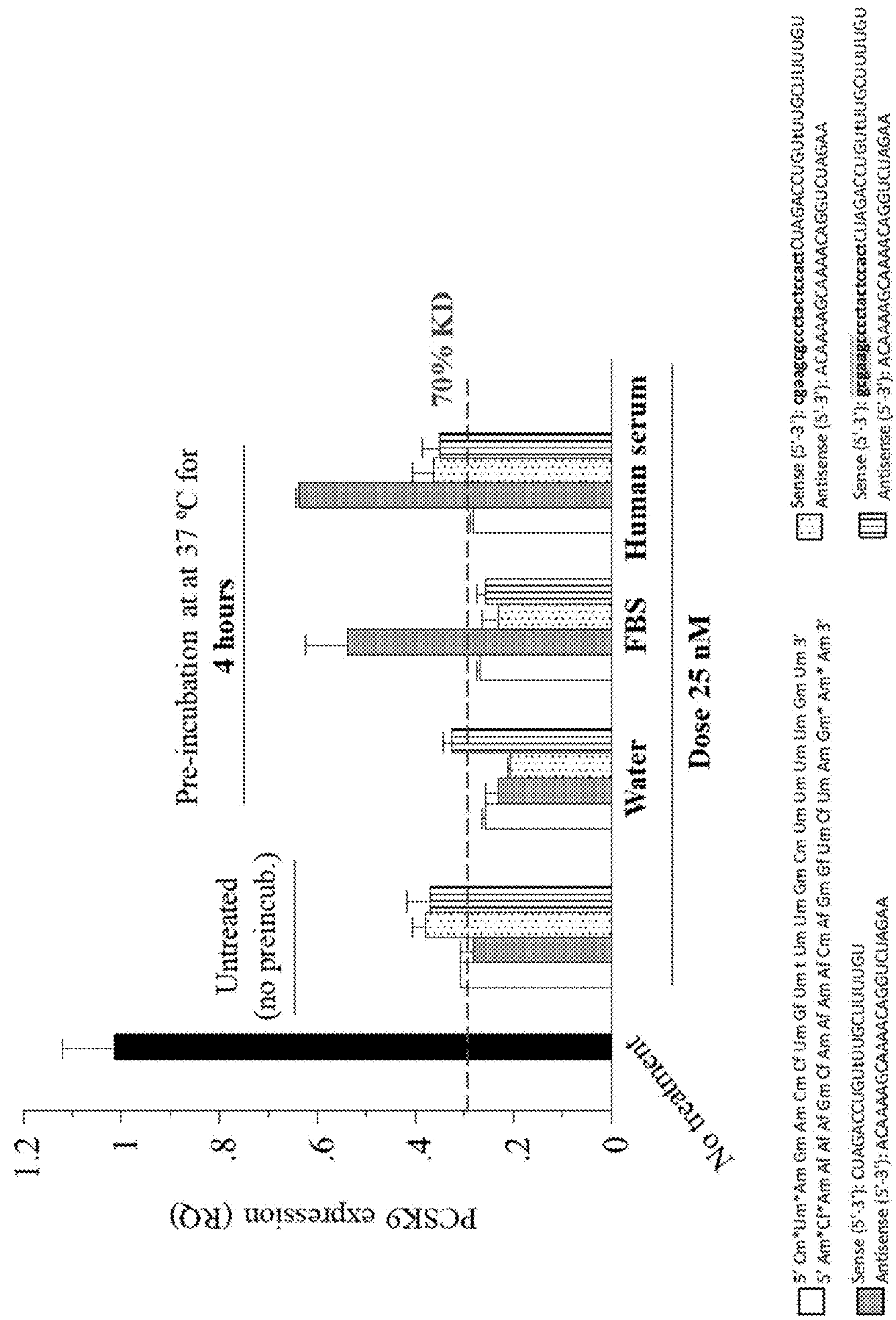

PCSK9 expression (RQ)
1.2
1
.8
.6
.4
.2
0
No treatment
Untreated
(no preincub.)
Pre-incubation at at 37 °C for
4 hours
Water
FBS
Human serum
70% KD
Dose 25 uM
5' Cm*Um*Am Gm Am Cm Cf Um Gf Um t Um Um Gm Cm Um Um Um Um Gm Um 3'
5' Am*Cf*Am Af Am Af Af Gm Cf Am Af Am Af Cm Af Gm Gf Um Cf Um Am Gm* Am* Am 3'
Sense (5'-3'): CUAGACCUGUUUUGCUUUUGU
Antisense (5'-3'): ACAAAAGCAAAACAGGUCUAGAA
Sense (5'-3'): cgaagcgccctactccactCUAGACCUGUUUUGCUUUUGU
Antisense (5'-3'): ACAAAAGCAAAACAGGUCUAGAA
Sense (5'-3'): gcgaagcgccctactccactCUAGACCUGUUUUGCUUUUGU
Antisense (5'-3'): ACAAAAGCAAAACAGGUCUAGAA

FIG. 4

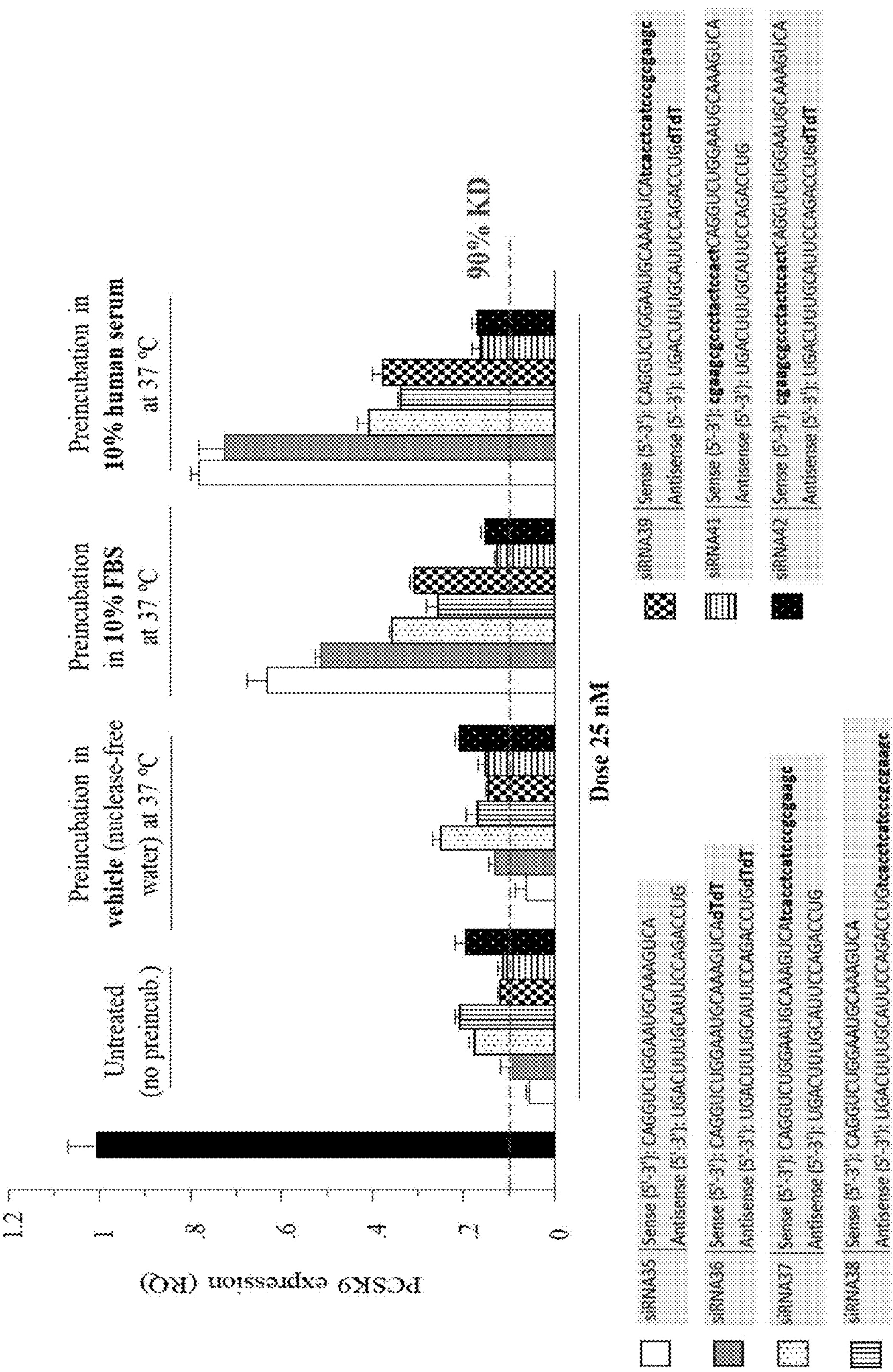

PCSK9 expression (RQ)
1.2
1
.8
.6
.4
.2
0
Untreated (no preincub.)
Preincubation in vehicle (nuclease-free water) at 37 °C
Preincubation in 10% FBS at 37 °C
Preincubation in 10% human serum at 37 °C
90% KD
Dose 25 nM
siRNA35 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCA
Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG
siRNA36 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAdTdT
Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdT
siRNA37 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAtcacctcatcccgcgaagc
Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG
siRNA38 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCA
Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGtcacctcatcccgcgaagc
siRNA39 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAtcacctcatcccgcgaagc
Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdT
siRNA41 Sense (5'-3'): cgaagcgcctactccactCAGGUCUGGAAUGCAAAGUCA
Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG
siRNA42 Sense (5'-3'): cgaagcgcctactccactCAGGUCUGGAAUGCAAAGUCA
Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdT

FIG. 5A

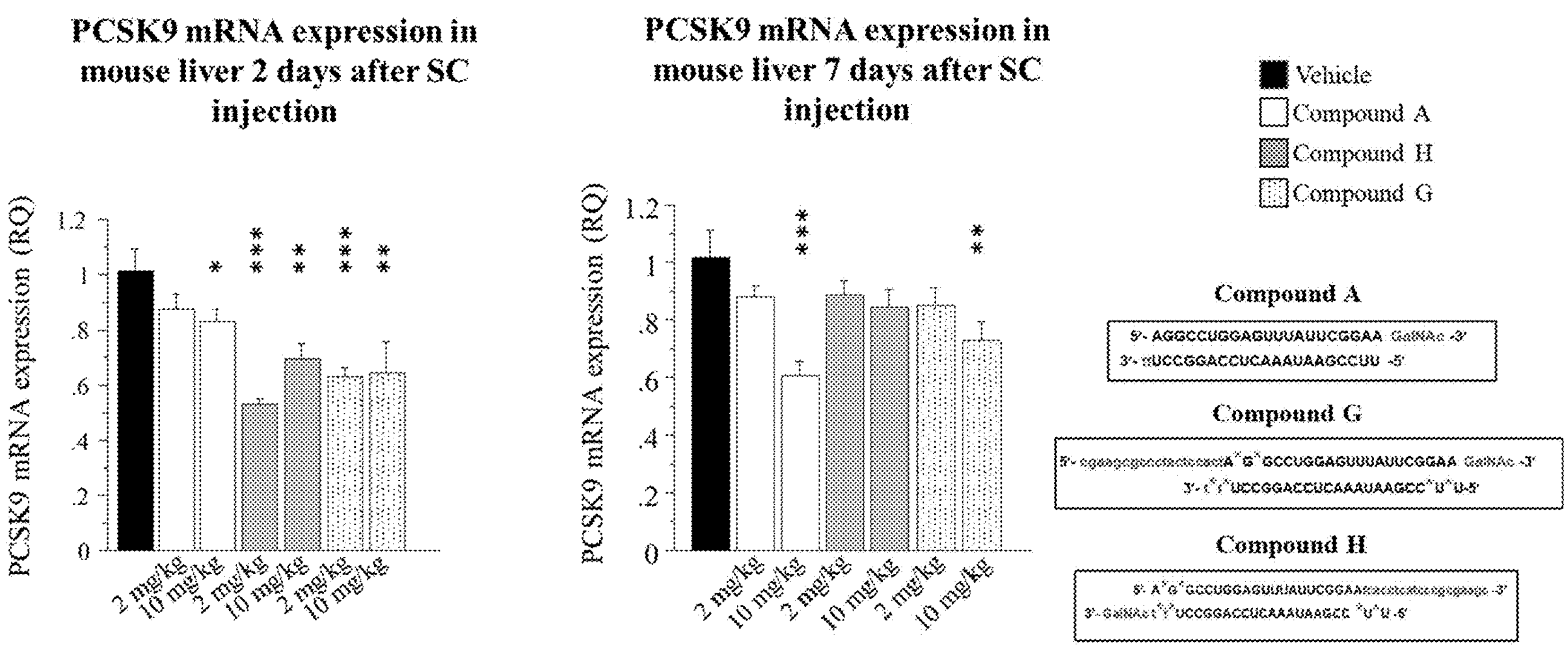

PCSK9 mRNA expression in mouse liver 2 days after SC injection
PCSK9 mRNA expression (RQ)
1.2
1
.8
.6
.4
.2
0
*
***
**
***
**
2 mg/kg
10 mg/kg
2 mg/kg
10 mg/kg
2 mg/kg
10 mg/kg
PCSK9 mRNA expression in mouse liver 7 days after SC injection
PCSK9 mRNA expression (RQ)
1.2
1
.8
.6
.4
.2
0
***
**
2 mg/kg
10 mg/kg
2 mg/kg
10 mg/kg
2 mg/kg
10 mg/kg
Vehicle
Compound A
Compound H
Compound G
Compound A
5'- AGGCCUGGAGUUUAUUCGGAA GalNAc -3'
3'- mUCCGGACCUCAAAUAAGCCUU -5'
Compound G
Compound H

FIG. 5B

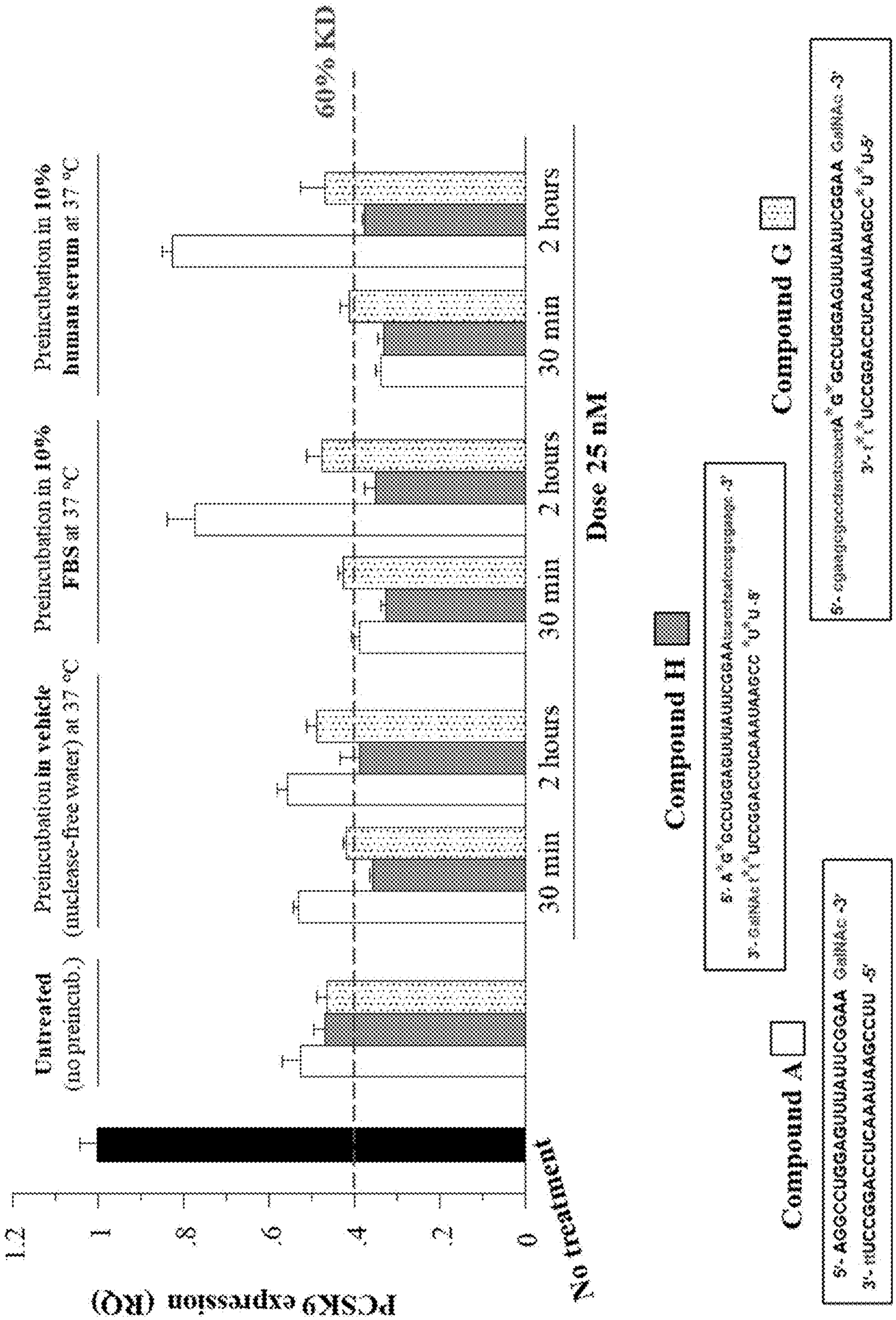
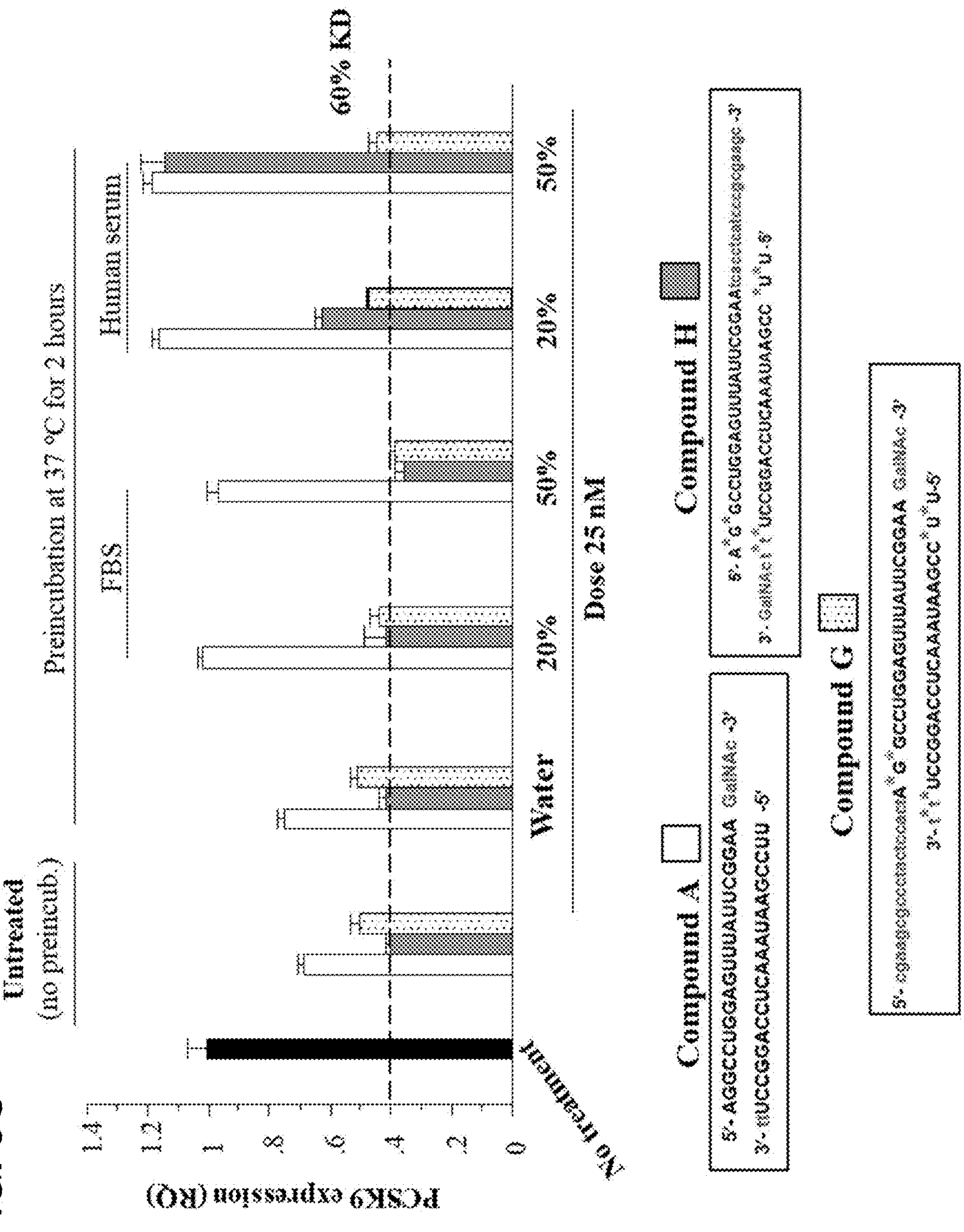

PCSK9 expression (RQ)
1.2
1
.8
.6
.4
.2
0
No treatment
Untreated (no preincub.)
Preincubation in vehicle (nuclease-free water) at 37 °C
Preincubation in 10% FBS at 37 °C
Preincubation in 10% human serum at 37 °C
30 min
2 hours
30 min
2 hours
30 min
2 hours
Dose 25 nM
60% KD
Compound A
5'- AGGCCUGGAGUUUAUUCGGAA GalNAc -3'
Compound H
Compound G

FIG. 5C


PCSK9 expression (RQ)
1.4
1.2
1
.8
.6
.4
.2
0
No treatment
Untreated
(no preincub.)
Preincubation at 37 °C for 2 hours
Water
FBS
Human serum
20%
50%
20%
50%
Dose 25 nM
60% KD
Compound A
Compound H
Compound G

TREATMENT OF CARDIOVASCULAR DISEASE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This is the U.S. National Stage of International Application No. PCT/EP2022/075355, filed Sep. 13, 2022, which was published in English under PCT Article 21(2), which claims the benefit of GB Application No. 2207239.1 filed May 18, 2022 and GB Application No. 2113104.0, filed Sep. 14, 2021.

FIELD OF THE DISCLOSURE

[0002] This disclosure relates to a nucleic acid comprising a double stranded RNA molecule comprising sense and antisense strands and further comprising a single stranded DNA molecule covalently linked to at least 5' end of either the sense or antisense RNA part of the molecule wherein the double stranded inhibitory RNA targets of cardiovascular disease genes; pharmaceutical compositions comprising said nucleic acid molecule and methods for the treatment of diseases associated with increased levels of expression of cardiovascular disease genes, for example hypercholesterolemia.

INCORPORATION OF ELECTRONIC SEQUENCE LISTING

[0003] The electronic sequence listing, submitted herewith as an XML file named 4860P-US.xml (772,353 bytes), created on Sep. 26, 2024, is herein incorporated by reference in its entirety.

BACKGROUND TO THE DISCLOSURE

[0004] Cardiovascular disease associated with hypercholesterolemia, for example ischaemic cardiovascular disease, is a common condition and results in heart disease and a high incidence of death and morbidity and can be a consequence of poor diet, obesity, or an inherited dysfunctional gene. For example, high levels of lipoprotein (a) and other lipoproteins, is associated with atherosclerosis. Cholesterol is essential for membrane biogenesis in animal cells. The lack of water solubility means that cholesterol is transported around the body in association with lipoproteins. Apolipoproteins form together with phospholipids, cholesterol and lipids which facilitate the transport of lipids such as cholesterol, through the bloodstream to the different parts of the body. Lipoproteins are classified according to size and can form HDL (High-density lipoprotein), LDL (Low-density lipoprotein), IDL (intermediate-density lipoprotein), VLDL (very low-density lipoprotein) and ULDL (ultra-low-density lipoprotein) lipoproteins.

[0005] Lipoproteins change composition throughout their circulation comprising different ratios of apolipoproteins A (ApoA), B (ApoB), C (ApoC), D(ApoD) or E (ApoE), triglycerides, cholesterol and phospholipids. For example, ApoB is the main apolipoprotein of ULDL and LDL and has two isoforms apoB-48 and apoB-100. Both ApoB isoforms are encoded by one single gene and wherein the shorter ApoB-48 gene is produced after RNA editing of the ApoB-100 transcript at residue 2180 resulting in the creation of a stop codon. ApoB-100 is the main structural protein of LDL and serves as a ligand for a cell receptor which allows transport of, for example, cholesterol into a cell.

[0006] Familial hypercholesterolemia is an orphan disease and results from elevated levels of LDL cholesterol (LDL-C) in the blood. The disease is an autosomal dominant disorder with both the heterozygous (350-550 mg/dL LDL-C) and homozygous (650-1000 mg/dL LDL-C) states resulting in elevated LDL-C. The heterozygous form of familial hypercholesterolemia is around 1:500 of the population. The homozygous state is much rarer and is approximately 1:1,000,000. The normal levels of LDL-C are in the region 130 mg/dL.

[0007] Hypercholesterolemia is particularly acute in paediatric patients which if not diagnosed early can result in accelerated coronary heart disease and premature death. If diagnosed and treated early the child can have a normal life expectancy. In adults, high LDL-C, either because of mutation or other factors, is directly associated with increased risk of atherosclerosis which can lead to coronary artery disease, stroke, or kidney disease. Lowering levels of LDL-C is known to reduce the risk of atherosclerosis and associated conditions. LDL-C levels can be lowered initially by administration of statins which block the de novo synthesis of cholesterol by inhibiting the HMG-CoA reductase. Some subjects can benefit from combination therapy which combines a statin with other therapeutic agents such as ezetimibe, colestipol or nicotinic acid. However, expression and synthesis of HMG-CoA reductase adapts in response to the statin inhibition and increases over time, thus the beneficial effects are only temporary or limited after statin resistance is established.

[0008] There is therefore a desire to identify alternative therapies that can be used alone or in combination with existing therapeutic approaches to control cardiovascular disease because of elevated LDL-C.

[0009] A technique to specifically ablate gene function is through the introduction of double stranded inhibitory RNA, also referred to as small inhibitory or interfering RNA (siRNA), into a cell which results in the destruction of mRNA complementary to the sequence included in the siRNA molecule. The siRNA molecule comprises two complementary strands of RNA (a sense strand and an antisense strand) annealed to each other to form a double stranded RNA molecule. The siRNA molecule is typically, but not exclusively, derived from exons of the gene which is to be ablated. Many organisms respond to the presence of double stranded RNA by activating a cascade that leads to the formation of siRNA. The presence of double stranded RNA activates a protein complex comprising RNase III which processes the double stranded RNA into smaller fragments (siRNAs, approximately 21-29 nucleotides in length) which become part of a ribonucleoprotein complex. The siRNA acts as a guide for the RNase complex to cleave mRNA complementary to the antisense strand of the siRNA thereby resulting in destruction of the mRNA.

[0010] The inhibition of expression of lipoprotein (a) is known and the use of inhibitory RNA to silence expression of lipoprotein (a) is also known. For example, WO2019/092283 discloses the identification of specific siRNA sequences that target knock down of mRNA encoding lipoprotein (a) and their use in the treatment of cardiovascular diseases linked to elevated lipoprotein (a) expression such as coronary heart disease, aortic stenosis or stroke. Similarly, U.S. Pat. No. 9,932,586 discloses specific siRNA sequences that target lipoprotein (a) expression and their use in the treatment of cardiovascular diseases linked to elevated lipoprotein (a) expression such as Buerger's disease, coronary heart disease, renal artery stenosis, hyperapobetalipoproteinemia, cerebrovascular atherosclerosis, cerebrovascular disease, and venous thrombosis.

[0011] Over expression of APOC III is associated with atherosclerosis and type 2 diabetes. For example, WO2003/020765 discloses a vaccination approach to the control of atherosclerosis using immunogens derived from ApoCIII polypeptide and its use in controlling atherosclerotic plaques in coronary and cerebrovascular disease. A similar vaccination approach is disclosed in WO2004/080375 and WO2001/064008. In WO2014/205449 and WO2014/179626 is disclosed the use of antisense oligonucleotides to improve insulin sensitivity and treat type II diabetes by targeting APOCIII expression.

[0012] Furthermore, WO2007/136989 and WO2005/019418 each disclose the use of antisense compounds directed to DGAT to regulate expression of DGAT2 and treat conditions that would benefit from reduction in DGAT2 expression in relation to conditions that would benefit from reduction in serum triglyceride levels such as hypercholesterolemia, cardiovascular disease, type II diabetes and metabolic syndrome. WO2018/093966 discloses the use of RNA silencing directed to DGAT2 and diglyceride acyltransferase 1(DGAT1) to treat obesity and obesity associated diseases such as hypercholesterolemia, cardiovascular disease, type II diabetes and metabolic syndrome. Similarly, WO2005/044981 discloses the use of siRNA to target DGAT2 amongst many other gene targets and their use in the treatment of diseases that would benefit from triglyceride regulation.

[0013] This disclosure relates to a nucleic acid molecule comprising a double stranded inhibitory RNA that is modified by the inclusion of a short DNA part linked to at least the 5' end of either the sense or antisense inhibitory RNA and which forms a hairpin structure. The double stranded inhibitory RNA uses solely or predominantly natural nucleotides and does not require modified nucleotides or sugars that prior art double stranded RNA molecules typically utilise to improve pharmacodynamics and pharmacokinetics. The disclosed double stranded inhibitory RNAs have activity in silencing cardiovascular gene targets with potentially fewer side effects.

STATEMENTS OF THE INVENTION

[0014] According to an aspect of the invention there is provided a nucleic acid molecule comprising

- **[0015]** a first part that comprises a double stranded inhibitory ribonucleic acid (RNA) molecule comprising a sense strand and an antisense strand; and
- **[0016]** a second part that comprises a single stranded deoxyribonucleic acid (DNA) molecule, wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule or wherein the 3' end of the single stranded DNA molecule is covalently linked to the 5' of the antisense strand of the double stranded inhibitory RNA molecule, characterized in that the double stranded inhibitory RNA comprises a sense nucleotide sequence that encodes a part of a cardiovascular gene target associated with cardiovascular disease and wherein said single stranded DNA molecule comprises a nucleotide sequence that is adapted over at least part of its length to anneal by complementary base pairing to a part of said single stranded DNA to form a double stranded DNA structure wherein said double stranded inhibitory RNA consists of natural nucleotides.

[0017] According to an aspect of the invention there is provided a nucleic acid molecule comprising

- **[0018]** a first part that comprises a double stranded inhibitory ribonucleic acid (RNA) molecule comprising a sense strand and an antisense strand; and
- **[0019]** a second part that comprises a single stranded deoxyribonucleic acid (DNA) molecule, wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule or wherein the 3' end of the single stranded DNA molecule is covalently linked to the 5' of the antisense strand of the double stranded inhibitory RNA molecule, characterized in that the double stranded inhibitory RNA comprises a sense nucleotide sequence that encodes a part of a cardiovascular gene target associated with cardiovascular disease, or a polymorphic sequence variant thereof, and wherein said single stranded DNA molecule comprises a nucleotide sequence that is adapted over at least part of its length to anneal by complementary base pairing to a part of said single stranded DNA to form a double stranded DNA structure comprising a stem and a loop domain, characterized in that said nucleic acid molecule comprises N-acetylgalactosamine and said double stranded inhibitory RNA consists of natural nucleotides.

[0020] A "polymorphic sequence variant" is a gene sequence that varies by one, two, three or more nucleotides.

[0021] In a preferred embodiment of the invention wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule.

[0022] In a preferred embodiment of the invention wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule.

[0023] In a preferred embodiment of the invention single stranded DNA molecule is covalently linked to the 5' end of said sense strand and the 5' end of said antisense strand.

[0024] In an alternative embodiment of the invention said single stranded DNA molecule is covalently linked to the 5' end of said sense strand, the 3' end of said sense strand.

[0025] In a preferred embodiment of the invention said loop portion comprises a region comprising the nucleotide sequence GNA or GNNA, wherein each N independently represents guanine (G), thymidine (T), adenine (A), or cytosine (C).

[0026] In a preferred embodiment of the invention said loop domain comprises G and C nucleotide bases.

[0027] In an alternative embodiment of the invention said loop domain comprises the nucleotide sequence GCGAAGC.

[0028] In a preferred embodiment of the invention said single stranded DNA molecule comprises the nucleotide sequence 5TCACCTCATCCCGCGAAGC 3' (SEQ ID NO 387 and 251).

[0029] In a preferred embodiment of the invention said single stranded DNA molecule comprises the nucleotide sequence 5' CGAAGCGCCCTACTCCACT 3'. (SEQ ID NO 130)

[0030] In a preferred embodiment of the invention said single stranded DNA molecule comprises the nucleotide sequence 5' GCGAAGCCCCTACTCCACT 3' (SEQ ID NO 400).

[0031] The inhibitory RNA molecules comprise or consist of natural nucleotide bases that do not require chemical modification. Moreover, in some embodiments of the invention, wherein the crook DNA molecule is linked to the 3' end of the sense strand of said double stranded inhibitory RNA, the antisense strand is optionally provided with at least a two-nucleotide base overhang sequence. The two-nucleotide overhang sequence can correspond to nucleotides encoded by the target or are non-encoding. The two-nucleotide overhang can be two nucleotides of any sequence and in any order, for example UU, AA, UA. AU. GG, CC, GC, CG, UG, GU, UC, CU, and TT.

[0032] In a preferred embodiment of the invention said inhibitory RNA molecule comprises a two-nucleotide overhang comprising or consisting of deoxythymidine dinucleotide (dTdT).

[0033] In a preferred embodiment of the invention said dTdT overhang is positioned at the 5' end of said antisense strand.

[0034] In an alternative preferred embodiment of the invention said dTdT overhang is positioned at the 3' end of said antisense strand.

[0035] In a preferred embodiment of the invention said dTdT overhang is positioned at the 5' end of said sense strand.

[0036] In an alternative preferred embodiment of the invention said dTdT overhang is positioned at the 3' end of said sense strand.

[0037] In a preferred embodiment of the invention said sense and/or said antisense strands comprises internucleotide phosphorothioate linkages.

[0038] In a preferred embodiment of the invention said sense strand comprises internucleotide phosphorothioate linkages.

[0039] Preferably, the 5' and/or 3' terminal two nucleotides of said sense strand comprises two internucleotide phosphorothioate linkage.

[0040] In a preferred embodiment of the invention said antisense strand comprises internucleotide phosphorothioate linkages.

[0041] Preferably, the 5' and/or 3' terminal two nucleotides of said antisense strand comprises two internucleotide phosphorothioate linkages.

[0042] In an alternative preferred embodiment of the invention said single stranded DNA molecule comprises one or more internucleotide phosphorothioate linkages.

[0043] In a preferred embodiment of the invention said nucleic acid molecule comprises a vinylphosphonate modification.

[0044] In a preferred embodiment of the invention said vinylphosphonate modification is to the 5' terminal phosphate of said sense RNA strand.

[0045] In a preferred embodiment of the invention said vinylphosphonate modification is to the 5' terminal phosphate of said antisense RNA strand.

[0046] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule is between 10 and 40 nucleotides in length.

[0047] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule is between 17 and 29 nucleotides in length.

[0048] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule is 19 to 21 nucleotides in length. Preferably, 19 nucleotides in length.

[0049] In a preferred embodiment of the invention said cardiovascular gene target is Human Lipoprotein (a).

[0050] In an alternative embodiment of the invention said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence selected from the group consisting of: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33 or 34.

[0051] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 41 and a sense nucleotide sequence comprising SEQ ID NO: 49, wherein said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule.

[0052] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 4 and a sense nucleotide sequence comprising SEQ ID NO: 44, wherein said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule.

[0053] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 5 and a sense nucleotide sequence comprising SEQ ID NO: 46, wherein said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule.

[0054] In an alternative preferred embodiment of the invention said cardiovascular gene target is Human Apolipoprotein C III (Apo C III).

[0055] Preferably, said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78 and 79.

[0056] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249 and 250.

[0057] Preferably said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 50, 51, 52, 53, 54, 55, 56, 57, 58, 80, 81, 82, 83, 84, 85, 86, 87, 88 and 89.

[0058] In an alternative preferred embodiment of the invention said cardiovascular gene target is Human diglyceride acyltransferase 2 (DGAT2).

[0059] Preferably, said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118 and 119.

[0060] Preferably, said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169 and 170.

[0061] Preferably said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 120, 121, 122, 123, 124, 125, 126, 127, 128 and 129.

[0062] In a preferred embodiment of the invention said cardiovascular gene target is Human PCSK9.

[0063] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 189 and 190.

[0064] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209 and 210.

[0065] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 255, 256, 257, 258, 259, 260, 261, 262, 263 and 264.

[0066] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 265, 266, 267, 268, 269, 270, 271, 272, 273 and 274.

[0067] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 292, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329 and 330.

[0068] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 285 and 386.

[0069] In a preferred embodiment of the invention said cardiovascular gene target is Human Apolipoprotein B.

[0070] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 499, 500, 453, 502, 503, 457, 505, 506, 462, 508, 509, 467, 511, 512, 472, 514, 515, 477, 517 518, 482, 520, 521, 487, 523, 524 and 492.

[0071] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising a nucleotide sequence selected from the group consisting of: 450, 501, 455, 504, 460, 507, 465, 510, 470, 513, 475, 516, 480, 519, 485, 522, 490 and 525.

[0072] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 1.

[0073] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 2.

[0074] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 3.

[0075] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 4.

[0076] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 5.

[0077] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 8.

[0078] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 10.

[0079] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 14.

[0080] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 15.

[0081] In a preferred embodiment of the invention said nucleic acid molecule comprises or consists of between 19 and 21 contiguous nucleotides of the nucleotide sequence set forth in SEQ ID NO:388.

[0082] In a preferred embodiment of the invention said nucleic acid molecule is covalently linked to N-acetylgalactosamine.

[0083] In a further embodiment of the invention N-acetylgalactosamine is linked to either the antisense part of said inhibitory RNA or the sense part of said inhibitory RNA.

[0084] Preferably, N-acetylgalactosamine is linked to the 5' terminus is of said sense RNA. In an alternative embodiment of the invention N-acetylgalactosamine is linked to the 3' terminus of said sense RNA.

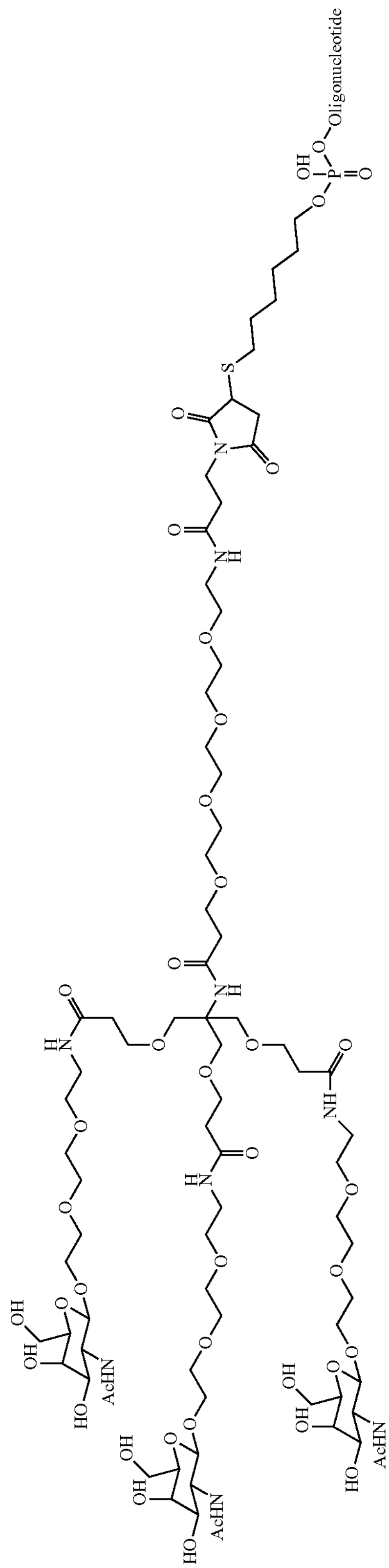
[0085] In an alternative preferred embodiment of the invention said N-acetylgalactosamine is linked to the 3' terminus of said antisense RNA.

[0086] In a preferred embodiment of the invention N-acetylgalactosamine is monovalent.

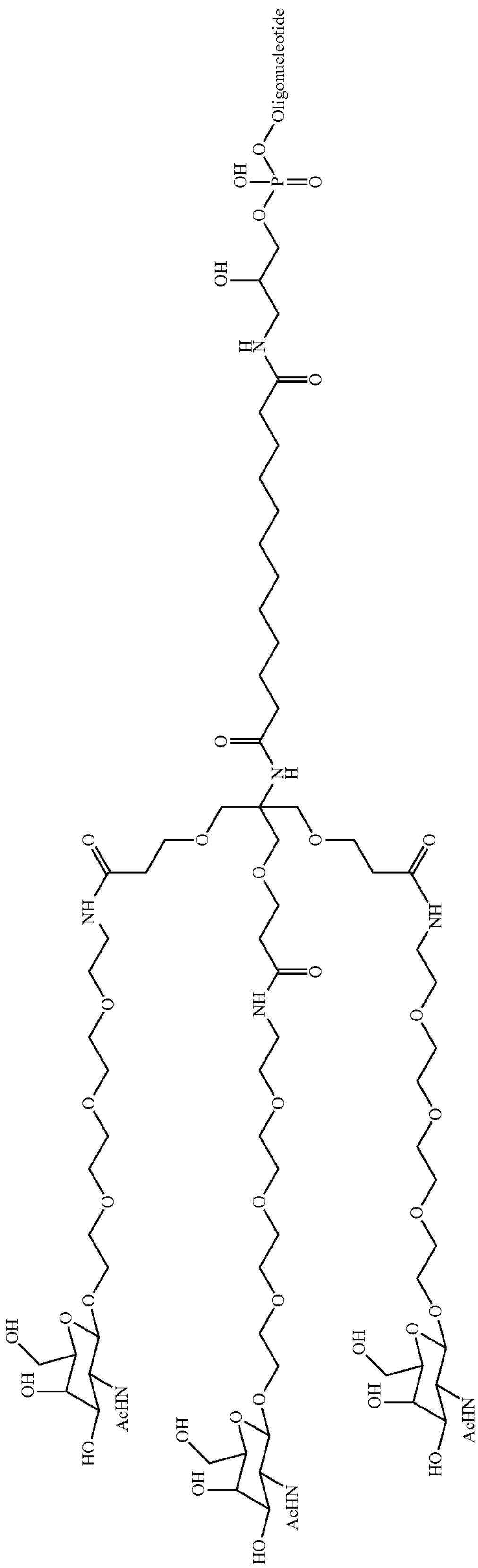
[0087] In a preferred embodiment of the invention N-acetylgalactosamine is divalent.

[0088] In an alternative embodiment of the invention N-acetylgalactosamine is trivalent.

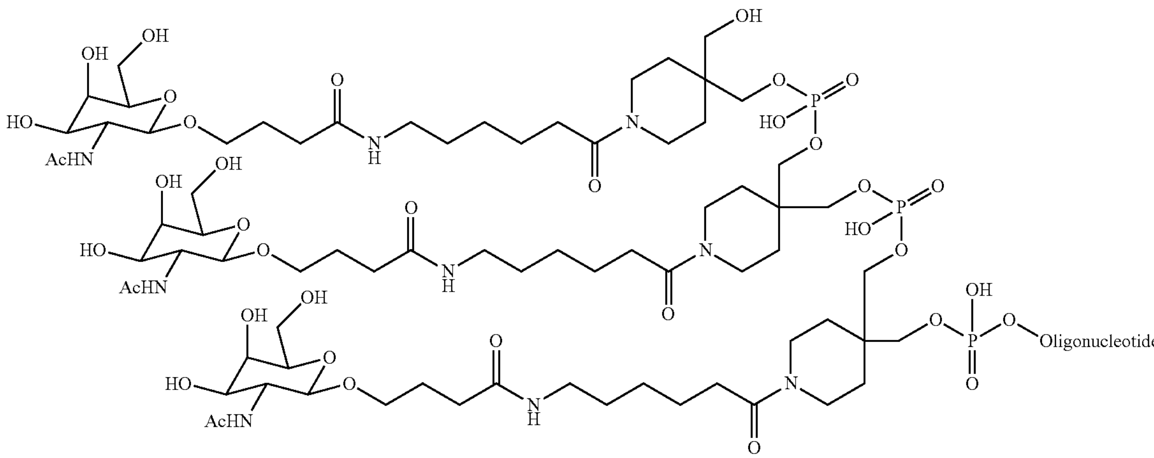
[0089] In a preferred embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising the structure:



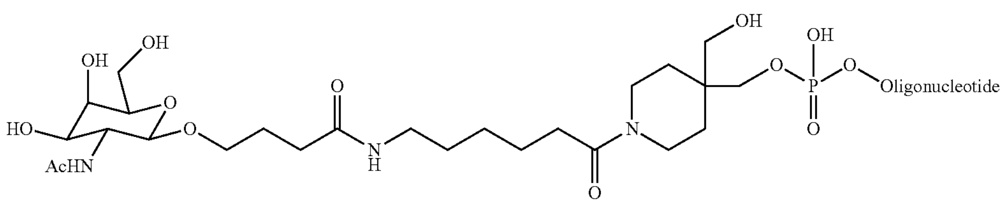
[0090] In an alternative embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising the structure:



[0091] In an alternative embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising the structure:



[0092] In an alternative embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising the structure:



[0093] In an alternative preferred embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising N-acetylgalactosamine 4-sulfate.

[0094] According to a further aspect of the invention there is provided a pharmaceutical composition comprising at least one nucleic acid molecule according to the invention.

[0095] In a preferred embodiment of the invention said composition further includes a pharmaceutical carrier and/or excipient.

[0096] When administered the compositions of the present invention are administered in pharmaceutically acceptable preparations. Such preparations may routinely contain pharmaceutically acceptable concentrations of salt, buffering agents, preservatives, compatible carriers and optionally other therapeutic agents, such as cholesterol lowering agents, which can be administered separately from the nucleic acid molecule according to the invention or in a combined preparation if a combination is compatible.

[0097] The combination of a nucleic acid according to the invention and the other, different therapeutic agent is administered as simultaneous, sequential or temporally separate dosages.

[0098] The therapeutics of the invention can be administered by any conventional route, including injection or by gradual infusion over time. The administration may, for example, be oral, intravenous, intraperitoneal, intramuscular, intracavity, subcutaneous, transdermal or transepithelial.

[0099] The compositions of the invention are administered in effective amounts. An "effective amount" is that amount of a composition that alone, or together with further doses, produces the desired response. In the case of treating a disease, such as cardiovascular disease, the desired response is inhibiting or reversing the progression of the disease. This may involve only slowing the progression of the disease temporarily, although more preferably, it involves halting the progression of the disease permanently. This can be monitored by routine methods.

[0100] Such amounts will depend, of course, on the particular condition being treated, the severity of the condition, the individual patient parameters including age, physical condition, size and weight, the duration of the treatment, the nature of concurrent therapy (if any), the specific route of administration and like factors within the knowledge and expertise of the health practitioner. These factors are well known to those of ordinary skill in the art and can be addressed with no more than routine experimentation. It is generally preferred that a maximum dose of the individual components or combinations thereof be used, that is, the highest safe dose according to sound medical judgment. It will be understood by those of ordinary skill in the art, however, that a patient may insist upon a lower dose or tolerable dose for medical reasons, psychological reasons or for virtually any other reasons.

[0101] The pharmaceutical compositions used in the foregoing methods preferably are sterile and contain an effective amount of a nucleic acid molecule according to the invention for producing the desired response in a unit of weight or volume suitable for administration to a patient. The response can, for example, be measured by determining regression of cardiovascular disease and decrease of disease symptoms etc.

[0102] The doses of the nucleic acid molecule according to the invention administered to a subject can be chosen in accordance with different parameters, in particular in accordance with the mode of administration used and the state of the subject. Other factors include the desired period of treatment. If a response in a subject is insufficient at the initial doses applied, higher doses (or effectively higher doses by a different, more localized delivery route) may be employed to the extent that patient tolerance permits. It will be apparent that the method of detection of the nucleic acid according to the invention facilitates the determination of an appropriate dosage for a subject in need of treatment.

[0103] In general, doses of the nucleic acid molecules herein disclosed of between 1 nM-1 μM generally will be formulated and administered according to standard procedures. Preferably doses can range from 1 nM-500 nM, 5 nM-200 nM, 10 nM-100 nM. Other protocols for the administration of compositions will be known to one of ordinary skill in the art, in which the dose amount, schedule of injections, sites of injections, mode of administration and the like vary from the foregoing. The administration of compositions to mammals other than humans, (e.g. for testing purposes or veterinary therapeutic purposes), is carried out under substantially the same conditions as described above. A subject, as used herein, is a mammal, preferably a human, and including a non-human primate or a transgenic mammal adapted for expression of human lipoprotein(a).

[0104] When administered, the pharmaceutical preparations of the invention are applied in pharmaceutically acceptable amounts and in pharmaceutically acceptable compositions. The term "pharmaceutically acceptable" means a non-toxic material that does not interfere with the effectiveness of the biological activity of the active ingredients. Such preparations may routinely contain salts, buffering agents, preservatives, compatible carriers, and optionally other therapeutic agents e.g., statins. When used in medicine, the salts should be pharmaceutically acceptable, but non-pharmaceutically acceptable salts may conveniently be used to prepare pharmaceutically acceptable salts thereof and are not excluded from the scope of the invention. Such pharmacologically and pharmaceutically acceptable salts include, but are not limited to, those prepared from the following acids: hydrochloric, hydrobromic, sulfuric, nitric, phosphoric, maleic, acetic, salicylic, citric, formic, malonic, succinic, and the like. Also, pharmaceutically acceptable salts can be prepared as alkaline metal or alkaline earth salts, such as sodium, potassium, or calcium salts.

[0105] Compositions may be combined, if desired, with a pharmaceutically acceptable carrier. The term "pharmaceutically acceptable carrier" as used herein means one or more compatible solid or liquid fillers, diluents or encapsulating substances which are suitable for administration into a human. The term "pharmaceutically acceptable carrier" in this context denotes an organic or inorganic ingredient, natural or synthetic, with which the active ingredient is combined to facilitate, for example, solubility and/or stability. The components of the pharmaceutical compositions also are capable of being co-mingled with the molecules of the present invention, and with each other, in a manner such that there is no interaction which would substantially impair the desired pharmaceutical efficacy.

[0106] The pharmaceutical compositions may contain suitable buffering agents, including acetic acid in a salt; citric acid in a salt; boric acid in a salt; and phosphoric acid in a salt. The pharmaceutical compositions also may contain, optionally, suitable preservatives.

[0107] The pharmaceutical compositions may conveniently be presented in unit dosage form and may be prepared by any of the methods well-known in the art of pharmacy. All methods include the step of bringing the active agent into association with a carrier which constitutes one or more accessory ingredients. In general, the compositions are prepared by uniformly and intimately bringing the active compound into association with a liquid carrier, a finely divided solid carrier, or both, and then, if necessary, shaping the product. Compositions suitable for oral administration may be presented as discrete units, such as capsules, tablets, lozenges, each containing a predetermined amount of the active compound.

[0108] Compositions suitable for parenteral administration conveniently comprise a sterile aqueous or non-aqueous preparation of nucleic acid, which is preferably isotonic with the blood of the recipient. This preparation may be formulated according to known methods using suitable dispersing or wetting agents and suspending agents. The sterile injectable preparation also may be a sterile injectable solution or suspension in a non-toxic parenterally acceptable diluent or solvent, for example, as a solution in 1, 3-butane diol. Among the acceptable solvents that may be employed are water, Ringer's solution, and isotonic sodium chloride solution. In addition, sterile, fixed oils are conventionally employed as a solvent or suspending medium. For this purpose, any bland fixed oil may be employed including synthetic mono- or di-glycerides. In addition, fatty acids such as oleic acid may be used in the preparation of injectables. Carrier formulation suitable for oral, subcutaneous, intravenous, intramuscular, etc. administrations can be found in Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, PA.

[0109] In a further preferred embodiment of the invention said pharmaceutical composition comprises at least one further, different, therapeutic agent.

[0110] In a preferred embodiment of the invention said further therapeutic agent is a statin.

[0111] Statins are commonly used to control cholesterol levels in subjects that have elevated LDL-C. Statins are effective in preventing and treating those subjects that are susceptible and those that have cardiovascular disease. The typical dosage of a statin is in the region 5 to 80 mg but this is dependent on the statin and the desired level of reduction of LDL-C required for the subject suffering from high LDL-C. However, expression and synthesis of HMG-CoA reductase, the target for statins, adapts in response to statin administration thus the beneficial effects of statin therapy are only temporary or limited after statin resistance is established.

[0112] Preferably said statin is selected from the group consisting of atorvastatin, fluvastatin, lovastatin, pitvastatin, pravastatin, rosuvastatin and simvastatin.

[0113] In a preferred embodiment of the invention said further therapeutic agent is ezetimibe.

[0114] Optionally, ezetimibe is combined with at least one statin, for example simvastatin.

[0115] In an alternative preferred embodiment of the invention said further therapeutic agent is selected from the group consisting of fibrates, nicotinic acid, cholestyramine.

[0116] In a further alternative preferred embodiment of the invention said further therapeutic agent is a therapeutic antibody, for example, evolocumab, bococizumab or alirocumab.

[0117] According to a further aspect of the invention there is provided a nucleic acid molecule or a pharmaceutical composition according to the invention for use in the treatment or prevention of a subject that has or is predisposed to hypercholesterolemia or diseases associated with hypercholesterolemia.

[0118] In a preferred embodiment of the invention said subject is a paediatric subject.

[0119] A paediatric subject includes neonates (0-28 days old), infants (1-24 months old), young children (2-6 years old) and prepubescent [7-14 years old] children.

[0120] In an alternative preferred embodiment of the invention said subject is an adult subject.

[0121] In a preferred embodiment of the invention the hypercholesterolemia is familial hypercholesterolemia.

[0122] In a preferred embodiment of the invention familial hypercholesterolemia is associated with elevated levels of lipoprotein (a) expression.

[0123] In a preferred embodiment of the invention said subject is resistant to statin therapy.

[0124] In a preferred embodiment of the invention said disease associated with hypercholesterolemia is selected from the group consisting of: stroke prevention, hyperlipidaemia, cardiovascular disease, atherosclerosis, coronary heart disease, aortic stenosis, cerebrovascular disease, peripheral arterial disease, hypertension, metabolic syndrome, type II diabetes, non-alcoholic fatty acid liver disease, non-alcoholic steatohepatitis, Buerger's disease, renal artery stenosis, hyperapobetalipoproteinemia, cerebrovascular atherosclerosis, cerebrovascular disease and venous thrombosis.

[0125] According to a further aspect of the invention there is provided a method to treat a subject that has or is predisposed to hypercholesterolemia comprising administering an effective dose of a nucleic acid or a pharmaceutical composition according to the invention thereby treating or preventing hypercholesterolemia.

[0126] In a preferred method of the invention said subject is a paediatric subject.

[0127] In an alternative preferred method of the invention said subject is an adult subject.

[0128] In a preferred method of the invention the hypercholesterolemia is familial hypercholesterolemia.

[0129] In a preferred method of the invention familial hypercholesterolemia is associated with elevated levels of lipoprotein (a) expression.

[0130] In a preferred method of the invention said subject is resistant to statin therapy.

[0131] In a preferred method of the invention said disease associated with hypercholesterolemia is selected from the group consisting of: stroke prevention, hyperiipidaemia, cardiovascular disease, atherosclerosis, coronary heart disease, aortic stenosis, cerebrovascular disease, peripheral arterial disease, hypertension, metabolic syndrome, type 11 diabetes, non-alcoholic fatty acid liver disease, non-alcoholic steatohepatitis, Buerger's disease, renal artery stenosis, hyperapobetalipoproteinemia, cerebrovascular atherosclerosis, cerebrovascular disease and venous thrombosis.

[0132] Throughout the description and claims of this specification, the words "comprise" and "contain" and variations of the words, for example "comprising" and "comprises", means "including but not limited to" and is not intended to (and does not) exclude other moieties, additives, components, integers or steps.

[0133] Throughout the description and claims of this specification, the singular encompasses the plural unless the context otherwise requires. In particular, where the indefinite article is used, the specification is to be understood as contemplating plurality as well as singularity, unless the context requires otherwise.

[0134] Features, integers, characteristics, compounds, chemical moieties or groups described in conjunction with an aspect, embodiment or example of the invention are to be understood to be applicable to any other aspect, embodiment or example described herein unless incompatible therewith.

[0135] An embodiment of the invention will now be described by example only and with reference to the following figures.

BRIEF SUMMARY OF THE DRAWINGS

[0136] FIG. **1** Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds. HepG2 cells were transfected with the following siRNAs after 30 mins or 2 hr incubation at 37 C in water, 10% FBS or 10% human serum: modified Inclisiran [white bar; SEQ ID NOs: 494 and 495], unmodified 'Inclisiran' with no Crook [grey bar; SEQ ID NOs: 389 and 390], unmodified Inclisiran with 3'SS Crook [hatched bar; SEQ ID NOs: 496 and 390], unmodified Inclisiran with 5' SS 'reversed hairpin' Crook [spotted bar; SEQ ID NOs: 497 and 390], or unmodified Inclisiran with 5'SS Crook [hatched bar; SEQ ID NOs: 498 and 390]. PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' treatment [black bar];

[0137] FIG. **2** Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds. HepG2 cells were transfected with the following siRNAs after a 2 hr incubation at 37 C in water, 10%, 20% or 50% FBS: modified Inclisiran [white bar; SEQ ID NOs: 494 and 495], unmodified 'Inclisiran' with no Crook [grey bar; SEQ ID NOs: 389 and 390], unmodified Inclisiran with 5' SS 'reversed hairpin' Crook [spotted bar; SEQ ID NOs: 497 and 390], or unmodified Inclisiran with 5'SS Crook [striped bar; SEQ ID NOs: 498 and 390]. PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' pre-treatment [black bar];

[0138] FIG. **3** Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds. HepG2 cells were transfected with the following siRNAs after a 4-hr incubation at 37 C in water, 10% FBS or 10% human serum: modified Inclisiran [white bar; SEQ ID NOs: 494 and 495], unmodified 'Inclisiran' with no Crook [grey bar; SEQ ID NOs: 389 and 390], unmodified Inclisiran with 5' SS 'reversed hairpin' Crook [spotted bar; SEQ ID NOs: 497 and 390], or unmodified Inclisiran with 5'SS Crook [striped bar; SEQ ID NOs: 498 and 390]. PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' pre-treatment [black bar];

[0139] FIG. **4** Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA (termed PC8-PC18) compounds. HepG2 cells were transfected with the following unmodified PC8-18 siRNAs after a 2-hr incubation at 37 C in water, 10% FBS or 10% human serum: siRNA35 with no Crook [white bar; SEQ ID NOs: 176 and 272], siRNA36 with no Crook but including dTdT overhangs on 3' SS & 3' AS [grey bar; SEQ ID NOs: 421 and 422], siRNA37 with Crook on 3' SS [spotted bar; SEQ ID NOs: 423 and 272], siRNA38 with Crook on 3' AS [vertical striped bar; SEQ ID NOs: 278 and 424], siRNA39 with Crook on 3' SS and dTdT overhang on 3' AS [hatched bar SEQ ID NOs: 423 and 422], siRNA41 with 5' SS 'reversed hairpin' Crook [horizontal stripe bar: SEQ ID NOs: 425 and 272], or siRNA42 with Crook on 5' SS and dTdT overhang on 3' AS [spots of black background bar; SEQ ID NOs: 425 and 422]. PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' pre-treatment [black bar];

[0140] FIG. **5**A In vivo silencing of liver PCSK9 mRNA following administration of unmodified siRNA compounds (termed PC2-PC12) Groups of 5 mice for each treatment group were injected subcutaneously (SC) with either vehicle [black bar], compound A (no Crook; white bar; SEQ ID NOs: 172 and 252), compound G (Crook on 5' end of sense strand (SS); spotted bar; SEQ ID NOs: 428 and 252), or compound H (Crook on 3' end of SS; grey bar; SEQ ID NOs: 253 and 252). Each compound was given at either 2 mg/kg or 10 mg/kg, and following sacrifice, levels of liver PCSK9 mRNA by RT-qPCR were measured at two time points (day 2 and day 7) and

[0141] FIG. **5**B Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds A, G and H, used in mouse in vivo study (FIG. **5**A). HepG2 cells were transfected with siRNA compounds A, G, or H after 30 min or 2 hr incubation at 37 C in water, 10% FBS or 10% human serum: compound A (no Crook; white bar' SEQ ID NOs: 172 and 252), compound G (Crook on 5' end of sense strand (SS); spotted bar; SEQ ID NOs: 428 and 252), or compound H (Crook on 3' end of SS; grey bar; SEQ ID NOs: 253 and 252). PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' [black bar], and 'no serum' pre-treatment.

[0142] FIG. **5**C Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds A, G and H, used in mouse in vivo study (FIG. **5**A). HepG2 cells were transfected with siRNA compounds A. G, or H after a 2 hr incubation at 37 C in water, 20% or 50% human serum: compound A (no Crook; white bar; SEQ ID NOs: 172 and 252), compound G (Crook on 5' end of sense strand (SS); spotted bar; SEQ ID NOs: 428 and 252), or compound H (Crook on 3' end of SS; grey bar; SEQ ID NOs: 253 and 252). PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' [black bar], and 'no serum' pre-treatment.

MATERIALS AND METHODS

Hepg2 Reverse Transfection

[0143] Duplex siRNAs synthesized by Bio-Synthesis (Lewisville, TX) (Table 1), were resuspended in Nuclease-free water (Invitrogen™ AM9932) to generate a stock solution of 10 μM. For serum stability assay, stock siRNAs were incubated at 37° C. in vehicle (nuclease-free water), 10% fetal bovine serum (FBS) or in various concentrations (10%-80%) of human serum (HS) for 2 hours. After pre-incubation in serum or vehicle, siRNAs were transfected into HepG2 cells in a 384-well plate (Thermo Scientific™ 164688) at a concentration of 25 nM using 0.15 μL of Lipofectamine RNAiMAX (Invitrogen™ 13778075) per well. Transfected cells were incubated at 37° C. and 5% CO_2. Cells receiving no siRNA treatment were used as control.

Free-Uptake and Transfection in Primary Mouse Hepatocytes

[0144] Mouse hepatocytes (MSCP10, Lonza) were thawed and seeded in a 384-well plate (Thermo Scientific™ 164688) in Williams E medium (Gibco™ A1217601) supplemented with Primary Hepatocyte Thawing and Plating Supplements (Gibco™ CM3000). Cells were treated with siRNAs at 25 nM using 0.15 μL of Lipofectamine per well or with 100 nM of GalNAc-siRNAs for free-uptake.

Duplex RT-qPCR

[0145] Cells were processed for RT-qPCR read-out using the Cells-to-CT 1-step TaqMan Kit (Invitrogen™ A25603). Briefly, cells were washed with 50 μL ice-cold PBS and lysed in 20 μl Lysis solution containing DNase I. Lysis was stopped after 5 minutes by addition of 2 μl STOP Solution for 2 min. For the RT-qPCR analysis, 1 μL of lysate was dispensed per well into a 96-well PCR plate in a 20 μL RT-qPCR reaction volume. RT-qPCR was performed using the TaqMan® 1-Step qRT-PCR Mix from the Cells-to-CT 1-step TaqMan Kit, with TaqMan probes for GAPDH (VIC_PL, Assay Id Hs00266705_g1) and PCSK9 (FAM, Assay Id Hs00545399-m1) or ApoB (FAM, Assay Id Mm01545150_m1). RT-qPCR was performed using a QuantStudio 5 thermocycling instrument (Applied BioSystems). Relative quantification was determined using the ΔΔCT method, where GAPDH was used as internal control and expression changes normalized to the reference sample (no siRNA treatment).

In Vivo Mouse Study Animals

[0146] Male C57BL/6J mice (20-25 g) were group housed in the Saretius animal unit at the University of Reading, and maintained under a 12 h light/dark cycle, at 23° C. with humidity controlled according to Home Office regulations. Mice were given access to standard rodent chow SDS rat expanded diet (RM3-E-FG) for the duration of the study.

Formulation of siRNA Compounds

[0147] Compound A, Compound G, and Compound H were each formulated in RNAase free PBS to concentrations of 0.4 and 2 mg/mL, to provide doses of 2 and 10 mg/kg when given subcutaneously (SC) in a 5 mL/kg dosing volume. Control groups (n=5) received Vehicle (RNase-free PBS) SC at 5 mL/kg dosing volume.

Liver Processing for RT-qPCR

[0148] At Day 2 (48 hrs) and Day 7 (168 hrs) following siRNA compound or Vehicle injection (n=5), each treatment group was terminally sampled by cardiac puncture under isoflurane. Liver tissue was excised and snap frozen in liquid N_2 Total RNA was extracted from homogenates of snap-frozen whole liver using GenElute™ Total RNA Purification Kit (RNB100-100RXN) Duplex RT-qPCR was performed using the ThermoFisher TaqMan Fast 1-Step Master Mix with TaqMan probes for GAPDH (VIC_PL), PCSK9 (FAM) and mTTR (FAM). Relative quantification (RQ) of PCSK9 was determined using the ΔΔCT method, where GAPDH was used as internal control and the expression changes of the target gene were normalized to the vehicle control.

Example 1

Testing 5' Versus 3' Positioning of Crook on the Sense Strand (SS) of Unmodified 'Inclisiran' Sequence in Serum Stability Assays

[0149] Following a 2 hr incubation in 10% FBS or 10% human serum, unmodified 'Inclisiran' with Crook positioned either at the 5' or 3' end of the SS, shows increased target mRNA (PCSK9) knockdown (KD) compared to the 'no crook' siRNA. However, superior KD is observed when crook is on the 5' end compared to 3' end of SS, following pre-treatment in human serum. This is demonstrated in FIG. 1, where 5' SS crook siRNA [striped bar] containing hairpin sequence GCGAAGC, maintains high levels of target KD (85%) in HepG2 cells following 2 hr treatment with 10% human serum comparable to that observed with modified Inclisiran (80%) [white bar]. Similar results can be shown when 'reversed' crook hairpin (CGAAGCG) is placed at the 5' end of the SS [spotted bar]. In contrast, 3' SS positioned Crook [hatched bar] shows −18.75% (loss of target KD) in HepG2 cells following pre-treatment in human serum; 65% KD (compared to 80% KD with no serum incubation). As expected, unmodified 'Inclisiran' with no Crook attached [grey bar] shows reduced levels of target KD after pre-treatment in either FBS or human serum: 50% and 60% KD, respectively, equating to −26.8% and −39% loss of KD.

Example 2

Testing 5' Positioning of Crook on the Sense Strand (SS) of Unmodified 'Incilsiran' Sequence in Serum Stability Assays with Increasing Concentrations of FBS

[0150] Following 2 hr incubations at 37 C in increasing concentrations of FBS, unmodified 'Inclisiran' sequence with Crook positioned at the 5' end of the SS [striped bar] shows sustained target mRNA (PCSK9) knockdown (70-80% KD) in all concentrations of FBS tested (10%, 20% and 50%), comparable to levels observed with modified Inclisiran (70-80% KD) [white bar]. Similarly, 'reversed' crook hairpin (CGAAGCG) on the 5' end of SS provides 65-75% KD with no loss of KD [spotted bar]. In contrast, the 'no crook' compound [grey bar] displays up to −85% loss of KD as only 20-50% target KD, is evident following serum treatment.

Example 3

Testing 5' Positioning of Crook on the Sense Strand (SS) of Unmodified 'Inclisiran' Sequence in Serum Stability Assays Over a 4-Hr Incubation Period

[0151] After a 4 hr incubation in either 10% FBS or 10% human serum, unmodified 'Inclisiran' with Crook positioned at the 5' end of the SS, shows sustained levels of approx. 75% target mRNA (PCSK9) knockdown (KD), and 65% KD, respectively [striped bar]. Similarly, there is no loss of KD evident for 'reversed' crook hairpin (CGAAGCG) on the 5' end of SS [spotted bar], comparable with modified 'Inclisiran', where approximately 70% KD is observed [white bar]. In contrast, the absence of Crook [grey bar] leads to substantially lower levels of KD following 4 hr pre-treatment in 10% FBS (45% KD) or 10% human serum (35% KD), equating to a −36% and −50% loss in KD, respectively.

Example 4

Testing 5' Versus 3' Positioning of Crook on an Unmodified siRNA Sequence Targeting Pcsk9, in Serum Stability Assays (Sequence Termed PC8-18)

[0152] Following a 2 hr incubation in 10% FBS or 10% human serum, PC8-18 with Crook positioned at the 5' end of the sense strand (SS), shows superior levels of knockdown (KD) of target mRNA (PCSK9) compared to 3' positioned Crook on either the SS or AS. This is shown in FIG. 4, where there is sustained target KD (approx. 85%) for PC8-18 siRNA with 5'SS Crook: [horizontal striped bar & spots on black background bar] compared to 60-70% KD (equating to a loss of 30% KD compared to no serum treatment) seen with 3' SS positioned Crook [spots on white background bar & hatched bar]. Similarly, when Crook is placed on 3' AS, loss of KD is 6-16% resulting in 65-75% target KD [vertical striped bar]. When Crook is not present on PC8-18 siRNA, target KD is reduced to only 35% following 2 hr incubation in FBS equating to a substantial loss of KD (−63% compared to no serum treatment), and to only 25% KD in human serum (−77%). Similarly, uncrooked molecules that contain 3' dTdT overhangs, show loss of KD levels of −44% and −72% (compared to no serum treatment) following pre-treatment in FBS and human serum, respectively.

Example 5 Testing the In Vivo Silencing Effect of 5' Versus 3' Positioning of Crook on an Unmodified siRNA Compound Targeting PCSK9 (PC2 Sequence)

[0153] Groups of 5 mice for each treatment group were injected subcutaneously (SC) with either vehicle (PBS), compound A (no Crook), compound G (Crook on 5' end of sense strand (SS)), or compound H (Crook on 3' end of SS). Each compound was given at either 2 mg/kg or 10 mg/kg, and following sacrifice, levels of liver PCSK9 mRNA were measured at two time points (day 2 and day 7).

[0154] Compound G (5' SS Crook) results in 40% KD of PCSK9 mRNA in the liver after 48 hours at 2 and 10 mg/kg and 30% KD at 10 mg/kg after 7 days, compared to vehicle controls (FIG. 5A). Comparable liver target KD is seen after 48 hrs for compound H (3' SS Crook) approx. 50% KD at 2 mg/kg (30% KD at 10 mg/kg), with no significant KD observed at day 7 (FIG. 5A). Compound A which contains no Crook, shows noticeably less target KD, with no silencing following SC injection of 2 mg/kg dose at either 2 or 7 days. At the 10 mg/kg dose, compound A shows and <20% KD after 48 hrs, and 40% after 7 days (FIG. 5A).

Example 6

Testing Compounds a, G and H in Serum Stability Assays (HepG2 Cells)

[0155] Comparable results are shown for both 5' and 3' positioned Crook on the SS. Compound G (5' SS Crook) and compound H (3' SS Crook) maintains PCSK9 mRNA KD of >50% following a 2 hr incubation in either 10% FBS or human serum (compared to no serum treatment). In contrast, there is loss of target KD seen for compound A (no Crook), from 50% to only 20% KD following a 2 hr serum treatment; FIG. 5B.

[0156] When these siRNA compounds were further challenged in increasing serum concentrations (20% and 50%) over a 2 hr period, compound G (5'SS Crook) displayed superior performance over 3'SS positioned Crook (H) in human serum. This is shown in FIG. 5C, where a sustained level of target mRNA KD (approx. 50%) is evident only in compound G [spotted bar] following 2 hrs incubation in 50% human serum. This equates to no loss of KD for G when compared to its 'no serum' treatment KD level. In contrast, compound H [grey bar] shows a complete loss in KD (0%) performing exactly as 'no crook' compound A [white bar] after 2 hrs in 50% human serum.

TABLE 1

Selection of Lp(a) candidate siRNA sequences to which crook is conjugated

siRNA ID (SEQ ID NO)	siRNA ID (SEQ ID NO)	Sense sequence (5'-3')	Antisense sequence (5'-3')
LP1 (43)	LP9 (2)	GCCCCUUAUUGUUAUACG (SEQ ID NO: 43)	CGUAUAACAAUAAGGGGC (SEQ ID NO: 2)
LP2 (44)	LP10 (3)	GCCCCUUAUUGUUAUACGA (SEQ ID NO: 44)	UCGUAUAACAAUAAGGGGC (SEQ ID NO: 3)
LP3 (45)	LP11 (4)	GCCCCUUAUUGUUAUACGA (SEQ ID NO: 45)	TCGUAUAACAAUAAGGGGC (SEQ ID NO: 4)
LP4 (46)	LP12 (5)	CCCCUUAUUGUUAUACGA (SEQ ID NO: 46)	UCGUAUAACAAUAAGGGG (SEQ ID NO: 5)
LPS (47)	LP13 (6)	CCCUUAUUGUUAUACGA (SEQ ID NO: 47)	UCGUAUAACAAUAAGGG (SEQ ID NO: 6)
LP6 (48)	LP14 (7)	CCCCUUAUUGUUAUACA (SEQ ID NO: 48)	UGUAUAACAAUAAGGGG (SEQ ID NO: 7)
LP7 (49)	LP15 (41)	CGGUAAUGGACAGAGUUAU (SEQ ID NO: 49)	AUAACUCUGUCCAUUACCG (SEQ ID NO: 41)
LP8 (42)	LP16 (35)	ACAGCCCCUUAUUGUUAUACGA (SEQ ID NO: 42)	CGUAUAACAAUAAGGGGC (SEQ ID NO: 35)

TABLE 2

Selection of APOC III and DGAT 2 siRNA sequences to which crook is conjugated

SEQ ID NO	Name	Sequence
50	APOC3_01	5'-ACGGGACAGUAUUCUCAGUNA
51	APOC3_02	5'-CCCAAUAAAGCUGGACAAGAA
52	APOC3_03	5'-CUGUAGGUUGCUUAAAAGGGA
53	APOC3_04	5'-CUGGAGCACCGUUAAGGACAA
54	APOC3_05	5'-UCCCAAUAAAGCUGGACAAGA
55	APOC3_06	5'-GCCCCUGUAGGUUGCUUAAAA
56	APOC3_07	5'-CCCUGAAAGACUACUGGAGCA
57	APOC3_08	5'-UGCUUAAAAGGGACAGUAUUC
58	APOC3_09	5'-GACCUCAAUACCCCAAGUCCA
59	APOC3_10	5'-GAGCACCGUUAAGGACAAGUU
60	APOC3_01	5'-ACGGGACAGUAUUCUCAGUNA
61	APOC3_02	5'-CCCAAUAAAGCUGGACAAGAA
62	APOC3_03	5'-CUGUAGGUUGCUUAAAAGGGA
63	APOC3_04	5'-CUGGAGCACCGUUAAGGACAA
64	APOC3_05	5'-UCCCAAUAAAGCUGGACAAGA
65	APOC3_06	5'-GCCCCUGUAGGUUGCUUAAAA
66	APOC3_07	5'-CCCUGAAAGACUACUGGAGCA

TABLE 2-continued

Selection of APOC III and DGAT 2 siRNA sequences to which crook is conjugated

SEQ ID NO	Name	Sequence
67	APOC3_08	5'-UGCUUAAAAGGGACAGUAUUC
68	APOC3_09	5'-GACCUCAAUACCCCAAGUCCA
69	APOC3_10	5'-GAGCACCGUUAAGGACAAGUUt
70	APOC3_01	5'-UCACUGAGAAUACUGUCCCGU-3'
71	APOC3_02	5'-UUCUUGUCCAGCUUUAUUGGG-3'
72	APOC3_03	5'-UCCCUUUUAAGCAACCUACAG-3'
73	APOC3_04	5'-UUGUCCUUAACGGUGCUCCAG-3'
74	APOC3_05	5'-UCUUGUCCAGCUUUAUUGGGA-3'
75	APOC3_06	5'-UUUUAAGCAACCUACAGGGGC-3'
76	APOC3_07	5'-UGCUCCAGUAGUCUUUCAGGG-3'
77	APOC3_08	5'-GAAUACUGUCCCUUUUAAGCA-3'
78	APOC3_09	5'-UGGACUUGGGGUAUUGAGGUC-3'
79	APOC3_10	5'-AACUUGUCCUUAACGGUGCUC-3'
80	APOC3_01	5'-UCACUGAGAAUACUGUCCCGU
81	APOC3_02	5'-UUCUUGUCCAGCUUUAUUGGG
82	APOC3_03	5'-UCCCUUUUAAGCAACCUACAG
83	APOC3_04	5'-UUGUCCUUAACGGUGCUCCAG

TABLE 2-continued

Selection of APOC III and DGAT 2 siRNA sequences to which crook is conjugated		
SEQ ID NO	Name	Sequence
84	APOC3_05	5'-UCUUGUCCAGCUUUAUUGGGA
85	APOC3_06	5'-UUUUAAGCAACCUACAGGGGC
86	APOC3_07	5'-UGCUCCAGUAGUCUUUCAGGG
87	APOC3_08	5'-GAAUACUGUCCCUUUUAAGCA
88	APOC3_09	5'-UGGACUUGGGGUAUUGAGGUC
89	APOC3_10	5'-AACUUGUCCUUAACGGUGCUC
90	DGAT2_01	5'-CUCUGUAAAUUUGGAAGUGUC
91	DGAT2_02	5'-CACCAUGAGCUAGGUGGAGUA
92	DGAT2_03	5'-UUCCUGAAGUGACAAAGGAAA
93	DGAT2_04	5'-GACCACCAGGAACUAUAUCUU
94	DGAT2_05	5'-GUUCCAGAAAUACAUUGGUUU
95	DGAT2_06	5'-AACCGCAAGGGCUUUGUGAAA
96	DGAT2_07	5'-GAGCAAGAAGUUCCCAGGCAU
97	DGAT2_08	5'-CAGUAGUAGGCAUCUGGAAUG
98	DGAT2_09	5'-GUCAUGGGUGUCUGUGGGUUA
99	DGAT2_10	5'-GCUCUGUAAAUUUGGAAGUGU
100	DGAT2_01	5'-CUCUGUAAAUUUGGAAGUGUC
101	DGAT2_02	5'-CACCAUGAGCUAGGUGGAGUA
102	DGAT2_03	5'-UUCCUGAAGUGACAAAGGAAA
103	DGAT2_04	5'-GACCACCAGGAACUAUAUCUU
104	DGAT2_05	5'-GUUCCAGAAAUACAUUGGUUU
105	DGAT2_06	5'-AACCGCAAGGGCUUUGUGAAA
106	DGAT2_07	5'-GAGCAAGAAGUUCCCAGGCAU

TABLE 2-continued

Selection of APOC III and DGAT 2 siRNA sequences to which crook is conjugated		
SEQ ID NO	Name	Sequence
107	DGAT2_08	5'-CAGUAGUAGGCAUCUGGAAUG
108	DGAT2_09	5'-GUCAUGGGUGUCUGUGGGUUA
109	DGAT2_10	5'-GCUCUGUAAAUUUGGAAGUGU
110	DGAT2_01	5'-GACACUUCCAAAUUUACAGAG-3'
111	DGAT2_02	5'-UACUCCACCUAGCUCAUGGUG-3'
112	DGAT2_03	5'-UUUCCUUUGUCACUUCAGGAA-3'
113	DGAT2_04	5'-AAGAUAUAGUUCCUGGUGGUC-3'
114	DGAT2_05	5'-AAACCAAUGUAUUUCUGGAAC-3'
115	DGAT2_06	5'-UUUCACAAAGCCCUUGCGGUU-3'
116	DGAT2_07	5'-AUGCCUGGGAACUUCUUGCUC-3'
117	DGAT2_08	5'-CAUUCCAGAUGCCUACUACUG-3'
118	DGAT2_09	5'-UAACCCACAGACACCCAUGAC-3'
119	DGAT2_10	5'-ACACUUCCAAAUUUACAGAGC-3'
120	DGAT2_01	5'-GACACUUCCAAAUUUACAGAG
121	DGAT2_02	5'-UACUCCACCUAGCUCAUGGUG
122	DGAT2_03	5'-UUUCCUUUGUCACUUCAGGAA
123	DGAT2_04	5'-AAGAUAUAGUUCCUGGUGGUC
124	DGAT2_05	5'-AAACCAAUGUAUUUCUGGAAC
125	DGAT2_06	5'-UUUCACAAAGCCCUUGCGGUU
126	DGAT2_07	5'-AUGCCUGGGAACUUCUUGCUC
127	DGAT2_08	5'-CAUUCCAGAUGCCUACUACUG
128	DGAT2_09	5'-UAACCCACAGACACCCAUGAC
129	DGAT2_10	5 -ACACUUCCAAAUUUACAGAGC

TABLE 3

Selection of DGAT2 siRNA sequences (SEQ ID NOs 131-170), PCSK9 (SEQ ID NO: 171-210 and ApoCIII (SEQ ID NO: 211 to 250)		
SEQ ID NO	Sequence	
131	GACCACCAGGAACUAUAUCUU	sense sequence
132	GUUCCAGAAAUACAUUGGUUU	sense sequence
133	AACCGCAAGGGCUUUGUGAAA	sense sequence
134	GAGCAAGAAGUUCCCAGGCAU	sense sequence
135	CUUUGGAGAGAAUGAAGUGUA	sense sequence
136	CUUCGACAAGCACAAGACCAA	sense sequence
137	GCCGAUGGGUCCAGAAGAAGU	sense sequence

TABLE 3-continued

Selection of DGAT2 siRNA sequences (SEQ ID NOs 131-170), PCSK9 (SEQ ID NO: 171-210 and ApoCIII (SEQ ID NO: 211 to 250)

SEQ ID NO	Sequence	
138	CUUCACUUGGCUGGUGUUUGA	sense sequence
139	CUCCUUUGGAGAGAAUGAAGU	sense sequence
140	UGCCAUCCUCAUGUACAUAUU	sense sequence
141	CCGCAAGGGCUUUGUGAAACU	sense sequence
142	AGCAAGAAGUUCCCAGGCAUA	sense sequence
143	AGUGUACAAGCAGGUGAUCUU	sense sequence
144	UGCUGACCACCAGGAACUAUA	sense sequence
145	CCGAUGGGUCCAGAAGAAGUU	sense sequence
146	UUUGGAGAGAAUGAAGUGUAC	sense sequence
147	UGGCGCUACUUUCGAGACUAC	sense sequence
148	AAUGCCUGUGUUGAGGGAGUA	sense sequence
149	AGUUCCAGAAAUACAUUGGUU	sense sequence
150	CAGAAGUGAGCAAGAAGUUCC	sense sequence
151	AAGAUAUAGUUCCUGGUGGUC	antisense sequence
152	AAACCAAUGUAUUUCUGGAAC	antisense sequence
153	UUUCACAAAGCCCUUGCGGUU	antisense sequence
154	AUGCCUGGGAACUUCUUGCUC	antisense sequence
155	UACACUUCAUUCUCUCCAAAG	antisense sequence
156	UUGGUCUUGUGCUUGUCGAAG	antisense sequence
157	ACUUCUUCUGGACCCAUCGGC	antisense sequence
158	UCAAACACCAGCCAAGUGAAG	antisense sequence
159	ACUUCAUUCUCUCCAAAGGAG	antisense sequence
160	AAUAUGUACAUGAGGAUGGCA	antisense sequence
161	AGUUUCACAAAGCCCUUGCGG	antisense sequence
162	UAUGCCUGGGAACUUCUUGCU	antisense sequence
163	AAGAUCACCUGCUUGUACACU	antisense sequence
164	UAUAGUUCCUGGUGGUCAGCA	antisense sequence
165	AACUUCUUCUGGACCCAUCGG	antisense sequence
166	GUACACUUCAUUCUCUCCAAA	antisense sequence
167	GUAGUCUCGAAAGUAGCGCCA	antisense sequence
168	UACUCCCUCAACACAGGCAUU	antisense sequence
169	AACCAAUGUAUUUCUGGAACU	antisense sequence
170	GGAACUUCUUGCUCACUUCUG	antisense sequence
171	CCUCAUAGGCCUGGAGUUUAU	sense sequence
172	AGGCCUGGAGUUUAUUCGGAA	sense sequence
173	CCCUCAUAGGCCUGGAGUUUA	sense sequence

TABLE 3-continued

Selection of DGAT2 siRNA sequences (SEQ ID NOs 131-170), PCSK9 (SEQ ID NO: 171-210 and ApoCIII (SEQ ID NO: 211 to 250)

SEQ ID NO	Sequence	
174	AGGUCUGGAAUGCAAAGUCAA	sense sequence
175	GGCCUGGAGUUUAUUCGGAAA	sense sequence
176	CAGGUCUGGAAUGCAAAGUCA	sense sequence
177	CCUCACCAAGAUCCUGCAUGU	sense sequence
178	ACCCUCAUAGGCCUGGAGUUU	sense sequence
179	CACCAGCAUACAGAGUGACCA	sense sequence
180	AUCUCCUAGACACCAGCAUAC	sense sequence
181	UCCUAGACACCAGCAUACAGA	sense sequence
182	CUGGAGUUUAUUCGGAAAAGC	sense sequence
183	GCCUGGAGUUUAUUCGGAAAA	sense sequence
184	GAGGCAGAGACUGAUCCACUU	sense sequence
185	UAGGCCUGGAGUUUAUUCGGA	sense sequence
186	CACUUCUCUGCCAAAGAUGUC	sense sequence
187	AUGCAAAGUCAAGGAGCAUGG	sense sequence
188	GGUCAUGGUCACCGACUUCGA	sense sequence
189	GGCAGCUGUUUUGCAGGACUG	sense sequence
190	GGGCAGGUUGGCAGCUGUUUU	sense sequence
191	AUAAACUCCAGGCCUAUGAGG	antisense sequence
192	UUCCGAAUAAACUCCAGGCCU	antisense sequence
193	UAAACUCCAGGCCUAUGAGGG	antisense sequence
194	UUGACUUUGCAUUCCAGACCU	antisense sequence
195	UUUCCGAAUAAACUCCAGGCC	antisense sequence
196	UGACUUUGCAUUCCAGACCUG	antisense sequence
197	ACAUGCAGGAUCUUGGUGAGG	antisense sequence
198	AAACUCCAGGCCUAUGAGGGU	antisense sequence
199	UGGUCACUCUGUAUGCUGGUG	antisense sequence
200	GUAUGCUGGUGUCUAGGAGAU	antisense sequence
201	UCUGUAUGCUGGUGUCUAGGA	antisense sequence
202	GCUUUUCCGAAUAAACUCCAG	antisense sequence
203	UUUUCCGAAUAAACUCCAGGC	antisense sequence
204	AAGUGGAUCAGUCUCUGCCUC	antisense sequence
205	UCCGAAUAAACUCCAGGCCUA	antisense sequence
206	GACAUCUUUGGCAGAGAAGUG	antisense sequence
207	CCAUGCUCCUUGACUUUGCAU	antisense sequence
208	UCGAAGUCGGUGACCAUGACC	antisense sequence
209	CAGUCCUGCAAAACAGCUGCC	antisense sequence

TABLE 3-continued

Selection of DGAT2 siRNA sequences (SEQ ID NOs 131-170), PCSK9 (SEQ ID NO: 171-210 and ApoCIII (SEQ ID NO: 211 to 250)

SEQ ID NO	Sequence	
210	AAAACAGCUGCCAACCUGCCC	antisense sequence
211	CUGGAGCACCGUUAAGGACAA	sense sequence
212	CCCUGAAAGACUACUGGAGCA	sense sequence
213	GAGCACCGUUAAGGACAAGUU	sense sequence
214	ACUGGAGCACCGUUAAGGACA	sense sequence
215	CCUGAAAGACUACUGGAGCAC	sense sequence
216	AAGACUACUGGAGCACCGUUA	sense sequence
217	CAGUUCCCUGAAAGACUACUG	sense sequence
218	GGUGACCGAUGGCUUCAGUUC	sense sequence
219	GGGUGACCGAUGGCUUCAGUU	sense sequence
220	ACUACUGGAGCACCGUUAAGG	sense sequence
221	GACUACUGGAGCACCGUUAAG	sense sequence
222	UUCAGUUCCCUGAAAGACUAC	sense sequence
223	GUUCCCUGAAAGACUACUGGA	sense sequence
224	UGGAGCACCGUUAAGGACAAG	sense sequence
225	CGCCACCAAGACCGCCAAGGA	sense sequence
226	GGGCUGGGUGACCGAUGGCUU	sense sequence
227	GCCACCAAGACCGCCAAGGAU	sense sequence
228	AGACUACUGGAGCACCGUUAA	sense sequence
229	CCACCAAGACCGCCAAGGAUG	sense sequence
230	UCCCUGAAAGACUACUGGAGC	sense sequence
231	UUGUCCUUAACGGUGCUCCAG	antisense sequence
232	UGCUCCAGUAGUCUUUCAGGG	antisense sequence
233	AACUUGUCCUUAACGGUGCUC	antisense sequence
234	UGUCCUUAACGGUGCUCCAGU	antisense sequence
235	GUGCUCCAGUAGUCUUUCAGG	antisense sequence
236	UAACGGUGCUCCAGUAGUCUU	antisense sequence
237	CAGUAGUCUUUCAGGGAACUG	antisense sequence
238	GAACUGAAGCCAUCGGUCACC	antisense sequence
239	AACUGAAGCCAUCGGUCACCC	antisense sequence
240	CCUUAACGGUGCUCCAGUAGU	antisense sequence
241	CUUAACGGUGCUCCAGUAGUC	antisense sequence
242	GUAGUCUUUCAGGGAACUGAA	antisense sequence
243	UCCAGUAGUCUUUCAGGGAAC	antisense sequence
244	CUUGUCCUUAACGGUGCUCCA	antisense sequence
245	UCCUUGGCGGUCUUGGUGGCG	antisense sequence

TABLE 3-continued

Selection of DGAT2 siRNA sequences (SEQ ID NOs 131-170), PCSK9 (SEQ ID NO: 171-210 and ApoCIII (SEQ ID NO: 211 to 250)

SEQ ID NO	Sequence	
246	AAGCCAUCGGUCACCCAGCCC	antisense sequence
247	AUCCUUGGCGGUCUUGGUGGC	antisense sequence
248	UUAACGGUGCUCCAGUAGUCU	antisense sequence
249	CAUCCUUGGCGGUCUUGGUGG	antisense sequence
250	GCUCCAGUAGUCUUUCAGGGA	antisense sequence

TABLE 4

siRNAs pairs used in silencing of APOC3 and DGAT 2 gene expression in HEPG2 cells in vitro

Name	Sense	Antisense
APOC3_01	5'-ACGGGACAGUAUUCUCAGUNAtcacctcatcccgcgaagc-3' (SEQ ID NO 401)	5'-UCACUGAGAAUACUGUCCCGU-3' (SEQ ID NO 70)
APOC3_02	5'-CCCAAUAAAGCUGGACAAGAAtcacctcatcccgcgaagc-3'(SEQ ID NO 402)	5'-UUCUUGUCCAGCUUUAUUGGG-3'(SEQ ID NO 71)
APOC3_03	5'-CUGUAGGUUGCUUAAAAGGGAtcacctcatcccgcgaagc-3'(SEQ ID NO 403)	5'-UCCCUUUUAAGCAACCUACAG-3(SEQ ID NO 72)
APOC3_04	5'-CUGGAGCACCGUUAAGGACAAtcacctcatcccgcgaagc-3'(SEQ ID NO 404)	5'-UUGUCCUUAACGGUGCUCCAG-3'(SEQ ID NO 73)
APOC3_05	5'-UCCCAAUAAAGCUGGACAAGAtcacctcatcccgcgaagc-3'(SEQ ID NO 405)	5'-UCUUGUCCAGCUUUAUUGGGA-3'(SEQ ID NO 74)
APOC3_06	5'-GCCCCUGUAGGUUGCUUAAAAtcacctcatcccgcgaagc-3'(SEQ ID NO 406)	5'-UUUUAAGCAACCUACAGGGGC-3'(SEQ ID NO 75)
APOC3_07	5'-CCCUGAAAGACUACUGGAGCAtcacctcatcccgcgaagc-3' (SEQ ID NO 407)	5'-UGCUCCAGUAGUCUUUCAGGG-3'(SEQ ID NO 76)
APOC3_08	5'-UGCUUAAAAGGGACAGUAUUCtcacctcatcccgcgaagc-3' (SEQ ID NO 408)	5'-GAAUACUGUCCCUUUUAAGCA-3'(SEQ ID NO 77)
APOC3_09	5'-GACCUCAAUACCCCAAGUCCAtcacctcatcccgcgaagc-3 (SEQ ID NO 409)	5'-UGGACUUGGGGUAUUGAGGUC-3'(SEQ ID NO 78)

TABLE 4-continued

siRNAs pairs used in silencing of APOC3 and DGAT 2 gene expression in HEPG2 cells in vitro

Name	Sense	Antisense
APOC3_10	5'-GAGCACCGUUAAGGACAAGUUtcacctcatcccgcgaagc-3'(SEQ ID NO 410)	5'-AACUUGUCCUUAACGGUGCUC-3'(SEQ ID NO 79)
DGAT2_01	5'-CUCUGUAAAUUUGGAAGUGUCtcacctcatcccgcgaagc-3' (SEQ ID NO 411)	5'-GACACUUCCAAAUUUACAGAG-3'(SEQ ID NO 110)
DGAT2_02	5'-CACCAUGAGCUAGGUGGAGUAtcacctcatcccgcgaagc-3'(SEQ ID NO 412)	5'-UACUCCACCUAGCUCAUGGUG-3'(SEQ ID NO 111)
DGAT2_03	5'-UUCCUGAAGUGACAAAGGAAAtcacctcatcccgcgaagc-3'(SEQ ID NO 413)	5'-UUUCCUUUGUCACUUCAGGAA-3'(SEQ ID NO 112)
DGAT2_04	5'-GACCACCAGGAACUAUAUCUUtcacctcatcccgcgaagc-3 (SEQ ID NO 414)	5'-AAGAUAUAGUUCCUGGUGGUC-3(SEQ ID NO 113)
DGAT2_05	5'-GUUCCAGAAAUACAUUGGUUUtcacctcatcccgcgaagc-3'(SEQ ID NO 415)	5'-AAACCAAUGUAUUUCUGGAAC-3'(SEQ ID NO 114)
DGAT2_06	5'-AACCGCAAGGGCUUUGUGAAAtcacctcatcccgcgaagc-3"(SEQ ID NO 416)	5'-UUUCACAAAGCCCUUGCGGUU-3'(SEQ ID NO 115)
DGAT2_07	5-GAGCAAGAAGUUCCCAGGCAUtcacctcatcccgcgaagc-3'(SEQ ID NO 417)	5'-AUGCCUGGGAACUUCUUGCUC-3'(SEQ ID NO 116)
DGAT2_08	5'-CAGUAGUAGGCAUCUGGAAUGtcacctcatcccgcgaagc-3'(SEQ ID NO 418)	5'-CAUUCCAGAUGCCUACUACUG-3'(SEQ ID NO 117)
DGAT2_09	5'-GUCAUGGGUGUCUGUGGGUUAtcacctcatcccgcgaagc-3'(SEQ ID NO 419)	5'-UAACCCACAGACACCCAUGAC-3'(SEQ ID NO 118)
DGAT2_10	5'-GCUCUGUAAAUUUGGAAGUGUtcacctcatcccgcgaagc-3 (SEQ ID NO 420)	5'-ACACUUCCAAAUUUACAGAGC-3'(SEQ ID NO 119)

TABLE 5

Crook structures tested in the serum stability assay for 5' crook siRNAs 14b to siRNA15-5'CR consist of unmodified 'inclisiran' sequence (C = crook; CR = reversed hairpin Crook). siRNAs 35-44 consist of PC8 sequence siRNAs A, G and H consist of PC2 sequence

Oligo name	Sequence
siRNA14m 'Inclisiran'	Sense: 5' Cm*Um*Am Gm Am Cm Cf Um Gf Um t Um Um Gm Cm Um Um Um Um Gm Um 3' (SEQ ID NO 494) Antisense: 5' Am*Cf*Am Af Af Af Gm Cf Am Af Am Af Cm Af Gm Gf Um Cf Um Am Gm* Am* Am 3' (SEQ ID NO 495)
siRNA14b	Sense (5'-3'): CUAGACCUGULUUGCUUUUGU (SEQ ID NO 389) Antisense (5'-3'): ACAAAAGCAAAACAGGUCUAGAA (SEQ ID NO 390)
siRNA15b	Sense (5'-3'): CUAGACCUGUtUUGCUUUUGU**tcacctcatcccgcgaagc** (SEQ ID NO 496) Antisense (5'-3'): ACAAAAGCAAAACAGGUCUAGAA (SEQ ID NO 390)
siRNA15-5'C	Sense (5'-3'): **cgaagcgccctactccact**CUAGACCUGUUUGCUUUUGU (SEQ ID NO 497) Antisense (5'-3'): ACAAAAGCAAAACAGGUCUAGAA(SEQ ID NO 390)

TABLE 5-continued

Crook structures tested in the serum stability assay for 5' crook
siRNAs 14b to siRNA15-5'CR consist of unmodified 'inclisiran' sequence (C = crook;
CR = reversed hairpin Crook).
siRNAs 35-44 consist of PC8 sequence
siRNAs A, G and H consist of PC2 sequence

Oligo name	Sequence
siRNA15-5'CR	Sense (5'-3'): **gcgaagcccctactccact**CUAGACCUGUTUUGCUUUUGU (SEQ ID NO 498) Antisense (5'-3'): ACAAAAGCAAAACAGGUCUAGAA(SEQ ID NO 390)
siRNA35	Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCA(SEQ ID NO 278 and 262 and 176) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG(SEQ ID NO 272, 196, 334)
siRNA36	Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAdTdT (SEQ ID NO 421) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdT (SEQ ID NO 422)
siRNA37	Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAdTdCdAdCdCdTdCdAdTdCdCdCdGdCdGdAdAd GdC (Seq ID NO 423) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334)
siRNA 38	Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO 278 and 262 and 176) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdCdAdCdCdTdCdAdTdCdCdCdGdCdGdAdAd GdC (SEQ ID NO 424)
siRNA39	Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAdTdCdAdCdCdTdCdAdTdCdCdCdGdCdGdAdAd GdC (Seq ID NO 423) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdT (SEQ ID NO 422)
siRNA40	Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAdTdT SEQ ID NO 421) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdCdAdCdCdTdCdAdTdCdCdCdGdCdGdAdAd GdC (SEQ ID NO 424)
siRNA41	Sense (5'-3'): dCdGdAdAdGdCdGdCdCdCdTdAdCdTdCdCdAdCdTCAGGUCUGGAAUGCAAAG UCA (SEQ ID NO 425) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334)
siRNA42	Sense (5'-3'): dCdGdAdAdGdCdGdCdCdCdTdAdCdTdCdCdAdCdTCAGGUCUGGAAUGCAAAG UCA (SEQ ID NO 425) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdT (SEQ ID NO 422)
siRNA43	Sense (5'-3'): dCdGdAdAdGdCdGdCdCdCdTdAdCdTdCdCdAdCdTCAGGUCUGGAAUGCAAAG UCAdTdT (SEQ ID NO 426) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334)
siRNA44	Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO 278 and 262 and 176) Antisense (5'-3'): dCdGdAdAdGdCdGdCdCdCdTdAdCdTdCdCdAdCdTUGACUUUGCAUUCCAGAC CUG (SEQ ID NO 427)
siRNA-A	5'-AGGCCUGGAGUUUAUUCGGAA GalNAc-3' (SEQ ID NO 172 and 256) 3'-**tt**UCCGGACCUCAAAUAAGCCUU-5' (SEQ ID NO 252 and 254)
siRNA-G	5'-cgaagcgccctactccactA*G*GCCUGGAGUUUAUUCGGAA GalNAc-3' (SEQ ID NO 428) 3'-t*t*UCCGGACCUCAAAUAAGCC*U*U-5' (SEQ ID NO: 527)
siRNA-H	5'-AGGCCUGGAGUUUAUUCGGAAtcacctcatcccgcgaagc-3' (SEQ ID NO 253) 3'-GalNAc UCCGGACCUCAAAUAAGCCUU-5' (SEQ ID NO 429)

Legend:
c, g, a, t or dT, dG, dA, dC: DNA bases
A, G, C, U: RNA bases
f: 2'-deoxy-2'-fluoro
m: 2'-O-methyl
*internucleotide linkage phosphorothioate (PS) GalNAc

Example 7 (Inclisiran, PCSK9 Sequence)

[0157] When crook was attached at the 5' end of the sense strand (siRNA15-5'C), the siRNA sequence maintained a full KD activity against the target PCSK9 comparable to the chemically modified version (siRNA4m) after 2-hour incubation in 10% FBS or human serum. Crook at the 3' end of the sense strand (siRNA5b) showed partial protection in HS. siRNA with short crook (harpin part only) at the 3' (siRNA15s7) and 5' end (Inc 03), as well as the stem 12-nt part only at the 5' end (INC_02), all showed significant loss of KD compared to the full 19-nt crook when transfected in HepG2 at 25 nM (Table 6 and 7).

TABLE 6

Sequence name	KD in no serum	KD after 10% FBS	KD after 10% HS	% KD loss in FBS	% KD loss in HS
siRNA14m	78.7	79.6	79.9	0.0	0.0
siRNA14b	82.0	51.4	59.1	37.3	28.0
siRNA15b	80.1	80.6	65.9	0.0	17.6
siRNA15-5'C	86.5	87.6	84.7	0.0	2.0
siRNA15s7	80.9	44.8	51.4	44.6	36.5

TABLE 7

siRNA name	KD in no serum	KD in 10% HS	% KD loss in HS
siRNA14m	50.4	62.2	0.0
siRNA14b	50.1	28.7	42.8
siRNA15-5'C	50.6	58.6	0.0
INC__02	34.2	0.0	100.0
INC__03	44.7	0.0	100.0

Example 8 (P8-18 PCSK9 Sequence)

[0158] When Crook was attached at the 5' end of the sense strand (PC8_05), the siRNA sequence maintained a full KD activity against the target PCSK9 after 2-hour incubation in 80% HS which was comparable to the level of KD observed with no serum pre-incubation. Crook at the 3' end of the sense strand (PC8_01) gave substantially reduced protection in HS showing 72% percentage loss of KD compared to no serum pre-incubation. siRNA with short crook (harpin part only) at the 3' (P8_03) and 5' end (PC8_11), as well as the stem 12-nt part only at the 5' end (PC8_10), all showed significant loss of KD compared to the full 19-nt crook when transfected in HepG2 at 25 nM (Table 9).

TABLE 9

siRNA name	KD in no serum	KD in 80% HS	% KD loss in 80% HS
PC8__00	72.9	0.0	100.0
PC8__01	51.6	14.5	72.0

TABLE 8

siRNA description		
siRNA name	Description	Sequence
siRNA14m	Fully chemically modified version	S (5'-3') Cm*Um*Am Gm Am Cm Cf Um Gf Um t Um Um Gm Cm Um Um Um Um Gm Um (SEQ ID NO: 494) AS (5'-3') Am*Cf*Am Af Af Af Gm Cf Am Af Am Af Cm Af Gm Gf Um Cf Um Am Gm* Am* Am (SEQ ID NO: 495)
siRNA14b	No crook	S (5'-3'): CUAGACCUGUtUUGCUUUUGU (SEQ ID NO: 389) AS (5'-3'): ACAAAAGCAAAACAGGUCUAGAA (SEQ ID NO: 390)
siRNA15b	Crook on 3' S strand	S (5'-3'): CUAGACCUGUtUUGCUUUUGU**tcacctcatcccgcgaagc** (SEQ ID NO: 496) AS (5'-3'): ACAAAAGCAAAACAGGUCUAGAA (SEQ ID NO: 390)
siRNA15-5'C	Crook on 5' S strand	S (5'-3'): **cgaagcgccctactccact**CUAGACCUGUtUUGCUUUUGU (SEQ ID NO: 497) AS (5'-3'): ACAAAAGCAAAACAGGUCUAGAA (SEQ ID NO: 390)
siRNA15s7	Harpin part only on 3' S strand	S (5'-3'): CUAGACCUGUtUUGCUUUUGU**gcgaagc** (SEQ ID NO 430) AS (5'-3'): ACAAAAGCAAAACAGGUCUAGAA (SEQ ID NO 390)
Inc_02	12-nt stem only, no hairpin	S (5'-3'): **ccctactccact**CUAGACCUGUtUUGCUUUUGU (SEQ ID NO 431) AS (5'-3'): ACAAAAGCAAAACAGGUCUAGAA (SEQ ID NO 390)
Inc 03	Hairpin on 5' S strand	S (5'-3'): **cgaagcg**CUAGACCUGUtUUGCUUUUGU (SEQ ID NO 432) AS (5'-3'): ACAAAAGCAAAACAGGUCUAGAA (SEQ ID NO 390)

TABLE 9-continued

siRNA name	KD in no serum	KD in 80% HS	% KD loss in 80% HS
PC8_03	50.0	1.4	97.3
PC8_05	63.3	60.4	4.6
PC8_10	54.0	25.8	52.2
PC8_11	72.6	17.3	76.2

TABLE 11

siRNA name	KD in no serum	KD in 80% HS	% KD loss in HS
siRNA_A	53.2	15.5	70.8
siRNA_G	59.5	58.5	1.7
siRNA_H	57.6	0.0	100.0

TABLE 10

siRNA name	Description	Sequence
PC8_00	No crook	S (5'-3'): CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO 262, 278, 176) AS (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334)
PC8_01	Crook on 3' S strand	S (5'-3'): CAGGUCUGGAAUGCAAAGUCA**tcacctcatcccgcgaagc** (SEQ ID NO 433) AS (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334)
PC8_03	Crook hairpin on 3' S strand	S (5'-3'): CAGGUCUGGAAUGCAAAGUCA**gcgaagc** (SEQ ID NO 434) AS (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334)
PC8_05	Crook on 5' S strand	S (5'-3'): **cgaagcgccctactccact**CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO 435) AS (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334)
PC8_10	12-nt stem only, no hairpin	S (5'-3'): **ccctactccact**CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO 436) AS (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334)
SIRNA_A	No crook	S (5'- 3') AGGCCUGGAGUUUAUUCGGAA GalNAc (SEQ ID NO 172, 256) AS (3'-5') ttUCCGGACCUCAAAUAAGCCUU (SEQ ID NO 252, 254)
SIRNA_G	Crook on 5' S strand	S (5'-3') **cgaagcgccctactccact**A*G*GCCUGGAGUUUAUUCGGAA GalNAc (SEQ ID NO 437) AS (3'-5') t*t*UCCGGACCUCAAAUAAGCC*U*U (SEQ ID NO: 527)
SIRNA_H	Crook on 3' S strand	S (5'-3) A*G*GCCUGGAGUUUAUUCGGAA**tcacctcatcccgcgaagc** (SEQ ID NO 438) AS (3'-5') GalNAc t*t*UCCGGACCUCAAAUAAGCC*U*U (SEQ ID NO: 528)
PC8_11	Hairpin on 5' S strand	S (5'-3'): **cgaagcg**CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO: 526) AS (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334)

Example 9 (Compound G—PCSK9 Sequence)

[0159] When Crook was attached at the 5' end of the sense strand (siRNA_G), the siRNA sequence maintained a full KD activity against PCSK9 after 8-hour incubation in 80% HS comparable to the level of KD observed with no serum pre-incubation (Table 11). In contrast, siRNA_A (no crook) or siRNA_H (crook at the 3' end of the sense strand) showed no protection in 80% HS and a loss of % KD of 70.8% and 100% respectively when transfected in HepG2 at 25 nM. In a free-uptake assay, siRNA_G showed better KD levels compared to siRNA_H in primary mouse hepatocytes cultured in 10% FBS and treated for 24, 48 and 72 hours with 100 nM of siRNA (Table 12).

TABLE 12

KD levels of PCSK9 following free-uptake of siRNA_A, siRNA_H and siRNA_G at 100 nM in primary mouse hepatocytes cultured in 10% FBS

Time from treatment	% KD siRNA_A	% KD siRNA_H	% KD siRNA_G
24 hours	0.0	0.0	15.0
48 hours	5.8	17.5	50.6
72 hours	0.0	0.0	36.5

Example 10 (ApoB Sequences)

[0160] When a total of 11 siRNAs carrying a sequence against mouse ApoB were exposed to 20% and 50% human serum, and subsequently transfected at 25 nM into primary mouse hepatocytes, the siRNA variants carrying 5' crook on the sense strand showed, overall, a better ability to induce KOD of ApoB after exposure to serum compared to the siRNA variants carrying crook on the 3' end (Table 13).

TABLE 13

siRNA Name	Crook position	KD % no serum	KD 20% HS	KD 50% HS	KD % loss in 20% HS	KD % loss in 50% HS
TS3_1	5'SS	70.5	55.1	0.0	21.8	100.0
TS3_2	3'SS	75.4	0.0	0.0	100.0	100.0
TS3_3	3'AS	73.1	0.0	0.0	100.0	100.0
TS4_1	5'SS	64.4	9.3	8.8	85.6	86.3
TS4_2	3'SS	63.4	10.9	0.0	82.7	100.0
TS4_3	3'AS	67.9	0.0	0.0	100.0	100.0
ApoB_C10_1	5'SS	58.0	67.3	62.3	0.0	0.0
ApoB_C10_2	3'SS	52.4	0.0	38.2	100.0	27.0
ApoB_C10_3	3'AS	56.6	56.6	18.9	0.0	66.6
ApoB_C3_1	5'SS	48.1	50.9	37.6	0.0	21.7
ApoB_C3_2	3'SS	58.1	49.8	48.0	14.2	17.3

TABLE 13-continued

siRNA Name	Crook position	KD % no serum	KD 20% HS	KD 50% HS	KD % loss in 20% HS	KD % loss in 50% HS
ApoB_C3_3	3'AS	54.9	41.2	0.0	25.0	100.0
ApoB_C2_1	5'SS	66.8	87.7	87.6	0.0	0.0
ApoB_C2_2	3'SS	70.9	87.1	86.8	0.0	0.0
ApoB_C2_3	3'AS	79.6	87.5	87.6	0.0	0.0
ApoB_DM2_1	5'SS	58.4	76.4	67.4	0.0	0.0
ApoB_DM2_2	3'SS	64.0	65.4	4.3	0.0	93.3
ApoB_DM2_3	3'AS	67.6	55.3	5.0	18.1	92.5
ApoB_DM3_1	5'SS	72.0	56.4	0.0	21.7	100.0
ApoB_DM3_2	3'SS	75.4	0.0	0.0	100.0	100.0
ApoB_DM3_3	3'AS	82.7	0.0	0.0	100.0	100.0
ApoB_DM5_1	5'SS	51.6	63.7	73.4	0.0	0.0
ApoB_DM5_2	3'SS	59.5	0.0	0.0	100.0	100.0
ApoB_DM5_3	3'AS	55.7	74.9	78.4	0.0	0.0
ApoB_DM13_1	5'SS	84.8	87.2	49.1	0.0	42.1
ApoB_DM13_2	3'SS	88.6	50.3	45.6	43.3	48.6
ApoB_DM13_3	3'AS	87.0	41.3	23.7	52.6	72.7
ApoB_DM18_1	5'SS	81.8	70.8	50.9	13.4	37.8
ApoB_DM18_2	3'SS	83.4	80.0	36.2	4.1	56.6
ApoB_DM18_3	3'AS	86.8	32.4	7.8	62.6	91.0
ApoB_DM19_1	5'SS	60.0	1.7	0.0	97.1	100.0
ApoB_DM19_2	3'SS	68.6	0.0	0.0	100.0	100.0
ApoB_DM19_3	3'AS	74.4	0.0	0.0	100.0	100.0

ApoB sequences Table 14

siRNA name	Description	Sequence
TS3_1	5'sense	S (5'-3'): **cgaagcgccctactccact**UAGACUUCCUGAAUAAC*U*A (SEQ ID NO 439) AS (5'-3'): U*A*GUUAUUCAGGAAGUCUA*U*U (SEQ ID NO 440)
TS3_2	3'sense	S (5'-3'): U*A*GACUUCCUGAAUAACUA**tcacctcatcccgcgaagc** (SEQ ID NO 441) AS (5'-3'): U*A*GUUAUUCAGGAAGUCUA*U*U (SEQ ID NO 440)
TS3_3	3'antisense	S (5'-3'): U*A*GACUUCCUGAAUAAC*U*A (SEQ ID NO 442) AS (5'-3'): U*A*GUUAUUCAGGAAGUCUA*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 443)
TS4_1	5'sense	S (5'-3'): cgaagcgccctactccactUCAUCACACUGAAUACC*A*A (SEQ ID NO 444) AS (5'-3'): U*U*GGUAUUCAGUGUGAUGA*U*U (SEQ ID NO 445)
TS4_2	3'sense	S (5'-3'): U*C*AUCACACUGAAUACCAA**tcacctcatcccgcgaagc** (SEQ ID NO 446) AS (5'-3'): U*U*GGUAUUCAGUGUGAUGA*U*U (SEQ ID NO 445)
TS4_3	3'antisense	S (5'-3'): U*C*AUCACACUGAAUACC*A*A (SEQ ID NO 447) AS (5'-3'): U*U*GGUAUUCAGUGUGAUGA*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 448)
ApoB_C10_1	5'sense	S (5'-3'): **cgaagcgccctactccact**GUCAUCACACUGAAUACCA*A*U (SEQ ID NO 449) AS (5'-3'): A*U*UGGUAUUCAGUGUGAUGAC*U*U (SEQ ID NO 450)

-continued

ApoB sequences Table 14

siRNA name	Description	Sequence
ApoB_C10_2	3'sense	S (5'-3'): G*U*CAUCACACUGAAUACCAAU**tcacctcatcccgcgaagc** (SEQ ID NO 451) AS (5'-3'): A*U*UGGUAUUCAGUGUGAUGAC*U*U (SEQ ID NO 450)
ApoB_C10_3	3'antisense	S (5'-3'): G*U*CAUCACACUGAAUACCA*A*U (SEQ ID NO 453) AS (5'-3'): A*U*UGGUAUUCAGUGUGAUGAC*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 452)
ApoB_C3_1	5'sense	S (5'-3'): **cgaagcgccctactccact**GGUGUAUGGCUUCAACCCU*G*A (SEQ ID NO 454) AS (5'-3'): U*C*AGGGUUGAAGCCAUACACC*U*U (SEQ ID NO 455)
ApoB_C3_2	3'sense	S (5'-3'): G*G*UGUAUGGCUUCAACCCUGA**tcacctcatcccgcgaagc** (SEQ ID NO 456) AS (5'-3'): U*C*AGGGUUGAAGCCAUACACC*U*U (SEQ ID NO 455)
ApoB_C3_3	3'antisense	S (5'-3'): G*G*UGUAUGGCUUCAACCCU*G*A (SEQ ID NO 457) AS (5'-3'): U*C*AGGGUUGAAGCCAUACACC*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 458)
ApoB_C2_1	5'sense	S (5'-3'): **cgaagcgccctactccact**CACCAACUUCUUCCACGAG*U*C (SEQ ID NO 459) AS (5'-3'): G*A*CUCGUGGAAGAAGUUGGUG*U*U (SEQ ID NO 460)
ApoB_C2_2	3'sense	S (5'-3'): C*A*CCAACUUCUUCCACGAGUC**tcacctcatcccgcgaagc** (SEQ ID NO 461) AS (5'-3'): G*A*CUCGUGGAAGAAGUUGGUG*U*U(SEQ ID NO 460)
ApoB_C2_3	3'antisense	S (5'-3'): C*A*CCAACUUCUUCCACGAG*U*C (SEQ ID NO 462) AS (5'-3'): G*A*CUCGUGGAAGAAGUUGGUG*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 463)
ApoB_DM2_1	5'sense	S (5'-3'): **cgaagcgccctactccact**AGGCAGAGCUAGUGGCA*A*A (SEQ ID NO 464) AS (5'-3'): U*U*UGCCACUAGCUCUGCCU*U*U (SEQ ID NO 465)
ApoB_DM2_2	3'sense	S (5'-3'): A*G*GCAGAGCUAGUGGCAAA**tcacctcatcccgcgaagc** (SEQ ID NO 466) AS (5'-3'): U*U*UGCCACUAGCUCUGCCU*U*U (SEQ ID NO 465)
ApoB_DM2_3	3'antisense	S (5'-3'): A*G*GCAGAGCUAGUGGCA*A*A (SEQ ID NO 467) AS (5'-3'): U*U*UGCCACUAGCUCUGCCU*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 468)
ApoB_DM3_1	5'sense	S (5'-3'): **cgaagcgccctactccact**GAGCAAAUCUCUUCAAU*A*A (SEQ ID NO 469) AS (5'-3'): U*U*AUUGAAGAGAUUUGCUC*U*U (SEQ ID NO 470)
ApoB_DM3_2	3'sense	S (5'-3'): G*A*GCAAAUCUCUUCAAUAA**tcacctcatcccgcgaagc** (SEQ ID NO 471) AS (5'-3'): U*U*AUUGAAGAGAUUUGCUC*U*U (SEQ ID NO 470)

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ApoB sequences Table 14		
siRNA name	Description	Sequence
ApoB_DM3_3	3'antisense	S (5'-3'): G*A*GCAAAUCUCUUCAAU*A*A (SEQ ID NO 472) AS (5'-3'): U*U*AUUGAAGAGAUUUGCUC*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 473)
ApoB_DM5_1	5'sense	S (5'-3'): **cgaagcgccctactccact**CCACAAAUGUCUACAGC*A*A (SEQ ID NO 474) AS (5'-3'): U*U*GCUGUAGACAUUUGUGG*U*U (SEQ ID NO 475)
ApoB_DM5_2	3'sense	S (5'-3'): C*C*ACAAAUGUCUACAGCAA**tcacctcatcccgcgaagc** (SEQ ID NO 476) AS (5'-3'): U*U*GCUGUAGACAUUUGUGG*U*U (SEQ ID NO 475)
ApoB_DM5_3	3'antisense	S (5'-3'): C*C*ACAAAUGUCUACAGC*A*A (SEQ ID NO 477) AS (5'-3'): U*U*GCUGUAGACAUUUGUGG*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 478)
ApoB_DM13_1	5'sense	S (5'-3'): **cgaagcgccctactccact**GAAACAGGCUUGAAAGA*A*U (SEQ ID NO 479) AS (5'-3'): A*U*UCUUUCAAGCCUGUUUC*U*U (SEQ ID NO 480)
ApoB_DM13_2	3'sense	S (5'-3'): G*A*AACAGGCUUGAAAGAAU**tcacctcatcccgcgaagc** (SEQ ID NO 481) AS (5'-3'): A*U*UCUUUCAAGCCUGUUUC*U*U (SEQ ID NO 480)
ApoB_DM13_3	3'antisense	S (5'-3'): G*A*AACAGGCUUGAAAGA*A*U (SEQ ID NO 482) AS (5'-3'): A*U*UCUUUCAAGCCUGUUUC*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 483)
ApoB_DM18_1	5'sense	S (5'-3'): **cgaagcgccctactccact**GAGAGAAAUCGAAGAGG*A*A (SEQ ID NO 484) AS (5'-3'): U*U*CCUCUUCGAUUUCUCUC*U*U (SEQ ID NO 485)
ApoB_DM18_2	3'sense	S (5'-3'): G*A*GAGAAAUCGAAGAGGAA**tcacctcatcccgcgaagc** (SEQ ID NO 486) AS (5'-3'): U*U*CCUCUUCGAUUUCUCUC*U*U (SEQ ID NO 485)
ApoB_DM18_3	3'antisense	S (5'-3'): G*A*GAGAAAUCGAAGAGG*A*A (SEQ ID NO 487) AS (5'-3'): U*U*CCUCUUCGAUUUCUCUC*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 488)
ApoB_DM19_1	5'sense	S (5'-3'): **cgaagcgccctactccact**AGUUAUAGUCCGUGAGC*U*A (SEQ ID NO 489) AS (5'-3'): U*A*GCUCACGGACUAUAACU*U*U (SEQ ID NO 490)
ApoB_DM19_2	3'sense	S (5'-3'): A*G*UUAUAGUCCGUGAGCUA**tcacctcatcccgcgaagc** (SEQ ID NO 491) AS (5'-3'): U*A*GCUCACGGACUAUAACU*U*U (SEQ ID NO 490)

-continued

ApoB sequences Table 14		
siRNA name	Description	Sequence
ApoB_DM19_3	3'antisense	S (5'-3'): A*G*UUAUAGUCCGUGAGC*U*A (SEQ ID NO 492) AS (5'-3'): U*A*GCUCACGGACUAUAACU*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 493)

TABLE 15

SEQ ID NO	
3	auaacucugu ccauuaccg
8	ucguauaaca auaaggggc
9	gauaacucug uccauuacc
10	auaacucugu ccauuacca
11	uaacucuguc cauuaccgu
12	agaaugugcc ucgauaacu
13	auaacucugu ccaucacca
14	auaacucugu ccaucaccu
15	uaacucuguc cauuaccau
16	augugccuug auaacucug
17	aguuggugcu gcuucagaa
18	aauaaggggc ugccacagg
19	uaacucuguc caucaccau
20	augagccucg auaacucug
21	aaugagccuc gauaacucu
22	aaugcuucca ggacauuuc
23	acaguggugg agaaugugc
24	guaugugccu cgauaacuc
25	ucgauaacuc uguccauca
26	Ugucacugga cauuguguc
27	cugggaucca uggugguaac
28	agaugaccaa gcuuggcag
29	uuaacucugu ccauuaccg
30	uuaacucugu ccauuaccc
31	uuaacucugu ccauuaccu
32	auaacucugu ccauuaccc
33	auaacucugu ccauuaccu
34	uuaacucugu ccauuacca
36	ucguauaaca auaaggggc
37	tcguauaaca auaaggggc

TABLE 15-continued

SEQ ID NO	
38	ucguauaaca auaagggg
39	ucguauaaca auaaggg
40	Uguauaacaa uaagggg
45	gccccuuauu guuauacga
251	tcacctcatc ccgcgaagc
255	ccucauaggc cuggaguuua u
257	cccucauagg ccuggaguuu a
258	acccucauag gccuggaguu u
259	uaggccugga guuuauucgg a
260	aggucuggaa ugcaaaguca a
261	ggccuggagu uuauucggaa a
263	ccucaccaag auccugcaug u
264	caccagcaua cagagugacc a
265	auaaacucca ggccuaugag g
266	uuccgaauaa acuccaggcc u
267	uaaacuccag gccuaugagg g
268	aaacuccagg ccuaugaggg u
269	uccgaauaaa cuccaggccu a
270	uugacuuugc auuccagacc u
271	uuuccgaaua aacuccaggc c
273	acaugcagga ucuuggugag g
274	uggucacucu guaugcuggu g
275	agcaagcaga cauuuaucuu u
276	aggucuggaa ugcaaaguca a
277	ggccuggagu uuauucggaa a
279	cccaagcaag cagacauuua u
280	ccucaccaag auccugcaug u
281	uuuucuagac cuguuuugcu u
282	acccaagcaa gcagacauuu a
283	caccagcaua cagagugacc a
284	auucuggguu uuguagcauu u
285	aucuccuaga caccagcaua c
286	uccuagacac cagcauacag a
287	gacauuuauc uuuugggucu g
288	uauucugggu uuuguagcau u
289	cuggaguuua uucggaaaag c
290	gccuggaguu uauucggaaa a
291	gaggcagaga cugauccacu u

TABLE 15-continued

SEQ ID NO	
292	aagcaagcag acauuuaucu u
293	uagaccuguu uugcuuuugu a
294	uuugcuuuug uaacuugaag a
295	cacuucucug ccaaagaugu c
296	uugcuuuugu aacuugaaga u
297	augcasaguc aaggagcaug g
298	cccacccaag caagcagaca u
299	ggguaacagu gaggcuggga a
300	ggucaugguc accgacuucg a
301	ggcagcuguu uugcaggacu g
302	gggcagguug gcagcuguuu u
303	uugaagauau uuauucuggg u
304	uggcagcugu uuugcaggac u
305	ccggggauac cucaccaaga u
306	acugauccac uucucugcca a
307	auccacuucu cugccaaaga u
308	acuucucugc caaagauguc a
309	gucuggaaug caaagucaag g
310	cuucucugcc aaagauguca u
311	gaguugaggc agagacugau
312	gaccuguuuu gcuuuuguaa c
313	cggggauacc ucaccaagau c
314	uuucuagacc uguuuugcuu u
315	ggucuggaau gcaaagucaa g
316	uaucuccuag acaccagcau a
317	agguuggcag cuguuuugca g
318	cuuuucuaga ccuguuuugc u
319	cuuuucuaga ccuguuuugc u
320	uccacuucuc ugccaaagau g
321	uggaguuuau ucggaaaagc c
322	ggcagguugg cagcuguuuu g
323	uggaggugua ucuccuagac a
324	gucaucaaug aggccugguu c
325	uucuagaccu guuuugcuuu u
326	uucuggguuu uguagcauuu u
327	gagacugauc cacuucucug c
328	agucaaggag cauggaaucc c

TABLE 15-continued

SEQ ID NO	
329	aucuuuuggg ucuguccucu c
330	cacccaagca agcagacauu u
331	aaagauaaau gucugcuugc u
332	uugacuuugc auuccagacc u
333	uuuccgaaua aacuccaggc c
335	auaaaugucu gcuugcuugg g
336	acaugcagga ucuuggugag g
337	aagcaaaaca ggucuagaaa a
338	uaaaugucug cuugcuuggg u
339	uggucacucu guaugcuggu g
340	aaaugcuaca aaacccagaa u
341	guaugcuggu gucuaggaga u
342	ucuguaugcu ggugucuagg a
343	cagacccaaa agauaaaugu c
344	aaugcuacaa aacccagaau a
345	gcuuuuccga auaaacucca g
346	uuuuccgaau aaacuccagg c
347	aaguggauca gucucugccu c
348	aagauaaaug ucugcuugcu u
349	uacaaaagca aaacaggucu a
350	ucuucaaguu acaaaagcaa a
351	gacaucuuug gcagagaagu g
352	aucuucaagu uacaaaagca a
353	ccaugcuccu ugacuuugca u
354	augucugcuu gcuugggugg g
355	uucccagccu cacuguuacc c
356	ucgaagucgg ugaccaugac c
357	caguccugca aaacagcugc c
358	aaaacagcug ccaaccugcc c
359	acccagaaua aauaucuuca a
360	aguccugcaa aacagcugcc a
361	aucuuggugа gguauccccg g
362	uuggcagaga aguggaucag u
363	aucuuuggca gagaagugga u
364	ugacaucuuu ggcagagaag u
365	ccuugacuuu gcauuccaga c
366	augacaucuu uggcagagaa g
367	gaucagucuc ugccucaacu c

TABLE 15-continued

SEQ ID NO	
368	guuacaaaag caaaacaggu c
369	gaucuuggug agguaucccc g
370	aaagcaaaac aggucuagaa a
371	cuugacuuug cauuccagac c
372	Uaugcuggug ucuaggagau a
373	cugcaaaaca gcugccaacc u
374	casaacaggu cuagaaaagu u
375	agcaaaacag gucuagaaaa g
376	caucuuuggc agagaagugg a
377	ggcuuuuccg aauaaacucc a
378	caaaacagcu gccaaccugc c
379	ugucuaggag auacaccucc a
380	gaaccaggcc ucauugauga c
381	aaaagcaaaa caggucuaga a
382	aaaaugcuac aaaacccaga a
383	gcagagaagu ggaucagucu c
384	gggauuccau gcuccuugac u
385	gagaggacag acccaaaaga u
386	aaaugucugc uugcuugggu g
387	tcacctcatc ccgcgaagc
388	agcgacgtcg aggcgctcat ggttgcaggc gggcgccgcc gttcagttca gggtctgagc 60 ctggaggagt gagccaggca gtgagactgg ctcgggcggg ccgggacgcg tcgttgcagc 120 agcggctccc agctcccagc caggattccg cgcgcccctt cacgcgccct gctcctgaac 180 ttcagctcct gcacagtcct ccccaccgca aggctcaagg cgccgccggc gtggaccgcg 240 cacggcctct aggtctcctc gccaggacag caacctctcc cctggccctc atgggcaccg 300 tcagctccag gcggtcctgg tggccgctgc cactgctgct gctgctgctg ctgctcctgg 360 gtcccgcggg cgcccgtgcg caggaggacg aggacggcga ctacgaggag ctggtgctag 420 ccttgcgttc cgaggaggac ggcctggccg aagcacccga gcacggaacc acagccacct 480 tccaccgctg cgccaaggat ccgtggaggt tgcctggcac ctacgtggtg gtgctgaagg 540 aggagaccca cctctcgcag tcagagcgca ctgcccgccg cctgcaggcc caggctgccc 600 gccggggata cctcaccaag atcctgcatg tcttccatgg ccttcttcct ggcttcctgg 660 tgaagatgag tagcgacctg ctggagctgg ccttgaagtt gccccatgtc gactacatcg 720 aggaggactc ctctqtcttt gcccagagca tcccgtggaa cctggagcgg attacccctc 480 cacggtaccg ggcggatgaa taccagcccc ccgacggagg cagcctggtg gaggtgtatc 840 tcctagacac cagcatacag agtgaccacc gggaaatcga gggcagggtc atggtcaccg 900 acttcgagaa tgtgcccgag gaggacggga cccgcttcca cagacaggcc agcaagtgtg 960 acagtcatgg cacccacctg gcagggtgg tcagcggccg ggatgccggc gtggccaagg 1020 gtgccagcat gcgcagcctg cgcgtgctca actgccaagg gaagggcacg gttagcggca 1080 ccctcatagg cctggagttt attcggaaaa gccagctggt ccagcctgtg gggccactgg 1140 tggtgctgct qcccctagcg gatgggtaca gccgcgtcct caacgccgcc tgccagcgcc 1200 tggcgagggc tggggtcgtg ctggtcaccg ctgccggcaa cttccgggac gatgcctgcc 1260 tctactcccc agcatcagct cccgagctca tcacagttgg ggccaccaat gcccaagacc 1320 agccggtgac cctggggact ttggggacca actttggccg ctgtgtggac ctctttgccc 1380 caggggagga catcattggt gcctccagcg actgcagcac ctgctttgtg tcacagagtg 1440 ggacatcaca ggctgctgcc cacgtggctc gcattgcagc catgatgctc tctgccgagc 1500 cggagctcac cctggccgag ttgaggcaga gactgatcca cttctctgcc aaagatgtca 1560 tcaatgaggc ctggttccct gaggaccagc gggtactgac cccaacctg gtggccgccc 1620 tgccccccag cacccatggg gcaggttggc agctgttttg caggactgta tggtcagcac 1680 actcggggcc tacacggatg gccacagccg tcgcccgctg cgccccagat gaggagctgc 1740 tgagctgctc cagtttctcc aggagtggga agcggcgggg cgagcgcatg gaggcccaag 1800 ggggcaagct ggtctgccgg gcccacaacg cttttggggg tgagggtgtc tacgccattg 1860 ccaggtgctg cctgctaccc caggccaact gcagcgtcca cacagctcca ccagctgagg 1920 ccagcatggg gacccgtgtc cactgccacc aacagggcca cgtcctcaca ggctgcagct 1980 cccactggga ggtggaggac cttggcaccc acaagccgcc tgtgctgagg ccacgaggtc 2040

TABLE 15-continued

SEQ ID NO	
	agcccaacca gtgcgtgggc cacagggagg ccagcatcca cgcttcctgc tgccatgccc 2100 caggtctgga atgcaaagtc aaggagcatg gaatcccggc ccctcaggag caggtgaccg 2160 tggcctgcga ggagggctgg accctgactg gctgcagtgc cctccctggg acctcccacg 2220 tcctgggggc ctacgccgta gacaacacgt gtgtagtcag gagccgggac gtcagcacta 2280 caggcagcac cagcgaaggg gccgtgacag ccgttgccat ctgctgccgg agccggcacc 2340 tggcgcaggc ctccaaggag ctccagtgac agccccatca caggatgggt gtctggggag 2400 ggtcaagggc tagggctgag ctttaaaatg gttccgactt gtccctctct cagccctcca 2460 tggcctgqca cgaggggatg gggatgcttc cgcctttccg ggctgctgg cctggccctt 2520 gagtggggca gcctccttgc ctggaactca ctcactctgg gtgcctcctc cccaggtgga 2580 ggtgccagga agctccctcc ctcactgtgg ggcatttcac cattcaaaca ggtcgagctg 2640 tgctcgggtg ctgccagctg ctcccaatgt gccgatgtcc gtgggcagaa tgacttttat 2700 tgagctcttg ttccgtgcca ggcattcaat cctcaggtct ccaccaagga ggcaggattc 2760 ttcccatgga taggggaggg ggcggtaggg gctgcaggga caaacatcgt tggggggtga 2820 gtgtgaaagg tgctgatggc cctcatctcc agctaactgt ggagaagccc ctgggggctc 2880 cctgattaat ggaggcttag ctttctggat ggcatctagc cagaggctgg agacaggtgc 2940 gcccctggty gtcacaggct gtgccttggt ttcctgagcc acctttactc tgctctatgc 3000 caggctgtgc tagcaacacc cacaggtggc ctgcggggag ccatcaccta ggactgactc 3060 ggcagtgtgc agtggtgcat gcactgtctc agcraacccg ctccactacc cggcagggta 3120 cacattcgca cccctacttc acagaggaag aaacctggaa ccagaggggg cgtgcctgcc 3180 aagctcacac agcaggaact gagccagaaa cgcagattgg gctggctctg aagccaagcc 3240 tcttcttact tcacccggct gggctcctca tttttacggg taacagtgag gctgggaagg 3300 ggaacacaga ccaggaagct cggtgagtga tggcagaacg atgcctgcag gcatggaact 3360 ttttccgtta tcacccaggc ctgattcact ggcctggcgg agatgcttct aaggcatggt 3420 cgggggagag ggccaacaac tgtccctcct tgagcaccag ccccacccaa gcaagcagac 3480 atttatcttt tgggtctgtc ctctctgtty cctttttaca gccaactttt ctagacctgt 3540 tttgcttttg taacttgaag atatttattc tgggttttgt agcattttta ttaatatggt 3600 gactttttaa aataaaaaca aacaaacgtt gtcctaa
391	gucaucacac ugaauaccaa u
392	auugguauuc agugugauga cac
393	uugauguguu uagucgcuau u
394	uagcgacuaa acacaucaau u
395	cuacacaaau cagcgauuu
396	aaaucgcuga uuuguguag
397	uaaggcuaug aagagauact t
398	aaguaucucu ucauagccuu a
399	acaaaagcaa aacaggucua gaa
400	gcgaagcccc tactccact
427	cgaagcgccc tactccactu gacuuugcau uccagaccug
473	uuauugaaga gauuugcucu utcacctcat cccgcgaagc
474	cgaagcgccc tactccactc cacaaauguc uacagcaa
475	uugcuguaga cauuuguggu u
499	GUCAUCACACUGAAUACCA*A*U
500	G*U*CAUCACACUGAAUACCAAU
501	A*U*UGGUAUUCAGUGUGAUGAC*U*U
502	GGUGUAUGGCUUCAACCCU*G*A
503	G*G*UGUAUGGCUUCAACCCUGA
504	U*C*AGGGUUGAAGCCAUACACC*U*U
505	CACCAACUUCUUCCACGAG*U*C
506	C*A*CCAACUUCUUCCACGAGUC
507	G*A*CUCGUGGAAGAAGUUGGUG*U*U
508	AGGCAGAGCUAGUGGCA*A*A

TABLE 15-continued

SEQ ID NO	
509	A*G*GCAGAGCUAGUGGCAAA
510	U*U*UGCCACUAGCUCUGCCU*U*U
511	GAGCAAAUCUCUUCAAU*A*A
512	G*A*GCAAAUCUCUUCAAUAA
513	U*U*AUUGAAGAGAUUUGCUC*U*U
514	CCACAAAUGUCUACAGC*A*A
515	C*C*ACAAAUGUCUACAGCAA(
516	U*U*GCUGUAGACAUUUGUGG*U*U
517	GAAACAGGCUUGAAAGA*A*U
518	G*A*AACAGGCUUGAAAGAAU
519	A*U*UCUUUCAAGCCUGUUUC*U*U
520	GAGAGAAAUCGAAGAGG*A*A
521	G*A*GAGAAAUCGAAGAGGAA
522	U*U*CCUCUUCGAUUUCUCUC*U*U
523	AGUUAUAGUCCGUGAGC*U*A
524	A*G*UUAUAGUCCGUGAGCUA
525	U*A*GCUCACGGACUAUAACU*U*U

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[0162] Soutschek, J., Akinc, A., Bramlage, B., Charisse, K., Constien, R., Donoghue, M., Elbashir, S., Geick, A., Hadwiger, P., Harborth, J. and John, M., 2004. Therapeutic silencing of an endogenous gene by systemic administration of modified siRNAs. *Nature.* 432(7014), p. 173

SEQUENCE LISTING

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Sequence total quantity: 528
SEQ ID NO: 1            moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = inhibitory RNA
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 1
ataactctgt ccattaccg                                                19

SEQ ID NO: 2            moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
misc_feature            1..18
                        note = inhibitory RNA
source                  1..18
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 2
cgtataacaa taaggggc                                                 18

SEQ ID NO: 3            moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
```

-continued

```
                          note = inhibitory RNA
source                    1..19
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 3
tcgtataaca ataaggggc                                                 19

SEQ ID NO: 4              moltype = RNA  length = 19
FEATURE                   Location/Qualifiers
modified_base             1
                          mod_base = OTHER
                          note = thymine
misc_feature              1..19
                          note = inhibitory RNA
source                    1..19
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 4
tcgtataaca ataaggggc                                                 19

SEQ ID NO: 5              moltype = RNA  length = 18
FEATURE                   Location/Qualifiers
misc_feature              1..18
                          note = inhibitory RNA
source                    1..18
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 5
tcgtataaca ataagggg                                                  18

SEQ ID NO: 6              moltype = RNA  length = 17
FEATURE                   Location/Qualifiers
misc_feature              1..17
                          note = inhibitory RNA
source                    1..17
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 6
tcgtataaca ataaggg                                                   17

SEQ ID NO: 7              moltype = RNA  length = 17
FEATURE                   Location/Qualifiers
misc_feature              1..17
                          note = inhibitory RNA
source                    1..17
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 7
tgtataacaa taagggg                                                   17

SEQ ID NO: 8              moltype = RNA  length = 19
FEATURE                   Location/Qualifiers
misc_feature              1..19
                          note = inhibitory RNA
source                    1..19
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 8
tcgtataaca ataaggggc                                                 19

SEQ ID NO: 9              moltype = RNA  length = 19
FEATURE                   Location/Qualifiers
misc_feature              1..19
                          note = inhibitory RNA
source                    1..19
                          mol_type = other RNA
                          organism = synthetic construct
SEQUENCE: 9
gataactctg tccattacc                                                 19

SEQ ID NO: 10             moltype = RNA  length = 19
FEATURE                   Location/Qualifiers
misc_feature              1..19
                          note = inhibitory RNA
source                    1..19
                          mol_type = other RNA
```

-continued

```
                        organism = synthetic construct
SEQUENCE: 10
ataactctgt ccattacca                                            19

SEQ ID NO: 11           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = inhibitory RNA
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 11
taactctgtc cattaccgt                                            19

SEQ ID NO: 12           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = inhibitory RNA
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 12
agaatgtgcc tcgataact                                            19

SEQ ID NO: 13           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = inhibitory RNA
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 13
ataactctgt ccatcacca                                            19

SEQ ID NO: 14           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = inhibitory RNA
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 14
ataactctgt ccatcacct                                            19

SEQ ID NO: 15           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = inhibitory RNA
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 15
taactctgtc cattaccat                                            19

SEQ ID NO: 16           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = inhibitory RNA
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 16
atgtgccttg ataactctg                                            19

SEQ ID NO: 17           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = inhibitory RNA
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 17
agttggtgct gcttcagaa                                            19

SEQ ID NO: 18           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
```

-continued

```
misc_feature          1..19
                      note = inhibitory RNA
source                1..19
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 18
aataaggggc tgccacagg                                        19

SEQ ID NO: 19         moltype = RNA  length = 19
FEATURE               Location/Qualifiers
misc_feature          1..19
                      note = inhibitory RNA
source                1..19
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 19
taactctgtc catcaccat                                        19

SEQ ID NO: 20         moltype = RNA  length = 19
FEATURE               Location/Qualifiers
misc_feature          1..19
                      note = inhibitory RNA
source                1..19
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 20
atgagcctcg ataactctg                                        19

SEQ ID NO: 21         moltype = RNA  length = 19
FEATURE               Location/Qualifiers
misc_feature          1..19
                      note = inhibitory RNA
source                1..19
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 21
aatgagcctc gataactct                                        19

SEQ ID NO: 22         moltype = RNA  length = 19
FEATURE               Location/Qualifiers
misc_feature          1..19
                      note = inhibitory RNA
source                1..19
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 22
aatgcttcca ggacatttc                                        19

SEQ ID NO: 23         moltype = RNA  length = 19
FEATURE               Location/Qualifiers
misc_feature          1..19
                      note = inhibitory RNA
source                1..19
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 23
acagtggtgg agaatgtgc                                        19

SEQ ID NO: 24         moltype = RNA  length = 19
FEATURE               Location/Qualifiers
misc_feature          1..19
                      note = inhibitory RNA
source                1..19
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 24
gtatgtgcct cgataactc                                        19

SEQ ID NO: 25         moltype = RNA  length = 19
FEATURE               Location/Qualifiers
misc_feature          1..19
                      note = inhibitory RNA
source                1..19
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 25
```

-continued

```
tcgataactc tgtccatca                                            19

SEQ ID NO: 26          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = inhibitory RNA
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 26
tgtcactgga cattgtgtc                                            19

SEQ ID NO: 27          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = inhibitory RNA
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 27
ctgggatcca tggtgtaac                                            19

SEQ ID NO: 28          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = inhibitory RNA
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 28
agatgaccaa gcttggcag                                            19

SEQ ID NO: 29          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = inhibitory RNA
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 29
ttaactctgt ccattaccg                                            19

SEQ ID NO: 30          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = inhibitory RNA
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 30
ttaactctgt ccattaccc                                            19

SEQ ID NO: 31          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = inhibitory RNA
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 31
ttaactctgt ccattacct                                            19

SEQ ID NO: 32          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = inhibitory RNA
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 32
ataactctgt ccattaccc                                            19

SEQ ID NO: 33          moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = inhibitory RNA
```

-continued

```
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 33
ataactctgt ccattacct                                        19

SEQ ID NO: 34           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = inhibitory RNA
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 34
ttaactctgt ccattacca                                        19

SEQ ID NO: 35           moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
misc_feature            1..18
                        note = passenger sequence
source                  1..18
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 35
cgtataacaa taaggggc                                         18

SEQ ID NO: 36           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = passenger sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 36
tcgtataaca ataaggggc                                        19

SEQ ID NO: 37           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
modified_base           1
                        mod_base = OTHER
                        note = thymine
misc_feature            1..19
                        note = passenger sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 37
tcgtataaca ataaggggc                                        19

SEQ ID NO: 38           moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
misc_feature            1..18
                        note = passenger sequence
source                  1..18
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 38
tcgtataaca ataagggg                                         18

SEQ ID NO: 39           moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
misc_feature            1..17
                        note = passenger sequence
source                  1..17
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 39
tcgtataaca ataaggg                                          17

SEQ ID NO: 40           moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
misc_feature            1..17
                        note = passenger sequence
source                  1..17
                        mol_type = other RNA
                        organism = synthetic construct
```

-continued

```
SEQUENCE: 40
tgtataacaa taagggg                                                    17

SEQ ID NO: 41           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = passenger sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 41
ataactctgt ccattaccg                                                  19

SEQ ID NO: 42           moltype = RNA  length = 22
FEATURE                 Location/Qualifiers
misc_feature            1..22
                        note = passenger sequence
source                  1..22
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 42
acagcccctt attgttatac ga                                              22

SEQ ID NO: 43           moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
misc_feature            1..18
                        note = passenger sequence
source                  1..18
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 43
gccccttatt gttatacg                                                   18

SEQ ID NO: 44           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = passenger sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 44
gccccttatt gttatacga                                                  19

SEQ ID NO: 45           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = passenger sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 45
gccccttatt gttatacga                                                  19

SEQ ID NO: 46           moltype = RNA  length = 18
FEATURE                 Location/Qualifiers
misc_feature            1..18
                        note = passenger sequence
source                  1..18
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 46
ccccttattg ttatacga                                                   18

SEQ ID NO: 47           moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
misc_feature            1..17
                        note = passenger sequence
source                  1..17
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 47
cccttattgt tatacga                                                    17

SEQ ID NO: 48           moltype = RNA  length = 17
FEATURE                 Location/Qualifiers
misc_feature            1..17
```

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```
                        note = passenger sequence
source                  1..17
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 48
ccccttattg ttataca                                                  17

SEQ ID NO: 49           moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = passenger sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 49
cggtaatgga cagagttat                                                19

SEQ ID NO: 50           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_01
misc_feature            20
                        note = n is a, c, g, or u
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 50
acgggacagt attctcagtn a                                             21

SEQ ID NO: 51           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_02
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 51
cccaataaag ctggacaaga a                                             21

SEQ ID NO: 52           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_03
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 52
ctgtaggttg cttaaaaggg a                                             21

SEQ ID NO: 53           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_04
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 53
ctggagcacc gttaaggaca a                                             21

SEQ ID NO: 54           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_05
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 54
tcccaataaa gctggacaag a                                             21

SEQ ID NO: 55           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_06
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
```

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```
SEQUENCE: 55
gcccctgtag gttgcttaaa a                                             21

SEQ ID NO: 56           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_07
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 56
ccctgaaaga ctactggagc a                                             21

SEQ ID NO: 57           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_08
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 57
tgcttaaaag ggacagtatt c                                             21

SEQ ID NO: 58           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_09
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 58
gacctcaata ccccaagtcc a                                             21

SEQ ID NO: 59           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = APOC3_10
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 59
gagcaccgtt aaggacaagt t                                             21

SEQ ID NO: 60           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
misc_feature            20
                        note = n is a, c, g, t or u
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 60
acgggacagt attctcagtn a                                             21

SEQ ID NO: 61           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 61
cccaataaag ctggacaaga a                                             21

SEQ ID NO: 62           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 62
ctgtaggttg cttaaaaggg a                                             21

SEQ ID NO: 63           moltype = RNA  length = 21
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 63
ctggagcacc gttaaggaca a                                                21

SEQ ID NO: 64           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 64
tcccaataaa gctggacaag a                                                21

SEQ ID NO: 65           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 65
gcccctgtag gttgcttaaa a                                                21

SEQ ID NO: 66           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 66
ccctgaaaga ctactggagc a                                                21

SEQ ID NO: 67           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 67
tgcttaaaag ggacagtatt c                                                21

SEQ ID NO: 68           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 68
gacctcaata ccccaagtcc a                                                21

SEQ ID NO: 69           moltype = RNA  length = 22
FEATURE                 Location/Qualifiers
modified_base           22
                        mod_base = OTHER
                        note = thymine
misc_feature            1..22
                        note = inhibitory RNA
source                  1..22
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 69
gagcaccgtt aaggacaagt tt                                               22

SEQ ID NO: 70           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
```

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```
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 70
tcactgagaa tactgtcccg t                                            21

SEQ ID NO: 71           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 71
ttcttgtcca gctttattgg g                                            21

SEQ ID NO: 72           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 72
tcccttttaa gcaacctaca g                                            21

SEQ ID NO: 73           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 73
ttgtccttaa cggtgctcca g                                            21

SEQ ID NO: 74           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 74
tcttgtccag ctttattggg a                                            21

SEQ ID NO: 75           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 75
ttttaagcaa cctacagggg c                                            21

SEQ ID NO: 76           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 76
tgctccagta gtctttcagg g                                            21

SEQ ID NO: 77           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 77
gaatactgtc ccttttaagc a                                            21
```

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```
SEQ ID NO: 78           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 78
tggacttggg gtattgaggt c                                              21

SEQ ID NO: 79           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 79
aacttgtcct taacggtgct c                                              21

SEQ ID NO: 80           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 80
tcactgagaa tactgtcccg t                                              21

SEQ ID NO: 81           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 81
ttcttgtcca gctttattgg g                                              21

SEQ ID NO: 82           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 82
tcccttttaa gcaacctaca g                                              21

SEQ ID NO: 83           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 83
ttgtccttaa cggtgctcca g                                              21

SEQ ID NO: 84           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 84
tcttgtccag ctttattggg a                                              21

SEQ ID NO: 85           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 85
ttttaagcaa cctacagggg c                                              21

SEQ ID NO: 86           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 86
tgctccagta gtctttcagg g                                              21
```

-continued

```
SEQ ID NO: 87           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 87
gaatactgtc ccttttaagc a                                           21

SEQ ID NO: 88           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 88
tggacttggg gtattgaggt c                                           21

SEQ ID NO: 89           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 89
aacttgtcct taacggtgct c                                           21

SEQ ID NO: 90           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = antisense or sense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 90
ctctgtaaat ttggaagtgt c                                           21

SEQ ID NO: 91           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = antisense or sense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 91
caccatgagc taggtggagt a                                           21

SEQ ID NO: 92           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = antisense or sense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 92
ttcctgaagt gacaaaggaa a                                           21

SEQ ID NO: 93           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = antisense or sense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 93
gaccaccagg aactatatct t                                           21

SEQ ID NO: 94           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = antisense or sense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 94
gttccagaaa tacattggtt t                                           21

SEQ ID NO: 95           moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
```

-continued

```
misc_feature           1..21
                       note = antisense or sense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 95
aaccgcaagg gctttgtgaa a                                              21

SEQ ID NO: 96          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = antisense or sense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 96
gagcaagaag ttcccaggca t                                              21

SEQ ID NO: 97          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = antisense or sense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 97
cagtagtagg catctggaat g                                              21

SEQ ID NO: 98          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = antisense or sense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 98
gtcatgggtg tctgtgggtt a                                              21

SEQ ID NO: 99          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = antisense or sense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 99
gctctgtaaa tttggaagtg t                                              21

SEQ ID NO: 100         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 100
ctctgtaaat ttggaagtgt c                                              21

SEQ ID NO: 101         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 101
caccatgagc taggtggagt a                                              21

SEQ ID NO: 102         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 102
```

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```
ttcctgaagt gacaaaggaa a                                             21

SEQ ID NO: 103          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 103
gaccaccagg aactatatct t                                             21

SEQ ID NO: 104          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 104
gttccagaaa tacattggtt t                                             21

SEQ ID NO: 105          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 105
aaccgcaagg gctttgtgaa a                                             21

SEQ ID NO: 106          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 106
gagcaagaag ttcccaggca t                                             21

SEQ ID NO: 107          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 107
cagtagtagg catctggaat g                                             21

SEQ ID NO: 108          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 108
gtcatgggtg tctgtgggtt a                                             21

SEQ ID NO: 109          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 109
gctctgtaaa tttggaagtg t                                             21

SEQ ID NO: 110          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
```

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```
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 110
gacacttcca aatttacaga g                                            21

SEQ ID NO: 111         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 111
tactccacct agctcatggt g                                            21

SEQ ID NO: 112         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 112
tttcctttgt cacttcagga a                                            21

SEQ ID NO: 113         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 113
aagatatagt tcctggtggt c                                            21

SEQ ID NO: 114         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 114
aaaccaatgt atttctggaa c                                            21

SEQ ID NO: 115         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 115
tttcacaaag cccttgcggt t                                            21

SEQ ID NO: 116         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 116
atgcctggga acttcttgct c                                            21

SEQ ID NO: 117         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = inhibitory RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 117
cattccagat gcctactact g                                            21
```

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```
SEQ ID NO: 118          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 118
taacccacag acacccatga c                                       21

SEQ ID NO: 119          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = inhibitory RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 119
acacttccaa atttacagag c                                       21

SEQ ID NO: 120          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 120
gacacttcca aatttacaga g                                       21

SEQ ID NO: 121          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 121
tactccacct agctcatggt g                                       21

SEQ ID NO: 122          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 122
tttcctttgt cacttcagga a                                       21

SEQ ID NO: 123          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 123
aagatatagt tcctggtggt c                                       21

SEQ ID NO: 124          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 124
aaaccaatgt atttctggaa c                                       21

SEQ ID NO: 125          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 125
tttcacaaag cccttgcggt t                                       21

SEQ ID NO: 126          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 126
atgcctggga acttcttgct c                                       21
```

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```
SEQ ID NO: 127          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 127
cattccagat gcctactact g                                            21

SEQ ID NO: 128          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 128
taacccacag acacccatga c                                            21

SEQ ID NO: 129          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 129
acacttccaa atttacagag c                                            21

SEQ ID NO: 130          moltype = DNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 130
cgaagcgccc tactccact                                               19

SEQ ID NO: 131          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 131
gaccaccagg aactatatct t                                            21

SEQ ID NO: 132          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 132
gttccagaaa tacattggtt t                                            21

SEQ ID NO: 133          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 133
aaccgcaagg gctttgtgaa a                                            21

SEQ ID NO: 134          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 134
gagcaagaag ttcccaggca t                                            21

SEQ ID NO: 135          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
```

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```
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 135
ctttggagag aatgaagtgt a                                              21

SEQ ID NO: 136        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 136
cttcgacaag cacaagacca a                                              21

SEQ ID NO: 137        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 137
gccgatgggt ccagaagaag t                                              21

SEQ ID NO: 138        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 138
cttcacttgg ctggtgtttg a                                              21

SEQ ID NO: 139        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 139
ctcctttgga gagaatgaag t                                              21

SEQ ID NO: 140        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 140
tgccatcctc atgtacatat t                                              21

SEQ ID NO: 141        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 141
ccgcaagggc tttgtgaaac t                                              21

SEQ ID NO: 142        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 142
agcaagaagt tcccaggcat a                                              21
```

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SEQ ID NO: 143          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 143
agtgtacaag caggtgatct t                                            21

SEQ ID NO: 144          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 144
tgctgaccac caggaactat a                                            21

SEQ ID NO: 145          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 145
ccgatgggtc cagaagaagt t                                            21

SEQ ID NO: 146          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 146
tttggagaga atgaagtgta c                                            21

SEQ ID NO: 147          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 147
tggcgctact ttcgagacta c                                            21

SEQ ID NO: 148          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 148
aatgcctgtg ttgagggagt a                                            21

SEQ ID NO: 149          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 149
agttccagaa atacattggt t                                            21

SEQ ID NO: 150          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
```

-continued

```
                         organism = synthetic construct
SEQUENCE: 150
cagaagtgag caagaagttc c                                          21

SEQ ID NO: 151           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = artificial sequence
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 151
aagatatagt tcctggtggt c                                          21

SEQ ID NO: 152           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = artificial sequence
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 152
aaaccaatgt atttctggaa c                                          21

SEQ ID NO: 153           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = artificial sequence
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 153
tttcacaaag cccttgcggt t                                          21

SEQ ID NO: 154           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = artificial sequence
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 154
atgcctggga acttcttgct c                                          21

SEQ ID NO: 155           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = artificial sequence
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 155
tacacttcat tctctccaaa g                                          21

SEQ ID NO: 156           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = artificial sequence
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 156
ttggtcttgt gcttgtcgaa g                                          21

SEQ ID NO: 157           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = artificial sequence
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 157
acttcttctg gacccatcgg c                                          21

SEQ ID NO: 158           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
```

-continued

```
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 158
tcaaacacca gccaagtgaa g                                            21

SEQ ID NO: 159         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 159
acttcattct ctccaaagga g                                            21

SEQ ID NO: 160         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 160
aatatgtaca tgaggatggc a                                            21

SEQ ID NO: 161         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 161
agtttcacaa agcccttgcg g                                            21

SEQ ID NO: 162         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 162
tatgcctggg aacttcttgc t                                            21

SEQ ID NO: 163         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 163
aagatcacct gcttgtacac t                                            21

SEQ ID NO: 164         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 164
tatagttcct ggtggtcagc a                                            21

SEQ ID NO: 165         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 165
```

-continued

```
aacttcttct ggacccatcg g                                            21

SEQ ID NO: 166          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 166
gtacacttca ttctctccaa a                                            21

SEQ ID NO: 167          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 167
gtagtctcga aagtagcgcc a                                            21

SEQ ID NO: 168          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 168
tactccctca acacaggcat t                                            21

SEQ ID NO: 169          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 169
aaccaatgta tttctggaac t                                            21

SEQ ID NO: 170          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 170
ggaacttctt gctcacttct g                                            21

SEQ ID NO: 171          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 171
cctcataggc ctggagttta t                                            21

SEQ ID NO: 172          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 172
aggcctggag tttattcgga a                                            21

SEQ ID NO: 173          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
```

-continued

```
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 173
ccctcatagg cctggagttt a                                    21

SEQ ID NO: 174         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 174
aggtctggaa tgcaaagtca a                                    21

SEQ ID NO: 175         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 175
ggcctggagt ttattcggaa a                                    21

SEQ ID NO: 176         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 176
caggtctgga atgcaaagtc a                                    21

SEQ ID NO: 177         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 177
cctcaccaag atcctgcatg t                                    21

SEQ ID NO: 178         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 178
accctcatag gcctggagtt t                                    21

SEQ ID NO: 179         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 179
caccagcata cagagtgacc a                                    21

SEQ ID NO: 180         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 180
atctcctaga caccagcata c                                    21
```

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```
SEQ ID NO: 181          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 181
tcctagacac cagcatacag a                                            21

SEQ ID NO: 182          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 182
ctggagttta ttcggaaaag c                                            21

SEQ ID NO: 183          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 183
gcctggagtt tattcggaaa a                                            21

SEQ ID NO: 184          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 184
gaggcagaga ctgatccact t                                            21

SEQ ID NO: 185          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 185
taggcctgga gtttattcgg a                                            21

SEQ ID NO: 186          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 186
cacttctctg ccaaagatgt c                                            21

SEQ ID NO: 187          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 187
atgcaaagtc aaggagcatg g                                            21

SEQ ID NO: 188          moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
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-continued

```
                       organism = synthetic construct
SEQUENCE: 188
ggtcatggtc accgacttcg a                                              21

SEQ ID NO: 189         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 189
ggcagctgtt ttgcaggact g                                              21

SEQ ID NO: 190         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 190
gggcaggttg gcagctgttt t                                              21

SEQ ID NO: 191         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 191
ataaactcca ggcctatgag g                                              21

SEQ ID NO: 192         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 192
ttccgaataa actccaggcc t                                              21

SEQ ID NO: 193         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 193
taaactccag gcctatgagg g                                              21

SEQ ID NO: 194         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 194
ttgactttgc attccagacc t                                              21

SEQ ID NO: 195         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 195
tttccgaata aactccaggc c                                              21

SEQ ID NO: 196         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
```

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```
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 196
tgactttgca ttccagacct g                                            21

SEQ ID NO: 197        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 197
acatgcagga tcttggtgag g                                            21

SEQ ID NO: 198        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 198
aaactccagg cctatgaggg t                                            21

SEQ ID NO: 199        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 199
tggtcactct gtatgctggt g                                            21

SEQ ID NO: 200        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 200
gtatgctggt gtctaggaga t                                            21

SEQ ID NO: 201        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 201
tctgtatgct ggtgtctagg a                                            21

SEQ ID NO: 202        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 202
gcttttccga ataaactcca g                                            21

SEQ ID NO: 203        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 203
```

-continued

```
ttttccgaat aaactccagg c                                            21

SEQ ID NO: 204          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 204
aagtggatca gtctctgcct c                                            21

SEQ ID NO: 205          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 205
tccgaataaa ctccaggcct a                                            21

SEQ ID NO: 206          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 206
gacatctttg gcagagaagt g                                            21

SEQ ID NO: 207          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 207
ccatgctcct tgactttgca t                                            21

SEQ ID NO: 208          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 208
tcgaagtcgg tgaccatgac c                                            21

SEQ ID NO: 209          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 209
cagtcctgca aaacagctgc c                                            21

SEQ ID NO: 210          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 210
aaaacagctg ccaacctgcc c                                            21

SEQ ID NO: 211          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
```

-continued

```
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 211
ctggagcacc gttaaggaca a                                                21

SEQ ID NO: 212          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 212
ccctgaaaga ctactggagc a                                                21

SEQ ID NO: 213          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 213
gagcaccgtt aaggacaagt t                                                21

SEQ ID NO: 214          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 214
actggagcac cgttaaggac a                                                21

SEQ ID NO: 215          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 215
cctgaaagac tactggagca c                                                21

SEQ ID NO: 216          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 216
aagactactg gagcaccgtt a                                                21

SEQ ID NO: 217          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 217
cagttccctg aaagactact g                                                21

SEQ ID NO: 218          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 218
ggtgaccgat ggcttcagtt c                                                21
```

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```
SEQ ID NO: 219          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 219
gggtgaccga tggcttcagt t                                              21

SEQ ID NO: 220          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 220
actactggag caccgttaag g                                              21

SEQ ID NO: 221          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 221
gactactgga gcaccgttaa g                                              21

SEQ ID NO: 222          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 222
ttcagttccc tgaaagacta c                                              21

SEQ ID NO: 223          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 223
gttccctgaa agactactgg a                                              21

SEQ ID NO: 224          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 224
tggagcaccg ttaaggacaa g                                              21

SEQ ID NO: 225          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 225
cgccaccaag accgccaagg a                                              21

SEQ ID NO: 226          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
```

-continued

```
                        organism = synthetic construct
SEQUENCE: 226
gggctgggtg accgatggct t                                              21

SEQ ID NO: 227          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 227
gccaccaaga ccgccaagga t                                              21

SEQ ID NO: 228          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 228
agactactgg agcaccgtta a                                              21

SEQ ID NO: 229          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 229
ccaccaagac cgccaaggat g                                              21

SEQ ID NO: 230          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 230
tccctgaaag actactggag c                                              21

SEQ ID NO: 231          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 231
ttgtccttaa cggtgctcca g                                              21

SEQ ID NO: 232          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 232
tgctccagta gtctttcagg g                                              21

SEQ ID NO: 233          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 233
aacttgtcct taacggtgct c                                              21

SEQ ID NO: 234          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
```

-continued

```
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 234
tgtccttaac ggtgctccag t                                                21

SEQ ID NO: 235         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 235
gtgctccagt agtctttcag g                                                21

SEQ ID NO: 236         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 236
taacggtgct ccagtagtct t                                                21

SEQ ID NO: 237         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 237
cagtagtctt tcagggaact g                                                21

SEQ ID NO: 238         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 238
gaactgaagc catcggtcac c                                                21

SEQ ID NO: 239         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 239
aactgaagcc atcggtcacc c                                                21

SEQ ID NO: 240         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 240
ccttaacggt gctccagtag t                                                21

SEQ ID NO: 241         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 241
```

-continued

```
cttaacggtg ctccagtagt c                                              21

SEQ ID NO: 242          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 242
gtagtctttc agggaactga a                                              21

SEQ ID NO: 243          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 243
tccagtagtc tttcagggaa c                                              21

SEQ ID NO: 244          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 244
cttgtcctta acggtgctcc a                                              21

SEQ ID NO: 245          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 245
tccttggcgg tcttggtggc g                                              21

SEQ ID NO: 246          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 246
aagccatcgg tcacccagcc c                                              21

SEQ ID NO: 247          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 247
atccttggcg gtcttggtgg c                                              21

SEQ ID NO: 248          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 248
ttaacggtgc tccagtagtc t                                              21

SEQ ID NO: 249          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
```

-continued

```
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 249
catccttggc ggtcttggtg g                                              21

SEQ ID NO: 250        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = artificial sequence
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 250
gctccagtag tctttcaggg a                                              21

SEQ ID NO: 251        moltype = DNA  length = 19
FEATURE               Location/Qualifiers
source                1..19
                      mol_type = other DNA
                      organism = synthetic construct
misc_feature          1..19
                      note = single stranded DNA
SEQUENCE: 251
tcacctcatc ccgcgaagc                                                 19

SEQ ID NO: 252        moltype = RNA  length = 23
FEATURE               Location/Qualifiers
source                1..23
                      mol_type = other RNA
                      organism = synthetic construct
modified_base         22..23
                      mod_base = OTHER
                      note = thymine
misc_feature          1..23
                      note = artificial sequence
SEQUENCE: 252
ttccgaataa actccaggcc ttt                                            23

SEQ ID NO: 253        moltype = DNA  length = 40
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = RNA
source                1..40
                      mol_type = other DNA
                      organism = synthetic construct
misc_feature          22..40
                      note = DNA
SEQUENCE: 253
aggcctggag tttattcgga atcacctcat cccgcgaagc                          40

SEQ ID NO: 254        moltype = RNA  length = 23
FEATURE               Location/Qualifiers
source                1..23
                      mol_type = other RNA
                      organism = synthetic construct
modified_base         22..23
                      mod_base = OTHER
                      note = thymine
misc_feature          1..23
                      note = artificial sequence
SEQUENCE: 254
ttccgaataa actccaggcc ttt                                            23

SEQ ID NO: 255        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 255
cctcataggc ctggagttta t                                              21

SEQ ID NO: 256        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
```

-continued

```
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 256
aggcctggag tttattcgga a                                             21

SEQ ID NO: 257        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 257
ccctcatagg cctggagttt a                                             21

SEQ ID NO: 258        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 258
accctcatag gcctggagtt t                                             21

SEQ ID NO: 259        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 259
taggcctgga gtttattcgg a                                             21

SEQ ID NO: 260        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 260
aggtctggaa tgcaaagtca a                                             21

SEQ ID NO: 261        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 261
ggcctggagt ttattcggaa a                                             21

SEQ ID NO: 262        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 262
caggtctgga atgcaaagtc a                                             21

SEQ ID NO: 263        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 263
```

-continued

```
cctcaccaag atcctgcatg t                                    21

SEQ ID NO: 264         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 264
caccagcata cagagtgacc a                                    21

SEQ ID NO: 265         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 265
ataaactcca ggcctatgag g                                    21

SEQ ID NO: 266         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 266
ttccgaataa actccaggcc t                                    21

SEQ ID NO: 267         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 267
taaactccag gcctatgagg g                                    21

SEQ ID NO: 268         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 268
aaactccagg cctatgaggg t                                    21

SEQ ID NO: 269         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 269
tccgaataaa ctccaggcct a                                    21

SEQ ID NO: 270         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 270
ttgactttgc attccagacc t                                    21

SEQ ID NO: 271         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
```

-continued

```
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 271
tttccgaata aactccaggc c                                    21

SEQ ID NO: 272         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 272
tgactttgca ttccagacct g                                    21

SEQ ID NO: 273         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 273
acatgcagga tcttggtgag g                                    21

SEQ ID NO: 274         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 274
tggtcactct gtatgctggt g                                    21

SEQ ID NO: 275         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 275
agcaagcaga catttatctt t                                    21

SEQ ID NO: 276         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 276
aggtctggaa tgcaaagtca a                                    21

SEQ ID NO: 277         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 277
ggcctggagt ttattcggaa a                                    21

SEQ ID NO: 278         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 278
caggtctgga atgcaaagtc a                                    21
```

-continued

```
SEQ ID NO: 279          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 279
cccaagcaag cagacattta t                                            21

SEQ ID NO: 280          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 280
cctcaccaag atcctgcatg t                                            21

SEQ ID NO: 281          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 281
ttttctagac ctgttttgct t                                            21

SEQ ID NO: 282          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 282
acccaagcaa gcagacattt a                                            21

SEQ ID NO: 283          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 283
caccagcata cagagtgacc a                                            21

SEQ ID NO: 284          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 284
attctgggtt ttgtagcatt t                                            21

SEQ ID NO: 285          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 285
atctcctaga caccagcata c                                            21

SEQ ID NO: 286          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
```

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```
                         organism = synthetic construct
SEQUENCE: 286
tcctagacac cagcatacag a                                            21

SEQ ID NO: 287           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 287
gacatttatc ttttgggtct g                                            21

SEQ ID NO: 288           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 288
tattctgggt tttgtagcat t                                            21

SEQ ID NO: 289           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 289
ctggagttta ttcggaaaag c                                            21

SEQ ID NO: 290           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 290
gcctggagtt tattcggaaa a                                            21

SEQ ID NO: 291           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 291
gaggcagaga ctgatccact t                                            21

SEQ ID NO: 292           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 292
aagcaagcag acatttatct t                                            21

SEQ ID NO: 293           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 293
tagacctgtt ttgcttttgt a                                            21

SEQ ID NO: 294           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
```

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```
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 294
tttgcttttg taacttgaag a                                            21

SEQ ID NO: 295        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 295
cacttctctg ccaaagatgt c                                            21

SEQ ID NO: 296        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 296
ttgcttttgt aacttgaaga t                                            21

SEQ ID NO: 297        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 297
atgcaaagtc aaggagcatg g                                            21

SEQ ID NO: 298        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 298
cccacccaag caagcagaca t                                            21

SEQ ID NO: 299        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 299
gggtaacagt gaggctggga a                                            21

SEQ ID NO: 300        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 300
ggtcatggtc accgacttcg a                                            21

SEQ ID NO: 301        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 301
```

-continued

```
ggcagctgtt ttgcaggact g                                              21

SEQ ID NO: 302          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 302
gggcaggttg gcagctgttt t                                              21

SEQ ID NO: 303          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 303
ttgaagatat ttattctggg t                                              21

SEQ ID NO: 304          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 304
tggcagctgt tttgcaggac t                                              21

SEQ ID NO: 305          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 305
ccggggatac ctcaccaaga t                                              21

SEQ ID NO: 306          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 306
actgatccac ttctctgcca a                                              21

SEQ ID NO: 307          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 307
atccacttct ctgccaaaga t                                              21

SEQ ID NO: 308          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 308
acttctctgc caaagatgtc a                                              21

SEQ ID NO: 309          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
```

-continued

```
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 309
gtctggaatg caaagtcaag g                                                   21

SEQ ID NO: 310          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 310
cttctctgcc aaagatgtca t                                                   21

SEQ ID NO: 311          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 311
gagttgaggc agagactgat c                                                   21

SEQ ID NO: 312          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 312
gacctgtttt gcttttgtaa c                                                   21

SEQ ID NO: 313          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 313
cggggatacc tcaccaagat c                                                   21

SEQ ID NO: 314          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 314
tttctagacc tgttttgctt t                                                   21

SEQ ID NO: 315          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 315
ggtctggaat gcaaagtcaa g                                                   21

SEQ ID NO: 316          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 316
tatctcctag acaccagcat a                                                   21
```

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```
SEQ ID NO: 317          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 317
aggttggcag ctgttttgca g                                            21

SEQ ID NO: 318          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 318
aacttttcta gacctgtttt g                                            21

SEQ ID NO: 319          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 319
cttttctaga cctgttttgc t                                            21

SEQ ID NO: 320          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 320
tccacttctc tgccaaagat g                                            21

SEQ ID NO: 321          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 321
tggagtttat tcggaaaagc c                                            21

SEQ ID NO: 322          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 322
ggcaggttgg cagctgtttt g                                            21

SEQ ID NO: 323          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 323
tggaggtgta tctcctagac a                                            21

SEQ ID NO: 324          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
```

-continued

```
                       organism = synthetic construct
SEQUENCE: 324
gtcatcaatg aggcctggtt c                                                21

SEQ ID NO: 325         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 325
ttctagacct gttttgcttt t                                                21

SEQ ID NO: 326         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 326
ttctgggttt tgtagcattt t                                                21

SEQ ID NO: 327         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 327
gagactgatc cacttctctg c                                                21

SEQ ID NO: 328         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 328
agtcaaggag catggaatcc c                                                21

SEQ ID NO: 329         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 329
atcttttggg tctgtcctct c                                                21

SEQ ID NO: 330         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 330
cacccaagca agcagacatt t                                                21

SEQ ID NO: 331         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 331
aaagataaat gtctgcttgc t                                                21

SEQ ID NO: 332         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
```

-continued

```
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 332
ttgactttgc attccagacc t                                            21

SEQ ID NO: 333         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 333
tttccgaata aactccaggc c                                            21

SEQ ID NO: 334         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 334
tgactttgca ttccagacct g                                            21

SEQ ID NO: 335         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 335
ataaatgtct gcttgcttgg g                                            21

SEQ ID NO: 336         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 336
acatgcagga tcttggtgag g                                            21

SEQ ID NO: 337         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 337
aagcaaaaca ggtctagaaa a                                            21

SEQ ID NO: 338         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 338
taaatgtctg cttgcttggg t                                            21

SEQ ID NO: 339         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 339
```

-continued

```
tggtcactct gtatgctggt g                                              21

SEQ ID NO: 340          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 340
aaatgctaca aaacccagaa t                                              21

SEQ ID NO: 341          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 341
gtatgctggt gtctaggaga t                                              21

SEQ ID NO: 342          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 342
tctgtatgct ggtgtctagg a                                              21

SEQ ID NO: 343          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 343
cagacccaaa agataaatgt c                                              21

SEQ ID NO: 344          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 344
aatgctacaa aacccagaat a                                              21

SEQ ID NO: 345          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 345
gcttttccga ataaactcca g                                              21

SEQ ID NO: 346          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 346
ttttccgaat aaactccagg c                                              21

SEQ ID NO: 347          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
```

-continued

```
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 347
aagtggatca gtctctgcct c                                     21

SEQ ID NO: 348          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 348
aagataaatg tctgcttgct t                                     21

SEQ ID NO: 349          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 349
tacaaaagca aaacaggtct a                                     21

SEQ ID NO: 350          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 350
tcttcaagtt acaaaagcaa a                                     21

SEQ ID NO: 351          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 351
gacatctttg gcagagaagt g                                     21

SEQ ID NO: 352          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 352
atcttcaagt tacaaaagca a                                     21

SEQ ID NO: 353          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 353
ccatgctcct tgactttgca t                                     21

SEQ ID NO: 354          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 354
atgtctgctt gcttgggtgg g                                     21
```

-continued

```
SEQ ID NO: 355          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 355
ttcccagcct cactgttacc c                                            21

SEQ ID NO: 356          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 356
tcgaagtcgg tgaccatgac c                                            21

SEQ ID NO: 357          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 357
cagtcctgca aaacagctgc c                                            21

SEQ ID NO: 358          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 358
aaaacagctg ccaacctgcc c                                            21

SEQ ID NO: 359          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 359
acccagaata aatatcttca a                                            21

SEQ ID NO: 360          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 360
agtcctgcaa aacagctgcc a                                            21

SEQ ID NO: 361          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 361
atcttggtga ggtatccccg g                                            21

SEQ ID NO: 362          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
```

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```
                         organism = synthetic construct
SEQUENCE: 362
ttggcagaga agtggatcag t                                          21

SEQ ID NO: 363           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 363
atctttggca gagaagtgga t                                          21

SEQ ID NO: 364           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 364
tgacatcttt ggcagagaag t                                          21

SEQ ID NO: 365           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 365
ccttgacttt gcattccaga c                                          21

SEQ ID NO: 366           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 366
atgacatctt tggcagagaa g                                          21

SEQ ID NO: 367           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 367
gatcagtctc tgcctcaact c                                          21

SEQ ID NO: 368           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 368
gttacaaaag caaaacaggt c                                          21

SEQ ID NO: 369           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
misc_feature             1..21
                         note = sense/antisense RNA
source                   1..21
                         mol_type = other RNA
                         organism = synthetic construct
SEQUENCE: 369
gatcttggtg aggtatcccc g                                          21

SEQ ID NO: 370           moltype = RNA  length = 21
FEATURE                  Location/Qualifiers
```

-continued

```
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 370
aaagcaaaac aggtctagaa a                                               21

SEQ ID NO: 371         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 371
cttgactttg cattccagac c                                               21

SEQ ID NO: 372         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 372
tatgctggtg tctaggagat a                                               21

SEQ ID NO: 373         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 373
ctgcaaaaca gctgccaacc t                                               21

SEQ ID NO: 374         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 374
caaaacaggt ctagaaaagt t                                               21

SEQ ID NO: 375         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 375
agcaaaacag gtctagaaaa g                                               21

SEQ ID NO: 376         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 376
catctttggc agagaagtgg a                                               21

SEQ ID NO: 377         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = sense/antisense RNA
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 377
```

-continued

```
ggcttttccg aataaactcc a                                              21

SEQ ID NO: 378          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 378
caaaacagct gccaacctgc c                                              21

SEQ ID NO: 379          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 379
tgtctaggag atacacctcc a                                              21

SEQ ID NO: 380          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 380
gaaccaggcc tcattgatga c                                              21

SEQ ID NO: 381          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 381
aaaagcaaaa caggtctaga a                                              21

SEQ ID NO: 382          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 382
aaaatgctac aaaacccaga a                                              21

SEQ ID NO: 383          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 383
gcagagaagt ggatcagtct c                                              21

SEQ ID NO: 384          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 384
gggattccat gctccttgac t                                              21

SEQ ID NO: 385          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = sense/antisense RNA
```

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```
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 385
gagaggacag acccaaaaga t                                            21

SEQ ID NO: 386        moltype = RNA  length = 21
FEATURE               Location/Qualifiers
misc_feature          1..21
                      note = sense/antisense RNA
source                1..21
                      mol_type = other RNA
                      organism = synthetic construct
SEQUENCE: 386
aaatgtctgc ttgcttgggt g                                            21

SEQ ID NO: 387        moltype = DNA  length = 19
FEATURE               Location/Qualifiers
misc_feature          1..19
                      note = crook sequence
source                1..19
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 387
tcacctcatc ccgcgaagc                                               19

SEQ ID NO: 388        moltype = DNA  length = 3637
FEATURE               Location/Qualifiers
misc_feature          1..3637
                      note = PCSK9 cDNA sequence
source                1..3637
                      mol_type = other DNA
                      organism = synthetic construct
SEQUENCE: 388
agcgacgtcg aggcgctcat ggttgcaggc gggcgccgcc gttcagttca gggtctgagc  60
ctggaggagt gagccaggca gtgagactgg ctcgggcggg ccgggacgcg tcgttgcagc  120
agcggctccc agctcccagc caggattccg cgcgcccctt cacgcgccct gctcctgaac  180
ttcagctcct gcacagtcct ccccaccgca aggctcaagg cgccgccggc gtggaccgcg  240
cacggcctct aggtctcctc gccaggacag caacctctcc cctggccctc atgggcaccg  300
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acttcgagaa tgtgcccgag gaggacggga cccgcttcca cagacaggcc agcaagtgtg  960
acagtcatgg cacccacctg gcaggggtgg tcagcggccg ggatgccggc gtggccaagg  1020
gtgccagcat gcgcagcctg cgcgtgctca actgccaagg gaagggcacg gttagcggca  1080
ccctcatagg cctggagttt attcggaaaa gccagctggt ccagcctgtg gggccactgg  1140
tggtgctgct gcccctggcg ggtgggtaca gccgcgtcct caacgccgcc tgccagcgcc  1200
tggcgagggc tggggtcgtg ctggtcaccg ctgccggcaa cttccgggac gatgcctgcc  1260
tctactcccc agcctcagct cccgaggtca tcacagttgg ggccaccaat gcccaagacc  1320
agccggtgac cctggggact ttggggacca actttggccg ctgtgtggac ctctttgccc  1380
caggggagga catcattggt gcctccagcg actgcagcac ctgctttgtg tcacagagtg  1440
ggacatcaca ggctgctgcc cacgtggctg gcattgcagc catgatgctg tctgccgagc  1500
cggagctcac cctggccgag ttgaggcaga gactgatcca cttctctgcc aaagatgtca  1560
tcaatgaggc ctggttccct gaggaccagc gggtactgac ccccaacctg gtggccgccc  1620
tgccccccag cacccatggg gcaggttggc agctgttttg caggactgta tggtcagcac  1680
actcggggcc tacacggatg gccacagccg tcgcccgctg cgccccagat gaggagctgc  1740
tgagctgctc cagtttctcc aggagtggga agcggcgggg cgagcgcatg gaggcccaag  1800
ggggcaagct ggtctgccgg gcccacaacg cttttggggg tgagggtgtc tacgccattg  1860
ccaggtgctg cctgctaccc caggccaact gcagcgtcca cacagctcca ccagctgagg  1920
ccagcatggg gacccgtgtc cactgccacc aacagggcca cgtcctcaca ggctgcagct  1980
cccactggga ggtggaggac cttggcaccc acaagccgcc tgtgctgagg ccacgaggtc  2040
agcccaacca gtgcgtgggc cacagggagg ccagcatcca cgcttcctgc tgccatgccc  2100
caggtctgga atgcaaagtc aaggagcatg gaatcccggc ccctcaggag caggtgaccg  2160
tggcctgcga ggagggctgg accctgactg gctgcagtgc cctccctggg acctcccacg  2220
tcctgggggc ctacgccgta gacaacacgt gtgtagtcag gagccgggac gtcagcacta  2280
caggcagcac cagcgaaggg gccgtgacag ccgttgccat ctgctgccgg agccggcacc  2340
tggcgcaggc ctcccaggag ctccagtgac agccccatcc caggatgggt gtctggggag  2400
ggtcaagggc tggggctgag ctttaaaatg gttccgactt gtccctctct cagccctcca  2460
tggcctggca cgaggggatg gggatgcttc cgcctttccg ggctgctgg cctggccctt  2520
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```
gagtggggca gcctccttgc ctggaactca ctcactctgg gtgcctcctc cccaggtgga  2580
ggtgccagga agctccctcc ctcactgtgg ggcatttcac cattcaaaca ggtcgagctg  2640
tgctcgggtg ctgccagctg ctcccaatgt gccgatgtcc gtgggcagaa tgacttttat  2700
tgagctcttg ttccgtgcca ggcattcaat cctcaggtct ccaccaagga ggcaggattc  2760
ttcccatgga taggggaggg ggcggtaggg gctgcaggga caaacatcgt tggggggtga  2820
gtgtgaaagg tgctgatggc cctcatctcc agctaactgt ggagaagccc ctgggggctc  2880
cctgattaat ggaggcttag ctttctggat ggcatctagc cagaggctgg agacaggtgc  2940
gcccctggtg gtcacaggct gtgccttggt ttcctgagcc acctttactc tgctctatgc  3000
caggctgtgc tagcaacacc caaaggtggc ctgcggggag ccatcaccta ggactgactc  3060
ggcagtgtgc agtggtgcat gcactgtctc agccaacccg ctccactacc cggcagggta  3120
cacattcgca cccctacttc acagaggaag aaacctggaa ccagaggggg cgtgcctgcc  3180
aagctcacac agcaggaact gagccagaaa cgcagattgg gctggctctg aagccaagcc  3240
tcttcttact tcacccggct gggctcctca tttttacggg taacagtgag gctgggaagg  3300
ggaacacaga ccaggaagct cggtgagtga tggcagaacg atgcctgcag gcatggaact  3360
ttttccgtta tcacccaggc ctgattcact ggcctggcgg agatgcttct aaggcatggt  3420
cgggggagag ggccaacaac tgtccctcct tgagcaccag ccccacccaa gcaagcagac  3480
atttatcttt tgggtctgtc ctctctgttg ccttttttaca gccaactttt ctagacctgt  3540
tttgcttttg taacttgaag atatttattc tgggttttgt agcattttta ttaatatggt  3600
gactttttaa aataaaaaca aacaaacgtt gtcctaa                          3637

SEQ ID NO: 389         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
modified_base          11
                       mod_base = OTHER
                       note = thymine
misc_feature           1..21
                       note = sense unmodified inclisiran
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 389
ctagacctgt tttgcttttg t                                            21

SEQ ID NO: 390         moltype = RNA  length = 23
FEATURE                Location/Qualifiers
misc_feature           1..23
                       note = antisense unmodified inclisiran
source                 1..23
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 390
acaaaagcaa aacaggtcta gaa                                          23

SEQ ID NO: 391         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = apoB-1 sense
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 391
gtcatcacac tgaataccaa t                                            21

SEQ ID NO: 392         moltype = RNA  length = 23
FEATURE                Location/Qualifiers
misc_feature           1..23
                       note = apoB-1 antisense
source                 1..23
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 392
attggtattc agtgtgatga cac                                          23

SEQ ID NO: 393         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = beta-Gal sense
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
SEQUENCE: 393
ttgatgtgtt tagtcgctat t                                            21

SEQ ID NO: 394         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
```

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```
                        note = beta-Gal antisense
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 394
tagcgactaa acacatcaat t                                         21

SEQ ID NO: 395          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = beta-gal 728 sense
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 395
ctacacaaat cagcgattt                                            19

SEQ ID NO: 396          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = beta-gal 728 antisense
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 396
aaatcgctga tttgtgtag                                            19

SEQ ID NO: 397          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
modified_base           20..21
                        mod_base = OTHER
                        note = thymine
misc_feature            1..21
                        note = Luc-siRNA sense; deoxythymidine (20)..(21); DNA/RNA
                         type sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 397
taaggctatg aagagatact t                                         21

SEQ ID NO: 398          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = Luc-siRNA antisense
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 398
aagtatctct tcatagcctt a                                         21

SEQ ID NO: 399          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
misc_feature            1..23
                        note = Antisense unmodified inclisiran
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
SEQUENCE: 399
acaaaagcaa aacaggtcta gaa                                       23

SEQ ID NO: 400          moltype = DNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = crook
source                  1..19
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 400
gcgaagcccc tactccact                                            19

SEQ ID NO: 401          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
```

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```
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 401
acgggacagt attctcagtn atcacctcat cccgcgaagc                        40

SEQ ID NO: 402         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 402
cccaataaag ctggacaaga atcacctcat cccgcgaagc                        40

SEQ ID NO: 403         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 403
ctgtaggttg cttaaaaggg atcacctcat cccgcgaagc                        40

SEQ ID NO: 404         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 404
ctggagcacc gttaaggaca atcacctcat cccgcgaagc                        40

SEQ ID NO: 405         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 405
tcccaataaa gctggacaag atcacctcat cccgcgaagc                        40

SEQ ID NO: 406         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 406
gcccctgtag gttgcttaaa atcacctcat cccgcgaagc                        40

SEQ ID NO: 407         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
```

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```
SEQUENCE: 407
ccctgaaaga ctactggagc atcacctcat cccgcgaagc                        40

SEQ ID NO: 408          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
SEQUENCE: 408
tgcttaaaag ggacagtatt ctcacctcat cccgcgaagc                        40

SEQ ID NO: 409          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
SEQUENCE: 409
gacctcaata ccccaagtcc atcacctcat cccgcgaagc                        40

SEQ ID NO: 410          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
SEQUENCE: 410
gagcaccgtt aaggacaagt ttcacctcat cccgcgaagc                        40

SEQ ID NO: 411          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
SEQUENCE: 411
ctctgtaaat ttggaagtgt ctcacctcat cccgcgaagc                        40

SEQ ID NO: 412          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
SEQUENCE: 412
caccatgagc taggtggagt atcacctcat cccgcgaagc                        40

SEQ ID NO: 413          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
SEQUENCE: 413
ttcctgaagt gacaaaggaa atcacctcat cccgcgaagc                        40

SEQ ID NO: 414          moltype = DNA  length = 40
```

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```
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 414
gaccaccagg aactatatct ttcacctcat cccgcgaagc                        40

SEQ ID NO: 415         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 415
gttccagaaa tacattggtt ttcacctcat cccgcgaagc                        40

SEQ ID NO: 416         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 416
aaccgcaagg gctttgtgaa atcacctcat cccgcgaagc                        40

SEQ ID NO: 417         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 417
gagcaagaag ttcccaggca ttcacctcat cccgcgaagc                        40

SEQ ID NO: 418         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 418
cagtagtagg catctggaat gtcacctcat cccgcgaagc                        40

SEQ ID NO: 419         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 419
gtcatgggtg tctgtgggtt atcacctcat cccgcgaagc                        40

SEQ ID NO: 420         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
```

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```
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
SEQUENCE: 420
gctctgtaaa tttggaagtg ttcacctcat cccgcgaagc                          40

SEQ ID NO: 421          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
modified_base           22..23
                        mod_base = OTHER
                        note = thymine
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..23
                        ncRNA_class = siRNA
SEQUENCE: 421
caggtctgga atgcaaagtc att                                            23

SEQ ID NO: 422          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
modified_base           22..23
                        mod_base = OTHER
                        note = thymine
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..23
                        ncRNA_class = siRNA
SEQUENCE: 422
tgactttgca ttccagacct gtt                                            23

SEQ ID NO: 423          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 423
caggtctgga atgcaaagtc atcacctcat cccgcgaagc                          40

SEQ ID NO: 424          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 424
tgactttgca ttccagacct gtcacctcat cccgcgaagc                          40

SEQ ID NO: 425          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..19
                        note = DNA
misc_feature            20..40
                        note = RNA
ncRNA                   20..40
                        ncRNA_class = siRNA
SEQUENCE: 425
cgaagcgccc tactccactc aggtctggaa tgcaaagtca                          40

SEQ ID NO: 426          moltype = DNA  length = 42
```

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```
FEATURE                 Location/Qualifiers
source                  1..42
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            order(1..19,41..42)
                        note = DNA
misc_feature            20..40
                        note = RNA
ncRNA                   20..40
                        ncRNA_class = siRNA
SEQUENCE: 426
cgaagcgccc tactccactc aggtctggaa tgcaaagtca tt                    42

SEQ ID NO: 427          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..19
                        note = DNA
misc_feature            20..40
                        note = RNA
ncRNA                   20..40
                        ncRNA_class = siRNA
SEQUENCE: 427
cgaagcgccc tactccactt gactttgcat tccagacctg                       40

SEQ ID NO: 428          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..19
                        note = DNA
misc_feature            20..40
                        note = RNA
ncRNA                   20..40
                        ncRNA_class = siRNA
modified_base           20..22
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 428
cgaagcgccc tactccacta ggcctggagt ttattcggaa                       40

SEQ ID NO: 429          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 429
ttccgaataa actccaggcc t                                           21

SEQ ID NO: 430          moltype = RNA  length = 28
FEATURE                 Location/Qualifiers
source                  1..28
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..28
                        ncRNA_class = siRNA
modified_base           11
                        mod_base = OTHER
                        note = thymine
SEQUENCE: 430
ctagacctgt tttgcttttg tgcgaagc                                    28

SEQ ID NO: 431          moltype = DNA  length = 33
FEATURE                 Location/Qualifiers
source                  1..33
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..12
                        note = DNA
misc_feature            13..33
                        note = RNA
```

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```
modified_base          23
                       mod_base = OTHER
                       note = thymine
ncRNA                  13..33
                       ncRNA_class = siRNA
SEQUENCE: 431
ccctactcca ctctagacct gttttgcttt tgt                                 33

SEQ ID NO: 432         moltype = DNA  length = 28
FEATURE                Location/Qualifiers
source                 1..28
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..7
                       note = DNA
misc_feature           8..28
                       note = RNA
ncRNA                  8..28
                       ncRNA_class = siRNA
modified_base          18
                       mod_base = OTHER
                       note = thymine
SEQUENCE: 432
cgaagcgcta gacctgtttt gcttttgt                                       28

SEQ ID NO: 433         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
ncRNA                  1..21
                       ncRNA_class = siRNA
misc_feature           22..40
                       note = DNA
SEQUENCE: 433
caggtctgga atgcaaagtc atcacctcat cccgcgaagc                          40

SEQ ID NO: 434         moltype = DNA  length = 28
FEATURE                Location/Qualifiers
source                 1..28
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..28
                       note = DNA
ncRNA                  1..21
                       ncRNA_class = siRNA
SEQUENCE: 434
caggtctgga atgcaaagtc agcgaagc                                       28

SEQ ID NO: 435         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..19
                       note = DNA
misc_feature           20..40
                       note = RNA
ncRNA                  20..40
                       ncRNA_class = siRNA
SEQUENCE: 435
cgaagcgccc tactccactc aggtctggaa tgcaaagtca                          40

SEQ ID NO: 436         moltype = DNA  length = 33
FEATURE                Location/Qualifiers
source                 1..33
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..12
                       note = DNA
misc_feature           13..33
                       note = RNA
```

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```
ncRNA                   13..33
                        ncRNA_class = siRNA
SEQUENCE: 436
ccctactcca ctcaggtctg gaatgcaaag tca                            33

SEQ ID NO: 437          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..19
                        note = DNA
misc_feature            20..40
                        note = RNA
modified_base           20..22
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   20..40
                        ncRNA_class = siRNA
SEQUENCE: 437
cgaagcgccc tactccacta ggcctggagt ttattcggaa                     40

SEQ ID NO: 438          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
ncRNA                   22..40
                        ncRNA_class = siRNA
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 438
aggcctggag tttattcgga atcacctcat cccgcgaagc                     40

SEQ ID NO: 439          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..19
                        note = DNA
misc_feature            20..38
                        note = RNA
ncRNA                   20..38
                        ncRNA_class = siRNA
modified_base           36..38
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 439
cgaagcgccc tactccactt agacttcctg aataacta                       38

SEQ ID NO: 440          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..21
                        ncRNA_class = siRNA
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 440
tagttattca ggaagtctat t                                         21

SEQ ID NO: 441          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
```

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```
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..19
                        note = RNA
misc_feature            20..38
                        note = DNA
ncRNA                   1..19
                        ncRNA_class = siRNA
SEQUENCE: 441
tagacttcct gaataactat cacctcatcc cgcgaagc                          38

SEQ ID NO: 442          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,17..19)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..19
                        ncRNA_class = siRNA
SEQUENCE: 442
tagacttcct gaataacta                                               19

SEQ ID NO: 443          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
ncRNA                   1..21
                        ncRNA_class = siRNA
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 443
tagttattca ggaagtctat ttcacctcat cccgcgaagc                        40

SEQ ID NO: 444          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
modified_base           36..38
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..19
                        note = DNA
misc_feature            20..38
                        note = RNA
ncRNA                   20..38
                        ncRNA_class = siRNA
SEQUENCE: 444
cgaagcgccc tactccactt catcacactg aataccaa                          38

SEQ ID NO: 445          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..21
                        ncRNA_class = siRNA
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 445
ttggtattca gtgtgatgat t                                            21

SEQ ID NO: 446          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
modified_base           1..3
```

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```
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..19
                        note = RNA
misc_feature            20..38
                        note = DNA
ncRNA                   1..19
                        ncRNA_class = siRNA
SEQUENCE: 446
tcatcacact gaataccaat cacctcatcc cgcgaagc                            38

SEQ ID NO: 447          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..19
                        ncRNA_class = siRNA
modified_base           order(1..3,17..19)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 447
tcatcacact gaataccaa                                                 19

SEQ ID NO: 448          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 448
ttggtattca gtgtgatgat ttcacctcat cccgcgaagc                          40

SEQ ID NO: 449          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
modified_base           38..40
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..19
                        note = DNA
misc_feature            20..40
                        note = RNA
ncRNA                   20..40
                        ncRNA_class = siRNA
SEQUENCE: 449
cgaagcgccc tactccactg tcatcacact gaataccaat                          40

SEQ ID NO: 450          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..23
                        ncRNA_class = siRNA
modified_base           order(1..3,21..23)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 450
attggtattc agtgtgatga ctt                                            23

SEQ ID NO: 451          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
```

-continued

```
modified_base          1..3
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
ncRNA                  1..21
                       ncRNA_class = siRNA
SEQUENCE: 451
gtcatcacac tgaataccaa ttcacctcat cccgcgaagc                        40

SEQ ID NO: 452         moltype = DNA  length = 42
FEATURE                Location/Qualifiers
misc_feature           1..23
                       note = RNA
source                 1..42
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           24..42
                       note = DNA
ncRNA                  1..23
                       ncRNA_class = siRNA
modified_base          order(1..3,21..23)
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
SEQUENCE: 452
attggtattc agtgtgatga ctttcacctc atcccgcgaa gc                     42

SEQ ID NO: 453         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
ncRNA                  1..21
                       ncRNA_class = siRNA
modified_base          order(1..3,19..21)
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
SEQUENCE: 453
gtcatcacac tgaataccaa t                                            21

SEQ ID NO: 454         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
modified_base          38..40
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
misc_feature           1..19
                       note = DNA
misc_feature           20..40
                       note = RNA
ncRNA                  20..40
                       ncRNA_class = siRNA
SEQUENCE: 454
cgaagcgccc tactccactg gtgtatggct tcaaccctga                        40

SEQ ID NO: 455         moltype = RNA  length = 23
FEATURE                Location/Qualifiers
source                 1..23
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          order(1..3,21..23)
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
ncRNA                  1..23
                       ncRNA_class = siRNA
SEQUENCE: 455
tcagggttga agccatacac ctt                                          23

SEQ ID NO: 456         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
```

-continued

```
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..21
                        note = RNA
ncRNA                   1..21
                        ncRNA_class = siRNA
misc_feature            22..40
                        note = DNA
SEQUENCE: 456
ggtgtatggc ttcaaccctg atcacctcat cccgcgaagc                        40

SEQ ID NO: 457          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 457
ggtgtatggc ttcaaccctg a                                            21

SEQ ID NO: 458          moltype = DNA  length = 42
FEATURE                 Location/Qualifiers
source                  1..42
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..23
                        note = RNA
misc_feature            24..42
                        note = DNA
ncRNA                   1..23
                        ncRNA_class = siRNA
modified_base           order(1..3,21..23)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 458
tcagggttga agccatacac ctttcacctc atcccgcgaa gc                     42

SEQ ID NO: 459          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
modified_base           38..40
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..19
                        note = DNA
misc_feature            20..40
                        note = RNA
ncRNA                   20..40
                        ncRNA_class = siRNA
SEQUENCE: 459
cgaagcgccc tactccactc accaacttct tccacgagtc                        40

SEQ ID NO: 460          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..23
                        ncRNA_class = siRNA
modified_base           order(1..3,21..23)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 460
gactcgtgga agaagttggt gtt                                          23

SEQ ID NO: 461          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
```

-continued

```
                        mol_type = other DNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 461
caccaacttc ttccacgagt ctcacctcat cccgcgaagc                    40

SEQ ID NO: 462          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 462
caccaacttc ttccacgagt c                                        21

SEQ ID NO: 463          moltype = DNA  length = 42
FEATURE                 Location/Qualifiers
source                  1..42
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..23
                        note = RNA
misc_feature            24..42
                        note = DNA
modified_base           order(1..3,21..23)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..23
                        ncRNA_class = siRNA
SEQUENCE: 463
gactcgtgga agaagttggt gtttcacctc atcccgcgaa gc                 42

SEQ ID NO: 464          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..19
                        note = DNA
misc_feature            20..38
                        note = RNA
modified_base           36..38
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   20..38
                        ncRNA_class = siRNA
SEQUENCE: 464
cgaagcgccc tactccacta ggcagagcta gtggcaaa                      38

SEQ ID NO: 465          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 465
tttgccacta gctctgcctt t                                        21

SEQ ID NO: 466          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
```

-continued

```
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..3
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..19
                        note = RNA
misc_feature            20..38
                        note = DNA
ncRNA                   1..19
                        ncRNA_class = siRNA
SEQUENCE: 466
aggcagagct agtggcaaat cacctcatcc cgcgaagc                              38

SEQ ID NO: 467          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,17..19)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..19
                        ncRNA_class = siRNA
SEQUENCE: 467
aggcagagct agtggcaaa                                                   19

SEQ ID NO: 468          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
ncRNA                   1..21
                        ncRNA_class = siRNA
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 468
tttgccacta gctctgcctt ttcacctcat cccgcgaagc                            40

SEQ ID NO: 469          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..19
                        note = DNA
misc_feature            20..38
                        note = RNA
ncRNA                   20..38
                        ncRNA_class = siRNA
modified_base           36..38
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 469
cgaagcgccc tactccactg agcaaatctc ttcaataa                              38

SEQ ID NO: 470          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 470
ttattgaaga gatttgctct t                                                21

SEQ ID NO: 471          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
```

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```
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..19
                        note = RNA
misc_feature            20..38
                        note = DNA
ncRNA                   1..19
                        ncRNA_class = siRNA
SEQUENCE: 471
gagcaaatct cttcaataat cacctcatcc cgcgaagc                      38

SEQ ID NO: 472          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..19
                        ncRNA_class = siRNA
modified_base           order(1..3,17..19)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 472
gagcaaatct cttcaataa                                           19

SEQ ID NO: 473          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 473
ttattgaaga gatttgctct ttcacctcat cccgcgaagc                    40

SEQ ID NO: 474          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..19
                        note = DNA
misc_feature            20..38
                        note = RNA
modified_base           36..38
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   20..38
                        ncRNA_class = siRNA
SEQUENCE: 474
cgaagcgccc tactccactc cacaaatgtc tacagcaa                      38

SEQ ID NO: 475          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 475
ttgctgtaga catttgtggt t                                        21

SEQ ID NO: 476          moltype = DNA  length = 38
```

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```
FEATURE                Location/Qualifiers
source                 1..38
                       mol_type = other DNA
                       organism = synthetic construct
modified_base          1..3
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
misc_feature           1..19
                       note = RNA
misc_feature           20..38
                       note = DNA
ncRNA                  1..19
                       ncRNA_class = siRNA
SEQUENCE: 476
ccacaaatgt ctacagcaat cacctcatcc cgcgaagc                          38

SEQ ID NO: 477         moltype = RNA  length = 19
FEATURE                Location/Qualifiers
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          order(1..3,17..19)
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
ncRNA                  1..19
                       ncRNA_class = siRNA
SEQUENCE: 477
ccacaaatgt ctacagcaa                                               19

SEQ ID NO: 478         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
modified_base          order(1..3,19..21)
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
ncRNA                  1..21
                       ncRNA_class = siRNA
SEQUENCE: 478
ttgctgtaga catttgtggt ttcacctcat cccgcgaagc                        40

SEQ ID NO: 479         moltype = DNA  length = 38
FEATURE                Location/Qualifiers
source                 1..38
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..19
                       note = DNA
misc_feature           20..38
                       note = RNA
ncRNA                  20..38
                       ncRNA_class = siRNA
SEQUENCE: 479
cgaagcgccc tactccactg aaacaggctt gaaagaat                          38

SEQ ID NO: 480         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
ncRNA                  1..21
                       ncRNA_class = siRNA
modified_base          order(1..3,19..21)
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
SEQUENCE: 480
attctttcaa gcctgtttct t                                            21

SEQ ID NO: 481         moltype = DNA  length = 38
FEATURE                Location/Qualifiers
source                 1..38
```

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```
                          mol_type = other DNA
                          organism = synthetic construct
modified_base             1..3
                          mod_base = OTHER
                          note = Phosphorothioate internucleotide linkage
misc_feature              1..19
                          note = RNA
misc_feature              20..38
                          note = DNA
ncRNA                     1..19
                          ncRNA_class = siRNA
SEQUENCE: 481
gaaacaggct tgaaagaatt cacctcatcc cgcgaagc                        38

SEQ ID NO: 482            moltype = RNA  length = 19
FEATURE                   Location/Qualifiers
source                    1..19
                          mol_type = other RNA
                          organism = synthetic construct
ncRNA                     1..19
                          ncRNA_class = siRNA
modified_base             order(1..3,17..19)
                          mod_base = OTHER
                          note = Phosphorothioate internucleotide linkage
SEQUENCE: 482
gaaacaggct tgaaagaat                                             19

SEQ ID NO: 483            moltype = DNA  length = 40
FEATURE                   Location/Qualifiers
source                    1..40
                          mol_type = other DNA
                          organism = synthetic construct
misc_feature              1..21
                          note = RNA
misc_feature              22..40
                          note = DNA
ncRNA                     1..19
                          ncRNA_class = siRNA
modified_base             order(1..3,19..21)
                          mod_base = OTHER
                          note = Phosphorothioate internucleotide linkage
SEQUENCE: 483
attctttcaa gcctgtttct ttcacctcat cccgcgaagc                      40

SEQ ID NO: 484            moltype = DNA  length = 38
FEATURE                   Location/Qualifiers
source                    1..38
                          mol_type = other DNA
                          organism = synthetic construct
misc_feature              1..19
                          note = DNA
misc_feature              20..38
                          note = RNA
modified_base             36..38
                          mod_base = OTHER
                          note = Phosphorothioate internucleotide linkage
ncRNA                     20..38
                          ncRNA_class = siRNA
SEQUENCE: 484
cgaagcgccc tactccactg agagaaatcg aagaggaa                        38

SEQ ID NO: 485            moltype = RNA  length = 21
FEATURE                   Location/Qualifiers
source                    1..21
                          mol_type = other RNA
                          organism = synthetic construct
ncRNA                     1..21
                          ncRNA_class = siRNA
modified_base             order(1..3,19..21)
                          mod_base = OTHER
                          note = Phosphorothioate internucleotide linkage
SEQUENCE: 485
ttcctcttcg atttctctct t                                          21

SEQ ID NO: 486            moltype = DNA  length = 38
FEATURE                   Location/Qualifiers
```

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```
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..19
                        note = RNA
misc_feature            20..38
                        note = DNA
ncRNA                   1..19
                        ncRNA_class = siRNA
SEQUENCE: 486
gagagaaatc gaagaggaat cacctcatcc cgcgaagc                          38

SEQ ID NO: 487          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..19
                        ncRNA_class = siRNA
modified_base           order(1..3,17..19)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 487
gagagaaatc gaagaggaa                                               19

SEQ ID NO: 488          moltype = DNA  length = 40
FEATURE                 Location/Qualifiers
source                  1..40
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..21
                        note = RNA
misc_feature            22..40
                        note = DNA
ncRNA                   1..21
                        ncRNA_class = siRNA
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 488
ttcctcttcg atttctctct ttcacctcat cccgcgaagc                        40

SEQ ID NO: 489          moltype = DNA  length = 38
FEATURE                 Location/Qualifiers
source                  1..38
                        mol_type = other DNA
                        organism = synthetic construct
modified_base           36..38
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
misc_feature            1..19
                        note = DNA
misc_feature            20..38
                        note = RNA
ncRNA                   20..38
                        ncRNA_class = siRNA
SEQUENCE: 489
cgaagcgccc tactccacta gttatagtcc gtgagcta                          38

SEQ ID NO: 490          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
ncRNA                   1..21
                        ncRNA_class = siRNA
SEQUENCE: 490
tagctcacgg actataactt t                                            21

SEQ ID NO: 491          moltype = DNA  length = 38
```

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```
FEATURE              Location/Qualifiers
source               1..38
                     mol_type = other DNA
                     organism = synthetic construct
misc_feature         1..19
                     note = RNA
misc_feature         20..38
                     note = DNA
modified_base        1..3
                     mod_base = OTHER
                     note = Phosphorothioate internucleotide linkage
ncRNA                1..19
                     ncRNA_class = siRNA
SEQUENCE: 491
agttatagtc cgtgagctat cacctcatcc cgcgaagc                         38

SEQ ID NO: 492       moltype = RNA  length = 19
FEATURE              Location/Qualifiers
source               1..19
                     mol_type = other RNA
                     organism = synthetic construct
modified_base        order(1..3,17..19)
                     mod_base = OTHER
                     note = Phosphorothioate internucleotide linkage
ncRNA                1..19
                     ncRNA_class = siRNA
SEQUENCE: 492
agttatagtc cgtgagcta                                              19

SEQ ID NO: 493       moltype = DNA  length = 40
FEATURE              Location/Qualifiers
source               1..40
                     mol_type = other DNA
                     organism = synthetic construct
misc_feature         1..21
                     note = RNA
misc_feature         22..40
                     note = DNA
ncRNA                1..21
                     ncRNA_class = siRNA
modified_base        order(1..3,19..21)
                     mod_base = OTHER
                     note = Phosphorothioate internucleotide linkage
SEQUENCE: 493
tagctcacgg actataactt ttcacctcat cccgcgaagc                       40

SEQ ID NO: 494       moltype = RNA  length = 21
FEATURE              Location/Qualifiers
source               1..21
                     mol_type = other RNA
                     organism = synthetic construct
modified_base        1..3
                     mod_base = OTHER
                     note = Phosphorothioate internucleotide linkage
modified_base        11
                     mod_base = OTHER
                     note = thymine
modified_base        order(1..6,8,10,12..21)
                     mod_base = OTHER
                     note = 2-Prime-O-methyl ribonucleotides
modified_base        order(7,9)
                     mod_base = OTHER
                     note = 2-Prime-deoxy-2-Prime-fluoro ribonucleotides
ncRNA                1..21
                     ncRNA_class = siRNA
SEQUENCE: 494
ctagacctgt tttgcttttg t                                           21

SEQ ID NO: 495       moltype = RNA  length = 23
FEATURE              Location/Qualifiers
source               1..23
                     mol_type = other RNA
                     organism = synthetic construct
modified_base        order(1..3,21..23)
                     mod_base = OTHER
                     note = Phosphorothioate internucleotide linkage
```

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```
modified_base          order(1,3,7,9,11,13,15,17,19..23)
                       mod_base = OTHER
                       note = 2-Prime-O-methyl ribonucleotides
modified_base          order(2,4..6,8,10,12,14,16,18)
                       mod_base = OTHER
                       note = 2-Prime-deoxy-2-Prime-fluoro ribonucleotides
ncRNA                  1..23
                       ncRNA_class = siRNA
SEQUENCE: 495
acaaaagcaa aacaggtcta gaa                                        23

SEQ ID NO: 496         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..21
                       note = RNA
misc_feature           22..40
                       note = DNA
modified_base          11
                       mod_base = OTHER
                       note = thymine
ncRNA                  1..21
                       ncRNA_class = siRNA
SEQUENCE: 496
ctagacctgt tttgcttttg ttcacctcat cccgcgaagc                      40

SEQ ID NO: 497         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
misc_feature           1..19
                       note = DNA
misc_feature           20..40
                       note = RNA
ncRNA                  20..40
                       ncRNA_class = siRNA
modified_base          30
                       mod_base = OTHER
                       note = thymine
SEQUENCE: 497
cgaagcgccc tactccactc tagacctgtt ttgcttttgt                      40

SEQ ID NO: 498         moltype = DNA  length = 40
FEATURE                Location/Qualifiers
source                 1..40
                       mol_type = other DNA
                       organism = synthetic construct
modified_base          30
                       mod_base = OTHER
                       note = thymine
misc_feature           1..19
                       note = DNA
misc_feature           20..40
                       note = RNA
ncRNA                  20..40
                       ncRNA_class = siRNA
SEQUENCE: 498
gcgaagcccc tactccactc tagacctgtt ttgcttttgt                      40

SEQ ID NO: 499         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          19..21
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
SEQUENCE: 499
gtcatcacac tgaataccaa t                                          21

SEQ ID NO: 500         moltype = RNA  length = 21
```

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```
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 500
gtcatcacac tgaataccaa t                                            21

SEQ ID NO: 501          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
misc_feature            1..23
                        note = artificial sequence
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,21..23)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 501
attggtattc agtgtgatga ctt                                          23

SEQ ID NO: 502          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           19..21
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 502
ggtgtatggc ttcaaccctg a                                            21

SEQ ID NO: 503          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 503
ggtgtatggc ttcaaccctg a                                            21

SEQ ID NO: 504          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
misc_feature            1..23
                        note = artificial sequence
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,21..23)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 504
tcagggttga agccatacac ctt                                          23

SEQ ID NO: 505          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           19..21
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 505
caccaacttc ttccacgagt c                                            21
```

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```
SEQ ID NO: 506          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 506
caccaacttc ttccacgagt c                                              21

SEQ ID NO: 507          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
misc_feature            1..23
                        note = artificial sequence
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,21..23)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 507
gactcgtgga agaagttggt gtt                                            23

SEQ ID NO: 508          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = artificial sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           17..19
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 508
aggcagagct agtggcaaa                                                 19

SEQ ID NO: 509          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = artificial sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 509
aggcagagct agtggcaaa                                                 19

SEQ ID NO: 510          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 510
tttgccacta gctctgcctt t                                              21

SEQ ID NO: 511          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = artificial sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           17..19
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
```

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SEQUENCE: 511
gagcaaatct cttcaataa                                           19

SEQ ID NO: 512         moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = artificial sequence
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          1..3
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
SEQUENCE: 512
gagcaaatct cttcaataa                                           19

SEQ ID NO: 513         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          order(1..3,19..21)
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
SEQUENCE: 513
ttattgaaga gatttgctct t                                        21

SEQ ID NO: 514         moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = artificial sequence
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          17..19
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
SEQUENCE: 514
ccacaaatgt ctacagcaa                                           19

SEQ ID NO: 515         moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = artificial sequence
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          1..3
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
SEQUENCE: 515
ccacaaatgt ctacagcaa                                           19

SEQ ID NO: 516         moltype = RNA  length = 21
FEATURE                Location/Qualifiers
misc_feature           1..21
                       note = artificial sequence
source                 1..21
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          order(1..3,19..21)
                       mod_base = OTHER
                       note = Phosphorothioate internucleotide linkage
SEQUENCE: 516
ttgctgtaga catttgtggt t                                        21

SEQ ID NO: 517         moltype = RNA  length = 19
FEATURE                Location/Qualifiers
misc_feature           1..19
                       note = artificial sequence
source                 1..19
                       mol_type = other RNA
                       organism = synthetic construct
modified_base          17..19
```

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                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 517
gaaacaggct tgaaagaat                                            19

SEQ ID NO: 518          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = artificial sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 518
gaaacaggct tgaaagaat                                            19

SEQ ID NO: 519          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 519
attctttcaa gcctgtttct t                                         21

SEQ ID NO: 520          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = artificial sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           17..19
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 520
gagagaaatc gaagaggaa                                            19

SEQ ID NO: 521          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = artificial sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 521
gagagaaatc gaagaggaa                                            19

SEQ ID NO: 522          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 522
ttcctcttcg atttctctct t                                         21

SEQ ID NO: 523          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = artificial sequence
source                  1..19
                        mol_type = other RNA
```

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```
                        organism = synthetic construct
modified_base           17..19
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 523
agttatagtc cgtgagcta                                                 19

SEQ ID NO: 524          moltype = RNA  length = 19
FEATURE                 Location/Qualifiers
misc_feature            1..19
                        note = artificial sequence
source                  1..19
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           1..3
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 524
agttatagtc cgtgagcta                                                 19

SEQ ID NO: 525          moltype = RNA  length = 21
FEATURE                 Location/Qualifiers
misc_feature            1..21
                        note = artificial sequence
source                  1..21
                        mol_type = other RNA
                        organism = synthetic construct
modified_base           order(1..3,19..21)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 525
tagctcacgg actataactt t                                              21

SEQ ID NO: 526          moltype = DNA  length = 28
FEATURE                 Location/Qualifiers
misc_feature            8..28
                        note = RNA
source                  1..28
                        mol_type = other DNA
                        organism = synthetic construct
misc_feature            1..7
                        note = DNA
SEQUENCE: 526
cgaagcgcag gtctggaatg caaagtca                                       28

SEQ ID NO: 527          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..23
                        ncRNA_class = siRNA
modified_base           order(1..3,21..23)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 527
ttccgaataa actccaggcc ttt                                            23

SEQ ID NO: 528          moltype = RNA  length = 23
FEATURE                 Location/Qualifiers
source                  1..23
                        mol_type = other RNA
                        organism = synthetic construct
ncRNA                   1..23
                        ncRNA_class = siRNA
modified_base           order(1..3,21..23)
                        mod_base = OTHER
                        note = Phosphorothioate internucleotide linkage
SEQUENCE: 528
ttccgaataa actccaggcc ttt                                            23
```

1. A nucleic acid molecule comprising:

a first part that comprises a double stranded inhibitory ribonucleic acid (RNA) molecule comprising a sense strand and an antisense strand; and

a second part that comprises a single stranded deoxyribonucleic acid (DNA) molecule, wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule or wherein the 3' end of the single stranded DNA molecule is covalently linked to the 5' of the antisense strand of the double stranded inhibitory RNA molecule, characterized in that the double stranded inhibitory RNA comprises a sense nucleotide sequence that encodes a part of a cardiovascular gene target associated with cardiovascular disease, or a polymorphic sequence variant thereof, and wherein said single stranded DNA molecule comprises a nucleotide sequence that is adapted over at least part of its length to anneal by complementary base pairing to a part of said single stranded DNA to form a double stranded DNA structure comprising a stem and a loop domain, characterized in that said nucleic acid molecule comprises N-acetylgalactosamine and said double stranded inhibitory RNA consists of natural nucleotides.

2-3. (canceled)

4. The nucleic acid molecule according to claim **1**, wherein said loop domain comprises the nucleotide sequence GCGAAGC.

5. The nucleic acid molecule according to claim **4** wherein said single stranded DNA molecule comprises the nucleotide sequence selected from the group:

```
                    (SEQ ID NO 387 and 251)
5' TCACCTCATCCCGCGAAGC 3';

                              (SEQ ID NO 130)
5' CGAAGCGCCCTACTCCACT 3'.;
and

                              (SEQ ID NO 400)
5' GCGAAGCCCCTACTCCACT 3'.
```

6. The nucleic acid molecule according to claim **1**, wherein:

said inhibitory RNA molecule comprises a two-nucleotide overhang comprising or consisting of at least one deoxythymidine dinucleotide (dTdT);

said sense and/or said antisense strands comprises at least one internucleotide phosphorothioate linkages; and/or

said nucleic acid molecule comprises a vinylphosphonate modification.

7-8. (canceled)

9. The nucleic acid molecule according to claim **1**, wherein said double stranded inhibitory RNA molecule is between 17 and 29 nucleotides or 19 to 21 nucleotides in length.

10. The nucleic acid molecule according to claim **1**, wherein said cardiovascular gene target is human Lipoprotein (a).

11. The nucleic acid molecule according to claim **10**, wherein said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence selected from the group consisting of: SEQ ID NO: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33 and 34.

12. The nucleic acid molecule according to claim **10**, wherein;

said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 41 and a sense nucleotide sequence comprising SEQ ID NO: 49, wherein said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule;

said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 4 and a sense nucleotide sequence comprising SEQ ID NO: 44, wherein said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule; or

said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 5 and a sense nucleotide sequence comprising SEQ ID NO: 46, wherein said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule.

13-14. (canceled)

15. The nucleic acid molecule according to claim **1**, wherein said cardiovascular gene target is human Apolipoprotein C III (Apo C III).

16. The nucleic acid molecule according to claim **15**, wherein;

said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78 and 79;

said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249 and 250; or

said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 50, 51, 52, 53, 54, 55, 56, 57, 58, 80, 81, 82, 83, 84, 85, 86, 87, 88 and 89.

17-18. (canceled)

19. The nucleic acid molecule according to claim **1**, wherein said cardiovascular gene target is human diglyceride acyltransferase 2 (DGAT2).

20. The nucleic acid molecule according to claim **19**, wherein;

said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118 and 119;

said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169 and 170: or

said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 120, 121, 122, 123, 124, 125, 126, 127, 128 and 129.

21-22. (canceled)

23. The nucleic acid molecule according to claim **1**, wherein said cardiovascular gene target is human PCSK9.

24. The nucleic acid molecule according to claim **23**, wherein;

said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 189, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209 and 210; or

said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209 and 210.

25. (canceled)

26. The nucleic acid molecule according to claim **1**, wherein said cardiovascular gene target is human Apolipoprotein B.

27. The nucleic acid molecule according to claim **26**, wherein;

said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 499, 500, 453, 502, 503, 457, 505, 506, 462, 508, 509, 467, 511, 512, 472, 514, 515, 477, 517 518, 482, 520, 521, 487, 523, 524 and 492; or

said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 450, 501, 455, 504, 460, 507, 465, 510, 470, 513, 475, 516, 480, 519, 485, 522, 490 and 525.

28. (canceled)

29. The nucleic acid molecule according to claim **1**, wherein said nucleic acid molecule is covalently linked to N-acetylgalactosamine.

30. A pharmaceutical composition comprising at least one nucleic acid molecule according to claim **1** and a pharmaceutically acceptable carrier or excipient.

31. The pharmaceutical composition according to claim **30** wherein said composition comprises at least one further therapeutic agent.

32. A method of treating a subject that has or is predisposed to hypercholesterolemia or diseases associated with hypercholesterolemia, comprising administering to the subject an effective dose of the nucleic acid molecule according to claim **1**.

33. The nucleic acid molecule according to claim **1**, wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule and said N-acetylgalactosamine is linked to the 3' end of said sense strand.

* * * * *